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An insight into Organic Farming in Nutricereals



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Foreword

The global quest for nutritious food, security of farmers, sustainable agriculture and conservation of environment is fuelling a revolution in organics and millets. Globally, over 43.7 million hectares are organic, with over 2 million producers from 172 countries. The total area under organic cultivation and the global food market are increasing at 10 and 12 per cent respectively. With climatic change wreaking havoc on agricultural land through increased temperature, water shortage, and the chemicals used in conventional farming practices only compounding the problems, sustainable organic farming and millet cultivation is the last hope to produce food without destroying the environment.

“Organic Agriculture - a production system that sustains the health of soils, ecosystem and people” and is based on the principles of health, ecology, fairness and care which primarily aims at cultivating the land and raising crops in such a way, as to keep the soil alive and in good health. Millets indeed have a great answer to all kinds of crises including that of water, nutrition and climate change and further carry the promise of mitigating the miseries of malnutrition prevalent over so widely in the country. More importantly, its better suitability to different cropping systems and mid season correction during vagaries of monsoon in the contingent plans made millets even more popular. Millet consumption has witnessed a steep decline in the past decades with a decrease in production from 40 to 20 per cent since the Green Revolution. The Loss of Millets is not just an agricultural loss, but a civilization loss. However, today with more people realizing the health and lifestyle benefits associated with millets, there is an increased global demand for this crop.

The publication entitled “**An Insight into Organic Farming in Nutricereals**” speaks about the global perspective of organic agriculture in these wonder crops- Millets, which seeks to pursue a holistic approach encompassing not only farming as such but also the wider implications it has for factors such as food, health, nutrition, livelihood and ecological securities. In this endeavour, I would like to commend the sincere efforts made by the scientists of Agricultural Research station, Vizianagaram in bringing out this publication which would be a snapshot for the progress in organic millet cultivation.



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Foreword

Millets are the ancient crops of the mankind and are important for rainfed agriculture. They are nutritionally rich and provide number of health benefits to the consumers. In recent times, the millets have become important owing to their good nutritional value, documented health benefits, versatile environmental adaptation, sustainability in low input agriculture and organic cultivation amenability. Millets are not just food, but are an integral part of the culture of thousands of communities all over the country. Tribal areas in central India and the Himalayas, are the home of millets, where they are the mainstay of agriculture, diet and the cultural systems of the regions. Moreover millets grown in these tribal areas are by default organic, cultivated with available organic manures only. Organic millet cultivation is a holistic production management system which promotes and enhances agro-ecosystem health, including biodiversity, biological cycles, and soil biological activity.

This book “**An Insight into Organic Farming in Nutricereals**” summarizes the broad information on the importance of organic agriculture in millets, explaining in detail the key themes of nutrient management, residue management, weed management and insect pest management supplemented with comprehensive case studies conducted in different AICRP centers all over the nation. This book also highlights once more the principles of organic agriculture, an environmentally sound and site-adapted agricultural approach. Further, it gives a clear picture to the reader, describing all the features that so amazingly combine and make Millets- The Miracle grains and elucidate, how sustainable organic farming and millet cultivation together remain our answer to the climatic crises that the world is encountering today.

The authors and contributors are highly appreciated for their efforts to make this book a unique and indispensable resource that offers a diverse range of valuable information, data and perspectives on organic agriculture in millets at a time when the farming community is increasingly aware of the problems of our current agricultural practices and the importance of creating sustainable agricultural systems for the long-term health of humankind and the biosphere as a whole.

Prabhakar Bhat

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CHAPTER I: INTRODUCTION TO ORGANIC FARMING

a. History of Organic Farming

Agriculture has been the basic source of subsistence for man over thousands of years and provides a livelihood to half of the world's population even today. Agriculture is usually described as a gamble with monsoons because of the great deal of uncertainty about crop prospects, as monsoon plays a decisive role in determining agricultural output and their failures resulted in wide spread famine and misery.

Indian agriculture before green revolution was characterized mainly by small and marginal farmers producing food and basic animal products for their family and local village communities. Farming was highly decentralised with individual farmers deciding on the types of crops to grow depending on climate, soil conditions, methods of controlling pests and diseases, building soil fertility and structure in their own ingenious ways, since farming did not include the use of chemical pesticides or fertilizers. Soil health and pests control were achieved using practices such as shifting cultivation, conservation, use of animal manure, farm wastes and introduction of legumes into crop rotations. Growing a mixture of crops in the fields has insulated the farmers from total crop failure caused by weather or pests epidemics. All these practices viz., intercropping, crop rotation, fallowing, composting and manuring has allowed continued farming on the same land for more than 2000 years without drop in yields and relatively free from pests and diseases.

During the British rule, Indian grain economy hinged on a unilateral relation of exploitation. A tax of 50-63 per cent revenue was collected, which has lead to more than 1/3rd of the irrigated land out of production. Consequently when India gained independence, the weakened country quickly became vulnerable to frequent famines, financial instabilities and low productivity. In 1964-65 and 1965-66, India experienced two severe droughts which led to food shortages and famines along the country's growing population. These factors formed a rationale for the implementation of Green Revolution as a development strategy in India.

After Green Revolution was launched in India, substantial increase in the production of food grains was achieved through the use of improved crop varieties and higher level of inputs of fertilizers and plant protection chemicals. The introduction of chemicals in farming was delirious at the sight of what they could accomplish. Yields exploded, at the start, the soil was healthy. Any damage brought about by the chemical fertilizer was hardly noticeable. Pests have not developed resistance to the chemicals and the technology spread across the world as it was considered the revolution in agriculture.

But over a period of time, it has now been realized that the increase in production was achieved at the cost of soil health and that sustainable production at higher levels is possible only by the proper use of factors which will help to maintain the fertility of the soil. About 60% of our agricultural land currently under cultivation suffer from indiscriminate use of chemical fertilizers and irrigation water. The gravity of environmental degradation resulting from faulty agricultural practices has caused alarm among the concerned farmers, scientists and conservationists and greater viable and sustainable farming systems have become a necessity. One such alternative agricultural system which will help to overcome the problems of soil degradation and declining soil fertility is Organic Farming.

b. Why Organic Farming ?

As the global population began to grow rapidly the need for a more sustainable approach in food production became more obvious. Unsustainable farm practices geared only towards higher yields and put additional pressure on global issues such as climate changes, loss of biodiversity, soil erosion and pollution of the most important natural resources (Soil and water).

In an attempt to accommodate global food demands and in efforts to mitigate the afore mentioned problems, many countries and farmers have been turning towards farm practices. As a completely natural and sustainable farm management practice, Organic farming is based on unique values. It is not only a farm practice but also a philosophy of working together with nature. As a holistic farm management approach organic farming aims to create a socially, environmentally and economically sustainable food production system. More precisely, organic farming is based on managing the agro-ecosystem rather than relying on external farming inputs, such as artificial fertilizers, additives and genetically modified organisms.

c. Main Principles of Organic Farming :

Organic farming is strictly regulated with few common principles.

1. **The Principles of Health:** Claims that organic farming is intended to protect and enhance the health of all organisms (this includes soil micro organisms, as well as humans.)
2. **The Principle of Ecology:** Which is based on the fact that organic farming should protect and benefit the environment (landscape, climate, natural habitats, bio diversity, air , water and soil)
3. **The Principle of Fairness:** Aims to provide socially and ecologically appropriate management over environmental resources, as well as provide a sufficient supply of quality food and other product.
4. **The Principle of Care:** Stresses precautions and responsibility as the key concerns in organic farm management.



d. Objectives of Organic Farming:

1. To develop a sustainable farming system, which maintains and /or improves soil fertility so as to ensure adequate food production into the foreseeable future.
2. A self sufficient or more realistically, a self sustaining agriculture, *i.e.* a system which relies as much as possible upon resources from within its own area and which is not reliant upon large quantities of imported resources.
3. An agriculture, which takes as its guide, the working of biological process as in natural ecosystem.

e. Organic Farm Management Practices:

1. Crop rotation: Used to maintain soil fertility and to improve crop protection from various pests.
2. Organic nutrient management: Based on improvement of soil organic matter through manuring, composting and mulching.
3. Growing cover crops: Beneficial practice for controlling insect pests and weeds, preventing soil erosion, as well as improving the nutrient content in the soil.
4. Preventive crop protection measures: choosing resistant varieties, adoption of planting or sowing and harvesting time
5. Relying on natural predators as a biological pest protection measure.
6. Weeding as a non chemical weed management practices.
7. Anaerobic soil disinfestations that eliminates or reduce soil borne pests.
8. Proper space between the crops.
9. Mechanical soil cultivation.
10. Recycling material.
11. Relying on renewable resource.

f. Advantages of Organic Farming:

1. Organic farming helps to prevent environmental degradation and can be used to regenerate degraded areas.
2. Organic farming helps to avoid chain reaction in the environment from chemical sprays and dusts.
3. Organically grown crops are believed to provide healthier and nutritionally superior food for man and animals than those grown with commercial fertilizers.
4. Organically grown crops are more resistant to diseases and insects and hence only a few chemical sprays or other protective treatment are required.
5. There is an increasing consumer demand for agricultural products which are free of toxic chemical residues.
6. Organically grown products gain importance because of high quality, premium price beside being free from toxic residues.
7. Organic fertilizers are considered as complete plant food. Organic matter restores the pH of the soil, which may become acidic due to continuous application of chemical fertilizers.
8. Organic manures, the principal component of Organic Farming produce optimal conditions in the soil for high yields and good quality crops by providing all the nutrients required by the plant (NPK, secondary and micronutrients), improving growth and physiological activities of plants.
9. Most of the organic manures are wastes or by-products which on accumulation may lead to pollution. By the way of utilizing them for organic farming, pollution is minimized.
10. Organic farming is labour intensive, which will also helps in generating more employment in rural areas.
11. As a whole, adoption of organic farming provides better and balanced environment, better products and living conditions to the human beings.

g. Can Organic Manures compensate Chemical Fertilizers?

In order to maintain high yields and protect the environment, the replacement of chemical fertilizers with organic manures has received increasing attention in recent years. Increase in grain yields over recent decades were largely dependent on heavy investments in chemical fertilizers. The nutrients required for any crop are available in the name of fertilizers in two forms namely organic and inorganic. Both the organic and inorganic fertilizers provide the plants with the essential nutrients needed for the plant to grow healthy and strong. The type of fertilizer used is a matter of personal choice. However, each contains different ingredients and supplies these nutrients in different ways. Organic fertilizers create a healthy growing environment, while inorganic fertilizers provide rapid nutrition.

h. Organic Vs Inorganic fertilizers

Organic fertilizers:

1. They contain only plant or animal based materials that are either a by product or end product of naturally occurring process, such as manures, leaves and compost.
2. The biggest advantage of an organic fertilizer is that the danger of over– fertilization is reduced as nutrients are released slowly and hence they are available over a longer period and only few applications are required.
3. Organic fertilizer improves the soil by escalating the soil's ability to hold water and nutrients and decreases the erosion and soil crusting caused by rain and wind.
4. Using organic fertilizers add more natural nutrients, feed important microbes in the soil and improve the structure of the soil.
5. Organic fertilizers continue to improve the quality of soil even after the plants have taken the required nutrients and therefore, the longer the soil is fed with organic fertilizer, the better its composition and texture will be.
6. Nutrients are not immediately available to plants because they need time to break down. You may have a period of time where the plants struggle before the fertilizer starts working its magic.
7. They can typically be more expensive and less available than inorganic options.
8. The final amount of nutrients going into the soil can only be a guess as there are many factors to consider such as the age of the raw material, where it came from, condition of soil and weather conditions.

Inorganic fertilizers:

1. They are referred to as synthetic fertilizers and are manufactured artificially and contain minerals or synthetic chemicals.
2. Synthetic nitrogen fertilizers are typically made from petroleum or natural gas, where as phosphorus, potassium and other trace elements in inorganic fertilizers are often mined from the earth.
3. Inorganic fertilizers or chemicals come in different forms like dry, liquid, slow release, pelletized and soluble solutions.

Types of Organic Manures



Farmyard Manure



Compost



Vermicompost



Poultry Manure



Azolla Biofertilizer



Green manure

Types of inorganic fertilizers



Urea



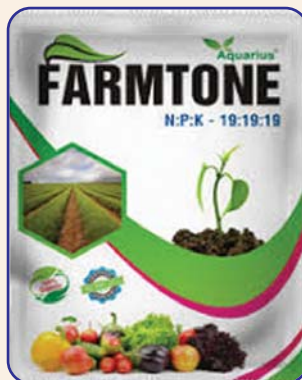
DAP



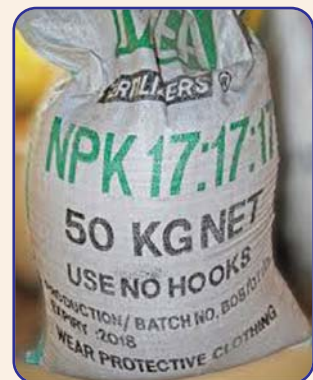
MOP



Single Super Phosphate



19-19-19



17-17-17

4. Inorganic fertilizers provide plant nutrients in a readymade form and release the nutrients as soon as possible.
5. Inorganic fertilizers have the necessary amounts of the nutrients that the plants require to help them to survive and flourish.
6. As inorganic fertilizers tend to leach, excessive use can lead to a build up of salts in the soil, causing damage to the plant .
7. Chemical fertilizers do not build up the soil quality and long term use of the same will deplete the soil of valuable microbes.

i. Main Demerits in use of inorganic fertilizers and other Agro-chemicals on soil and plant:

Excessive use of chemical fertilizers and other agro-chemicals which are the important inputs in modern farming creates depletion of soil fertility and causes pollution in surface water bodies.

1. Water soluble fertilizers when applied to soil, a good portion of the added nutrients does not become available to the crop plants and cost either to the atmosphere up to the hydrosphere due to non stimulation of the activities of heterotrophic soil organisms but facilitate that of the autotrophic nitrifying organisms, there by hindering the immobilization of nutrients.
2. As a matter of fact, it results in rapid rate of nutrients loss in different forms and increases the soil acidity with nitrification.
3. Emission of ammonia, Methane, nitrous oxide and elemental nitrogen from the soil system as a result of denitrification.
4. Depletion of secondary and micronutrients especially sulphur and zinc.
5. Deficiency of these nutrients (S & Zn) along with that of Mg, Mn, Fe, Mo, B and Cu limits productivity of many fields crops especially in rice.
6. Addition of large doses of N-fertilizers in modern agriculture without the use of organic manures, there is always the danger of humus depletion and fall in crop production, which can be avoided only by adding amounts of organic residues and manures.
7. When high levels of N-fertilizers especially nitrate forms are applied to the soil, nitrate pollution of drinking water is a serious health hazard found in extensively irrigated coarse textured highly percolating soils, where 40-50% of applied nitrogen is lost in leaching.
8. Alarming issue to human health is regular use of phosphatic fertilizer in large quantities often causes the build up of trace metal contamination such as arsenic, fluoride, cadmium etc. in soil and plants. Cadmium in single super phosphate is available to plants as the Cd in cadmium chloride. Similarly, chloride contained in MOP and NH_4Cl creates toxicity to many crops like beans, citrus, grapes lettuce, potatoes etc. These trace metal toxic contaminants reach the human body, through food chain and cause health problems.
9. The water soluble nutrients when carried to lakes and stream through leaching and surface run off cause eutrophication as manifested by the luxuriant growth of algae and other water weeds on the water surface leading to oxygen deficient condition. This situation is not conducive to healthy aquatic life.

CHAPTER II: MILLETS- THE WONDER GRAINS

a. Millets as Miracle Grains:

India is the largest producer of millets in the world, and accounts for more than 40 per cent of the global consumption. Millet cultivation is the main stay of rainfed farming which provide livelihood to nearly 50 per cent of total rural work force and sustain 60 per cent of cattle population in India. Millets are most unique among cereals. The term Millets includes a number of small grained cereal grasses. Based on the grain size, millets have been classified as major millets which include Sorghum [*Sorghum bicolor*(L.)] and Pearl millet /Bajra [*Pennisetum glaucum* (L.) R.Br.] and the small millets which include Finger millet [*Eleusine coracana* (L.) Gaertn.], Foxtail millet [*Setaria italica*(L.) P. Beauv.], Little millet(*Panicum sumatrense*), Kodo millet [*Paspalum scrobiculatum* (L.)], Proso Millet [*Panicum miliaceum* (L.)] and Barnyard millet (*Echinochloa frumentacea*).



Generally, Millet production is not dependent on the use of synthetic fertilizers. Most millet farmers use farm yard manure, household produces and biofertilizers. Therefore, they can significantly reduce the huge burden of fertilizer subsidy borne by the government. Grown under traditional methods, no millet attracts any pest. They can be termed as pest- free crops. A majority of them are not affected by storage pests either. Therefore, their need for pesticides is close to nil. Thus, they are a great boon to the agricultural environment. In addition, millets sequesterate carbon, thereby adding to CO₂ abatement opportunities, contribute to improved agro - biodiversity by their varietal diversity, allow for mutually beneficial intercropping with other vital crops and have significant cultural values due to their long history.

i) Millets as Foods of Future:

Millets grow under dry conditions, can perform well in relatively poor soil and require low inputs. Compared to irrigated commodity crops currently promoted by policy measures, millets need very less irrigation and require just around 25 per cent of the rainfall regime demanded by crops such as sugarcane and banana. Thus, they do not burden the state with demands for irrigation or power. Millets are adapted to wide range of ecological conditions often growing on skeletal soil that are less than 15cm deep. It does not demand rich soil for their survival and growth. Hence millets are a boon for vast dryland area.

Moreover millets are a staple food with superior nutritional qualities compared to other cereals for the poor and farmers in dry areas including the Deccan, Central India, Western Indian states such as Gujarat and Rajasthan. Used as a dual purpose crop (food and fodder) they make strong economic sense in mixed farming systems.



Comparative Rainfall requirement of various crops

Crop	Rainfall requirement (in mm)
Sugarcane	2000-2200
Banana	2000-2200
Rice	1200-1300
Chillies	600
Cotton	600-650
Maize	500-550
Groundnut	450-500
Sorghum	400-500
Bajra	350-400
Ragi	350-400
Pulses	300-350
Sesame	300-350

ii) Millets as Store-Houses of Nutrition:

Millets are called as the Nutri-cereals/ Nutri-grains as they are highly nutritious, non-glutinous, non acid forming and easily digestible foods. Millet ingestion helps in a slower release of glucose over a longer period of time, thus due to low glycemic index(GI), their habitual intake reduces the risk of *diabeties mellitus*. Further, millets are rich sources of minerals like iron, calcium, zinc, magnesium, phosphorus and potassium. By any nutritional parameter, millets are miles ahead of rice and wheat. Finger millet is very rich in calcium, It is thirty times more than rice, while every other millet has at least twice the amount of calcium compared to rice. In iron content, little millet and Pearl millet are so rich that rice is nowhere in the race. They also contain appreciable amounts of dietary fibre, Vitamins, folic acid and high amounts of lecithin which are useful for strengthening of the nervous system. While the essential micronutrient Beta Carotene which is consumed as pharmaceutical pills and capsules is abundantly found in millets, while the much privileged rice, ironically, has zero quantity of this precious micronutrient. Millets are also rich in phytochemicals (polyphenols, tannins, phytosterols) and antioxidants; however, they do contain some anti-nutritional factors that can be reduced by certain processing treatments.



In this fashion, nutrient to nutrient, every single millet is extraordinarily superior to rice and wheat and therefore, a regular consumption can help to overcome malnutrition that affects a vast majority of the Indian population.

Despite numerous qualities, utilization of millets as food is confined to the traditional consumers, particularly the tribal populations. This is mainly due to the non-availability of consumer friendly, ready-to-use/ready-to-eat millet based products. Recently, millets have gained attention and efforts are under way to obtain their convenient and value added processed products. Although among the food crops, millets occupy relatively a lower position in Indian agriculture, they are quite important from the point of food security at regional/household level.

Nutritional Content of Millets: (In 100gm)

Food grain	Carbohydrates (gm)	Protein (gm)	Fat (gm)	Energy (gm)	Crude fibre (gm)	Mineral Matter (gm)	Ca (mg)	P (mg)	Fe (mg)
Finger millet	72.0	7.3	1.3	328	3.6	2.7	344	283	3.9
Kodo millet	65.9	8.3	1.4	309	9.0	2.6	27	188	0.5
Proso millet	70.4	12.5	1.1	341	2.2	1.9	14	206	0.8
Fox millet	60.9	12.3	4.3	331	8.0	3.3	31	290	2.8
Little millet	67.0	7.7	4.7	341	7.6	1.5	17	220	9.3
Barnyard millet	65.5	6.2	2.2	307	9.8	4.4	20	280	5.0
Sorghum	72.6	10.4	1.9	349	1.6	1.6	25	222	4.1
Bajra	67.5	11.6	5.0	361	1.2	2.3	42	296	8.0
Wheat (Whole)	71.2	11.8	1.5	346	1.2	1.5	41	306	5.3
Rice (raw, milled)	78.2	6.8	0.5	345	0.2	0.6	10	160	0.7

(Source: Nutritive value of Indian foods, NIN, 2007)

iii) Millets grow in Poor of the Poorest soils:

Most millets can be grown on low fertility soils, some in acidic soils, some on saline soils. Millets such as Pearl millet can also be grown on sandy soils, as is done in Rajasthan. In fact, finger millet grows well in saline soils. Barnyard millet too thrives in problem soils, where as other crops like rice, and struggle to grow in such soils. Many of them are also grown to reclaim soils. Poor farmers especially in dryland India are owners of very poor lands. The only crops that sustain agriculture and food security on these lands are millets.

In fact, the capacity of millets to grow on poor soils can be gauged from the fact that they grow in Sahelian soil conditions in West Africa which produces 74% of all the millets in Africa and 28% of the world production. If they flourish in such ecological zones where average rainfall can be less than 500 mm using soils that are sandy and slightly acid, it is a testimony for their hardiness and extraordinary capacity to survive very harsh conditions. That is why millets can withstand drought like conditions in the Deccan and Rajasthan and produce food and fodder for people and livestock, respectively.

iv) Millets do not demand synthetic fertilizers:

Millets do not demand chemical fertilizers. In fact, under dry land conditions, millets grow better in the absence of chemical fertilizers. Therefore, most millet farmers grow them using farmyard manure under purely ecofriendly conditions. In recent years farmers have also started using biofertilisers such as vermicompost produced in their backyard and growth promoters such as panchagavya, aamritpani etc. These practices make millet production not only ecofriendly but stays under the control of farmers.



v) Millets are mostly pest free:

Growing traditional local land races and under ecological conditions, most millets such as foxtail are mostly pest free, and hence do not need much pesticides. Even in storage conditions, most millets such as foxtail not only need any fumigants, but act as anti pest agents to store delicate pulses such as green gram.

vi) Millets are not just crops but a cropping system:

Most millets grown under traditional practices are a Farming System and not just a crop. Most millet fields are inherently biodiverse. This is the tradition of millet farming in the country. Six to twenty crops are planted on the same space at the same time. The famous “Baranaja” cropping systems in the Himalayas are a testimony to this. In this millet led system are embedded 12 different crop varieties. “Saat Dhan” in Rajasthan also is a host to a large variety of millets. The “Pannendu Pantalu” system of the South, grow millets in combination with pulses and oilseeds, thus making it a holistic farming system.

vii) Millets produce multiple security:

Major cereal crops such as rice and wheat can succeed in producing only food security for India, whereas millets can provide multiple securities (Food, Nutrition and Financial security) including security of livelihood and ecology. Most millets have edible stalks which are the most favoured fodder for cattle. Many a time, crops such as sorghum and pearl millet are grown only for their fodder value. Besides fodder, millets are storehouses of nutrition and hence produce nutrition security. Being hosts to diverse crops such as red gram, millet fields also produce fuelwood and fibre. The legume crops that are companion crops for millets are also prolific leaf shedders. This leaf fall acts as natural manure and maintains soil fertility. Thus, millet farms not just use soil fertility for their growth but also return this fertility to the soil.



viii) Millets as climate Resilient crops:

Due to all the qualities mentioned above, Millets remain our answer to the climatic crises that the world is facing. Climate change is expected to confront us with the challenges viz., Less rainfall, Increase in temperatures upto 2-5 degree Celsius, increased water stress and severe malnutrition. If there is any cropping system that can with stand these challanges, survive and flourish, It is the millet system. It is important to note that rice grown under standing water makes it a dangerous crop under climate change conditions. Methane emanating from water, drenched rice fields, is a green house gas, that severely threatens our environment. Wheat being a thermally sensitive crop, with increasing temperatures, its production is liable to be adversely affected. Thus, in due course, wheat might disappear from our farms. Besides, millets help in reducing the atmospheric CO₂ and thus contribute in mitigating the climate change. Hence, wheat and rice might provide only food security, Whereas Millets produce multiple securities (food, fodder, health, nutrition, livelihood, and ecological.) which make them the crops of agricultural security.



In view of all these features that they so amazingly combine, Millets are harbingers of evergreen revolution and they can also be called as Miracle Grains.

b) Disappearing Millet System:

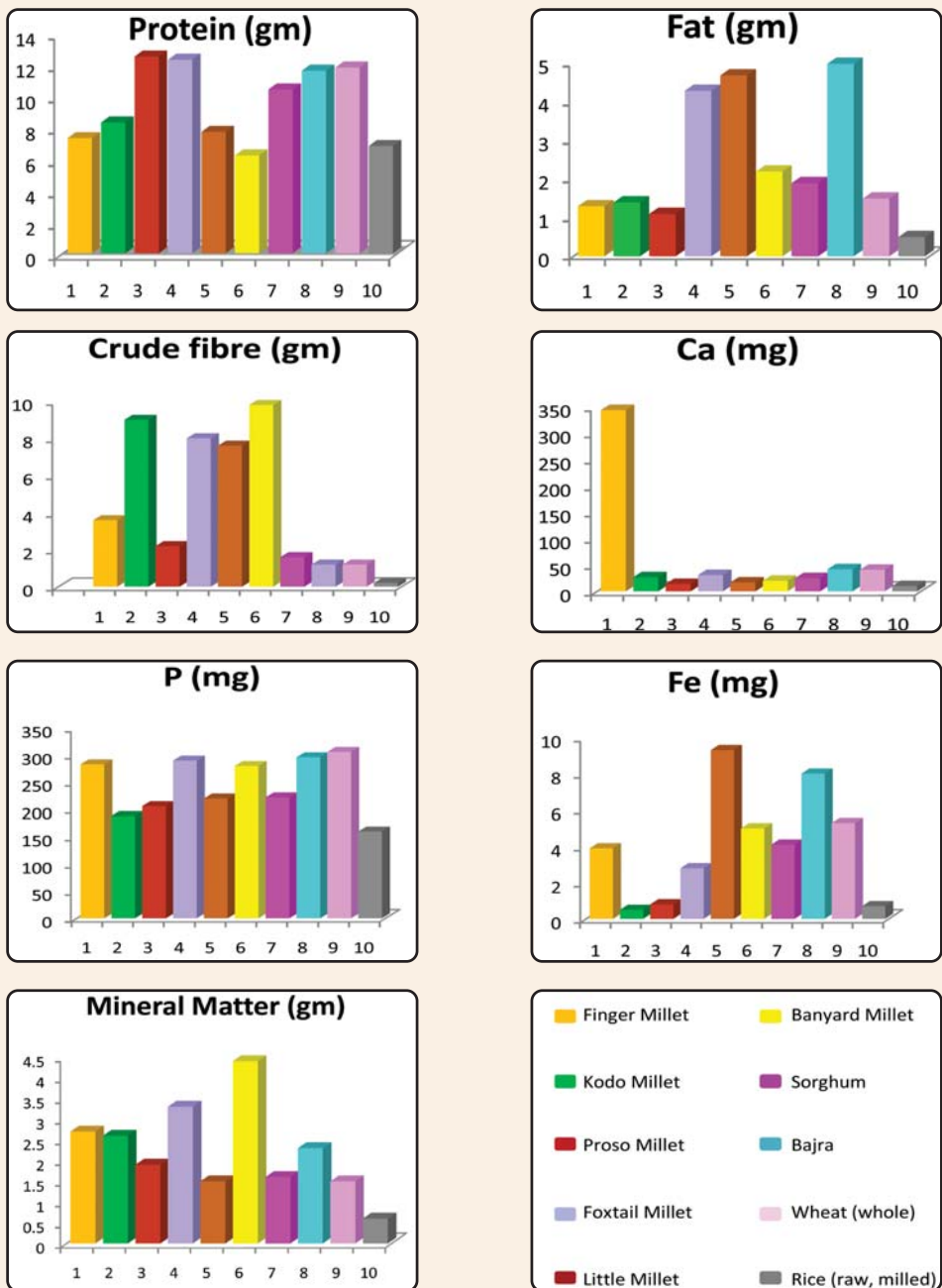
In spite of all these extraordinary qualities and capabilities of millet farming systems, the area under millet production has been shrinking rapidly over the last five decades, since the Green Revolution. Between 1966 and 2006, 44% of millet cultivation areas were occupied by other crops signifying an extraordinary loss in India's food and farming systems. A combination of factors like low remuneration as compared to other food crops, lack of input subsidies and price incentives, subsidized supply of fine cereals through Public Distribution System(PDS), change in consumer preference, declining state support in terms of crop loans and crop insurance has significantly contributed to this decline and fall of millets in Indian agriculture .

Fifty years of cultivation of millets vis a vis other crops in India (Area in Million ha)

Years										
Crops	1956 - 1961	1961- 1966	1966- 1971	1971- 1976	1976- 1981	1981- 1986	1986- 1991	1991- 1996	1996 2001	2001- 2006
Millets	36.2	36.81	37.89	35.27	34.16	33.39	30.88	25.9	23.17	21.31
Rice	33.14	35.63	36.79	38.02	39.77	40.5	41.31	42.53	44.31	42.85
Wheat	12.84	13.33	15.73	19.13	21.89	23.39	23.59	24.74	26.67	26.2
Cereals	89.65	93.29	98.89	100.95	103.58	104.54	102.7	100.04	101.29	98.13
Gross Cropped Area	150.5	156.89	161.73	166.54	171.32	176.77	179.5	186.01	189.67	185.7

Millets are termed as the last standing crop in times of drought and as wonder grain that has a capability to enhance nutritional security in the country. Despite this, the consumption and cultivation of millets have been on the decline, greatly due to lack of awareness about them. Bringing millets into main stream of agriculture and diet is the challenge ahead. Need of the hour is to focus on awareness, creation of consumption demand, enhancing cultivation, conservation and commercialization of small millet in an integrated approach is very crucial.

Comparative Nutrition Chart of Millets with Wheat & Rice



c) Importance of Organic Agriculture in Small Millets:

The Green Revolution of the sixties heralded a new era for agriculture in India. The need for large-scale food production was met by using high yielding varieties of wheat and rice and implementing modern farming methods, which entailed the extensive use of chemical fertilizers and pesticides. While this movement helped India meet its high demand for food, the practices it popularised had a severe deteriorating impact on the environment. The use of chemicals and the rapid cultivation of successive crops, without giving the soil time to regenerate its nutritional quality, resulted in the poisoning of water bodies and the widespread destruction of agricultural land. The ceaseless cultivation of cereals and other crops, all over the nation (especially Punjab) with extensive use of chemical fertilizers to blaster the crop yields, has spelled devastation for the country's rivers and water, several of which are so contaminated that their water cannot be used for any purpose. The same problems extended to the once fertile soils of the country, most tracts of which are now unfit for growing any crop.

This problem has taken a turn for the worse in recent years, with the effect of climate change being felt across the country. The sudden temperature increase and corresponding water shortage have impacted agricultural communities across the length and breadth of India. Successive heat waves have severely impacted the wheat production in Madhya Pradesh, recognized as the "Wheat Bowl" of India. Meanwhile, the recent monsoon failure in the North Eastern States of India has had a devastating effect on the farmers of the region, who mainly rely on rainfed agriculture to grow their crops. In a similar way, the water shortage of the Kaveri river resulted in the failure of around 70 % of the crops being cultivated in agricultural belt in Tamil Nadu.

The adverse effects of climate change aren't restricted only to India, but with many countries in the world, are suffering from crop failures due to the unpredictable fluctuations in seasons. With droughts becoming more common in some of India's most prominent agricultural regions, the countries farmers need to adopt farming practices, which have the least impact on the environment while still producing a large enough quantity of crops to sustain their livelihood and country's food demand, That's where sustainable Organic Farming and Millets cultivation come into play.

d) Millets to the Rescue:

There are several reasons why millets are the crops that need to be cultivated in earnest today. They are Climate-Smart and can be grown in drought-hit, arid regions more successfully than any other crop. They also have an extremely low water foot print, with a crop of millets requiring around 80% less water than crops like rice, wheat or sugarcane. Another often overlooked positive nature of millets is that they are excelled for soil preservation.

"Millets as Organic Matter, are slow to break down in soil just as in our digestive systems. Their slow composting nature helps in maintaining soil structure and retaining water, thus preserving soil health for extended duration"- By Lynette Abbott, an emeritus professor at University of Western Australia's Institute of Agriculture.

Another reason why millets are good soil preserves is their root network. Millets like most grasses, have a fibrous root network that helps maintain soil integrity due to the extensively branched nature. One cannot realistically expect all the farmers in the country to start cultivating only millets. Rice and wheat are an integral part of the Indian population staple diet, and replacing them entirely with millets

is impossible as it is food hardy. Millets are, however, the ideal crop for cultivation during fallow periods – the time between the cultivation of one batch of crops and the sowing of the next. This is because millets have a high tendency to form symbiotic relationship with mycorrhizal fungi and these fungi colonize the root system of the plant and provide increased water and nutrient absorption capabilities. These fungi can also easily absorb phosphorous and nitrogen- the two elements that are vital for a crop's growth but are present in small quantities in the soil during fallow period.

There are however, a few roadblocks that stand in the way of millet cultivation viz., millets having a lower yield per hectare than rice and wheat, making them less profitable than those popular crops for farmers. Moreover farmers in possession of fertile lands that receive steady rainfall, for instance have no reason to start cultivating millets as that would just reduce their profits. In spite of all these stumbling blocks millets have been going through its journey with ups and downs for about 7000 years with pivotal role in the raise of millet crop agriculture in the farming societies.

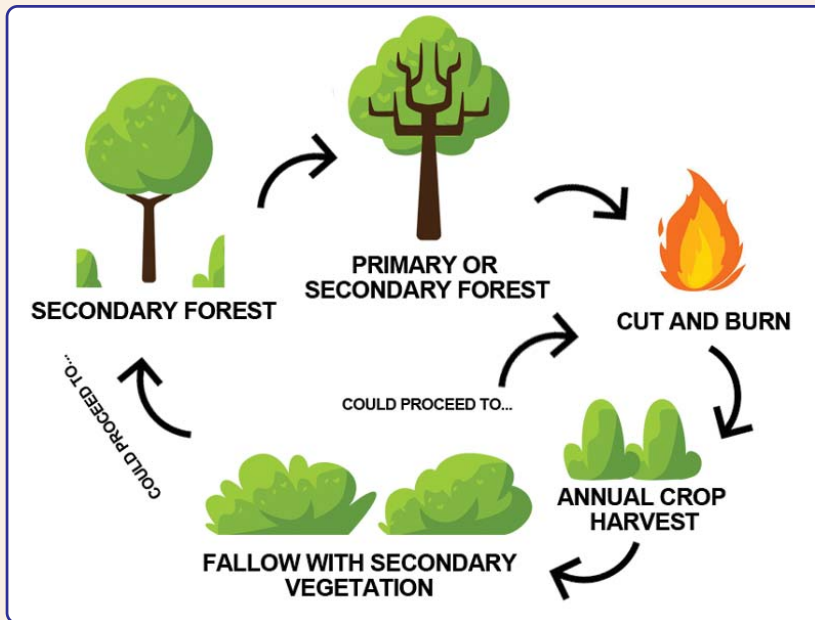
d) Journey of millets in India:

Millets are the first crops to be domesticated by the mankind in Asia and Africa which later on spread across the globe as critical food sources to the evolving civilizations. In India, millets have been among the first cultivated crops grown in the “Hoe Age” preceding the “Plow Age” and were consumed as staple cereals and brewed from prehistoric times. It occurred in Harappa levels (2500-2200 BC) at Shikarpur (Kutch) and late Harappa levels (1900-1400 BC) at Punjab. Later, these millets, are being grown in Karnataka, Andhra Pradesh, Maharashtra, Tamil Nadu, Odisha, Jarkhand, Chattisgarh, Madhya Pradesh and Uttarakhand, spread in a variety of agro-ecological situations viz., plains, coast and hills as well as in diverse soils(from very poor to very fertile) and varying rainfall, widely differing in thermo and photo periods. Millets are considered as one of the most important crops in hilly areas, through shifting cultivation where organic farming still remains a viable option and has a potential for quality production. Further, millets cultivation has also spread to the drylands in the plain areas of many states of the country where they are grown with no fertilizers or very less fertilizers.

i) Shifting cultivation/ Podu cultivation of millets in hill/tribal areas:

Tribal communities of India living on the hilly tracts follow a method of cultivation by cutting and burning forest patches on the hills of the neighboring areas called shifting cultivation which is also called as “Slashed burn” or “Swidden”. In various local terms it is described in Andhra Pradesh and Odisha as “Podu”, a term equivalent to “Bewar” in the usage of Madhya Pradesh and “Jhum” in that of Northeast India. This Agricultural system allows growing of two or almost three annual crops and then abandoning that land until the trees have grown again sufficiently to allow a second filling. This process continues until the land gets washed so bare of soil and see that no more forest growth is possible. It is then finally abandoned and there remains a bare hill side useless for any purpose producing only thorns, creepers and coarse grains. Under Shifting/Podu cultivation, the crops generally grown on podu lands are mostly millets viz., finger millet, little millet, foxtail millet, proso, kodo millets , pearl millet and sorghum. In fact, no one particular crop is grown in any one season, as the growing of mixed crops is the rule rather than exception. A variety of grains are broadcasted and beans are raised in their midst. In addition to millets valuable commercial crops such as turmeric, castor, plantain, chilies, red gram, horse gram etc were raised. The slash and burn/podu cultivation in the tribal regions is significant for the

Shifting/Podu cultivation



biodiversity of crops it has helped to sustain and also the diversity of cultivation practices it has generated viz., crop rotation, intercropping and other agricultural practices as a part of the inherited knowledge system of the podu farmers.

Growing of over a 50 varieties of crops is almost a standard with any single farmer. In case any single crop fails, they have many more to survive on. Millet seed (finger millet, foxtail millet, pearl millet, barnyard millet, little millet and sorghum) are broadcasted on the hill slopes during the summer months and sowing of the paddy seeds is usually done with arrival of monsoon, simultaneously vegetables and other crops are also grown. The tribal farmers don't need to do anything for pest and diseases control as they do not use any kind of chemical fertilizers or pesticides. By default, podu cultivation is completely under organic farming. The land is often cleared by slash-and-burn methods - trees, bushes and forests are cleared by slashing, and the remaining vegetation is burnt and the ashes add potash to the soil. Due to the slope and topography hill soils are subjected to soil erosion and many other natural processes which reduces the soil fertility. Moreover animal manure is the vital source in hilly areas for restoring and maintaining fertility of these soils. Live stock sector has remained as an integral part of various farming systems in hills by way of symbolic relationship with crops. Besides manure, live stock sector provides major products like meat and milk and also a main source of draught power for hill farms where the scope of mechanization is least. But attempts have been made to restrict podu cultivation in order to conserve forests and permit for growth of commercial tree species such as teak. This large-scale commercial exploitation of forests not only destroyed the source of livelihood for tribal but also adversely affected the ecology of the area.

ii) Cultivation of millets in plain areas:

In plain uplands/ drylands, millets are grown under rainfed conditions especially by marginal farmers in *kharif* and under irrigated conditions in *rabi* season. Millets are generally grown either as a sole crop or mixed crop in regions of low rainfall, the other crops grown with them being usually one of the legumes. In *kharif* season, they are sown between May-July and harvested between October–December. The *rabi* crop sown between September and November and harvested during January –March. Millets have wide adaptability to different soils from very poor to very fertile and can tolerate a certain degree of alkalinity. The best soils are alluvial, loamy and sandy soils with good drainage. Some millets like kodo millet can be grown in gravelly and stony soils such as in the hilly regions.

In some tropical dryland regions, along with rice, wheat, maize, pulses etc. Millets are also grown as a major crop because of their resilience and adaptability to withstand adverse conditions. Millets are either grown under completely organic or Integrated Nutrient Management method. The nutrient requirement of millets is generally less than other cereals crops, hence cultivation under complete organic nutrient management practices was adopted by farmers. Application of farm yard manure to crops is being practiced for long period. Farm yard manure alone or in combination with compost, oilcakes, green manure, biofertilizers could improve the physicochemical and biological properties of the soil and improve the production of millets. But in due course of time, the use of organic manures has been continuously declining in Indian agriculture due to several causes. Decrease in cattle population in recent years and utilization of agricultural wastes for preparation of valuable by products have made the availability of organic manure in agriculture questionable. Due to this the millet growing farmers of the uplands are mostly following Integrated Nutrient Management practices for sustainable production.

CHAPTER III: NUTRIENT MANAGEMENT IN ORGANIC MILLET CULTIVATION

“Soil fertility” can be considered as a measure of the soil ability to sustain satisfactory crop growth, both in the short and long term. Organic matter is essential for soil fertility. It maintains good soil physical conditions (eg. soil structure, aeration and water holding capacity). Plants require 18 mineral nutrient elements for their growth. The nutrients which come from the soil, are dissolved in water and absorbed through plant roots. These nutrients are not sufficient to the soil for a plant to grow healthy. Under organic cultivation, organic manures/organic matter add nutrients to the soil. The primary nutrients, Nitrogen, Phosphorus and Potassium usually are deficient in the soil because plants use large amounts of these nutrients for their growth and survival. Maintaining or enhancing long term soil productivity is a key provision of organic cultivation. The organic matter that is added to the soil should either maintain or improve the physical, chemical and biological properties of soil and minimize soil erosion, manage crop nutrients and soil fertility, maintain or improve soil organic matter, but minimize contamination of crops, soil or water with nutrients, pathogenic organisms, heavy metals or residues of prohibited substances.

Nutrient management in organic millet farms should economically meet crop nutrient needs and avoid nutrient depletion, while maintaining or improving soil productivity without excessive nutrient losses. The soil nutrient availability is dependent on diverse soil chemical, physical and biological properties and their interaction with the cropping system. Crop performance is considered as the best indicator of soil productivity and is a measuring yard for many soil properties

Millet farmers typically manage to improve the soil physical and chemical properties of the soil for sustainable productivity through practices such as :

1. Applying organic material viz., farm yard manure, compost, vermicompost and bio fertilizers to supply nutrients and maintain soil organic matter.
2. Growing cover crops to cycle soil nutrients and biologically fix atmospheric nitrogen.
3. Growing green manure legume crops *insitu* or by green leaf manuring and incorporation into the soil.
4. Mixed farming or intercropping
5. Crop rotation.
6. Crop residue management



1. Application of organic matter to the soil:

a. Manures: Manure application to the millet crop is often a valuable source of nutrients in organic production. However applying of manures to meet all of the crop nutrient demands can lead to excess of certain nutrient like phosphorus, because millet crops remove more nitrogen and potassium than phosphorus. This excessive soil phosphorus added through organic manures is not likely to be harmful to crops but contribute to phosphorus loss in runoff and erosion. Therefore manure nitrogen needs to be complemented by Biological Nitrogen Fixation or nitrogen sources in organic systems. Raw organic manures must be applied more than 120 days before harvest if the edible portion of the crop is in direct contact with soil and 90 days before harvest if the edible portion of the crop is not in direct contact with the soil. Typical nutrient concentration estimates for different manure source are given in table below. But the actual nutrient content varies with the methods of manure storage and handling, live stock ration, animal maturity and weather conditions.

Average Nutrient content of organic manures:

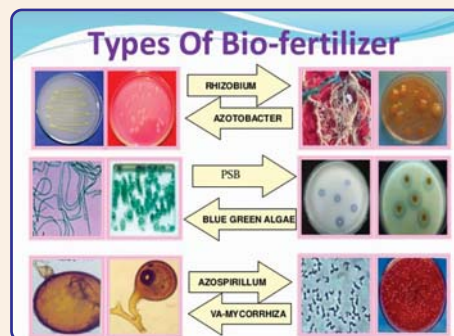
Manure	Nutrient Composition(%)		
	N	P2O5	K2O
Animal refuse	0.3-0.4	0.1-0.2	0.1-0.3
Cattle dung, fresh	0.4-0.5	0.3-0.4	0.3-0.4
Horse dung, fresh	0.5	0.4-0.6	0.3-1.0
Poultry manure, fresh	1.0-1.8	1.4-1.8	0.8-0.9
Sewage sludge, dry	2.0-3.5	1.0-5.0	0.2-0.5
Sewage sludge, activate dry	4.0-7.0	2.1-4.2	0.5-0.7
Cattle urine	0.9-1.2	trace	0.5-1.0
Human urine	0.6-1.0	0.1-0.2	0.2-0.3
Sheep urine	1.5-1.7	trace	1.8-2.0
Ash, coal	0.73	0.45	0.53
Ash, house hold	0.5-1.9	1.6-4.2	2.3-12.0
Ash,wood	0.1-0.2	0.8-5.9	1.5-36.0
Rural compost,dry	0.5-1.0	0.4-0.8	0.8-1.2
Urban compost,dry	0.7-2.0	0.9-3.0	1.0-2.0
Farmyard,dry	0.4-1.5	0.3-0.9	0.3-1.9
Filter-Press cake	1.0-1.5	4.0-5.0	2.0-7.0
Rice hulls	0.3-0.5	0.2-0.5	0.3-0.5
Groundnut husks	1.6-1.8	0.3-0.5	1.1-1.7
Banana dry	0.61	0-12	1.0
Non edible oil cakes			
Castor cake	4.3	1.8	1.3
Cotton cake	3.9	1.8	1.6
Karanj cake	3.9	0.9	1.6
Mahua cake	2.5	0.8	1.8
Neem cake	5.2	1.0	1.4
Safflower cake	4.9	1.4	1.2
Edible oil cakes			
Coconut cake	3.0	1.9	1.8
Groundnut cake	7.3	1.5	1.3
Niger cake	4.7	1.8	1.3
Rape seed cake	5.2	1.8	1.2
Sesame cake	6.2	2.0	1.2

Source: Organic farming for Sustainable Agriculture by A.K.Dhana, 1996 Agro Beneficial Publishers
(India)

Manure nitrogen is primarily in organic and ammonium forms. Ammonia-Nitrogen is immediately available to the crop for uptake, as it is not lost through volatilization as ammonia. Ammonia–Nitrogen availability depends on the method of application of the manure, the time before incorporation, temperature, precipitation and irrigation. Instead of depending on only manures to meet annual crop nitrogen demand, organic farmers should build diverse fertility strategies into this farming system. In conventional method of farming crop may receive majority of nitrogen from mineral fertilizers applied prior to sowing of the crop. However, organic crop receive the nutrients from a greater number of sources at a greater number of intervals to ensure regular mineralization of organic nitrogen.

b. Compost: Composting is a process by which organic wastes are converted into organic fertilizers of higher nutrient concentration by means of biological activities under controlled conditions and reduces the bulk of organic materials through carbohydrates and water loss during decomposition. Compost is often easier to handle than the bulk organic material and also composting process kills some pathogens and weed seeds. Compost has less odour and fewer microbial pathogens with less risk of microbial contamination of fresh produce than with raw manure. In finished composts with a Carbon : Nitrogen ratio (C:N) less than 20:1, nearly all of the nitrate and ammonium in the compost will be available to the plants in the first season after application and only 15% of the organic nitrogen in the compost may be available during the next season. In addition to supplying nitrogen, compost is an important source of other micro and macro nutrients. As with other organic materials compost application can improve soil organic matter content, Cation Exchange Capacity, soil porosity, aggregate stability and water holding capacity, although the magnitude of improvement will depend on current soil organic matter levels. Similarly, soil biological properties, including microbial biomass, microbial enzymatic activity and nitrification potential are often increased following compost application. Despite of many benefits of compost, raw manure is still popular because of the added cost of composting and the nitrogen volatilization loss that occurs during composting.

c. Biofertilizers: Biofertilizers are biologically active products or microbial inoculants of bacteria, fungi, algae that may help either directly or indirectly in the enrichment of soil fertility. Several microorganisms influence nutrient uptake in the plants, those that directly benefit the plant metabolism. These biofertilizers are products designed to provide enhanced nutrient availability and uptake, stimulation of crop growth, biological nitrogen fixation and protection against insect pests and diseases. Depending on the purpose, biofertilizer products, can be applied to soil, seeds or foliar tissue. Biofertilizers are most beneficial in soils of low to moderate soil organic matter and nutrient availability or with foliar application. In contrast, biofertilizer application to soils greater than 3% organic matter rarely result in measurable crop or soil benefits.



Several categories of biofertilizers that are applied to millet crops include nitrogen fixers, phosphorus, solubilizers and potassium solubilizers. Nitrogen fixers such as *Azospirillum* convert atmospheric nitrogen into ammonia. *Bacillus* and *Pseudomonas* are examples of microbes found in phosphorus solubilising

biofertilizers that lower the soil pH to dissolve soil bound phosphate for plant availability and may be most effective for calcareous soils. Arbuscular mycorrhizal fungi take up soil phosphorus, zinc and copper and transfer these to plant roots.

2. Cover crops: Cover crops can improve soil physical properties, nutrient cycling and soil microbial activities. In addition, cover crops can scavenge residual nitrogen mineralized from soil and organic amendments before it is lost through volatilization, runoff or leaching. However, like other sources of organic nitrogen, nitrogen contained in cover crops biomass is not entirely available to the next crops.



It is important to consider the C: N of the cover crop residue. Species with high C:N (>20:1 eg. Grasses) result in net immobilization of soil nitrogen in the short term, whereas nitrogen will be more readily available following decomposition of species with low C: N (< 20:1 eg. Legumes). Nitrogen fixation by legume cover crops can be especially useful for increasing soil nitrogen availability and balancing the abundant phosphorous and potassium supplied through manure or compost application. However, the cover crop needs to be terminated early enough to avoid increased C: N as the plant matures, and to allow sufficient time for residue decomposition, mineralization of organic nitrogen and soil water recharge. To this end, it is recommended that legume cover crops should be terminated at the flowering stage and atleast one or two weeks prior to planting. Cover crops when combined with the organic amendments, meet plant phosphorus, potassium and micro nutrient demand, and can be a sustainable and cost effective nutrient source under organic production.

3. Green Manures: Green, undecomposed plant material used as manure is called green manure. It is obtained in two ways: growing green manure crops or by collecting green leaf (along with twigs) from plants grown in wastelands, field bunds and forest. Green manures grown in the fields usually belong to leguminous family and are incorporated into the soil after sufficient growth. The most important green manure crops are sunhemp, dhaincha, pillipesara, clusterbeans and *Sesbania rostrana*.

Nutrient content of Green Manure Crops and Green Leaf Manure

Plant	Scientific Name	Nutrient Content (%) air dry basis		
		N	P2O5	K2O
Green Manure Crops				
Sunhemp	<i>Crotalaria juncea</i>	2.30	0.50	1.80
Dhaincha	<i>Sesbania aculeata</i>	3.50	0.60	1.20
Sesbania	<i>Sesbania speciosa</i>	2.71	0.53	2.21
Green Leaf Manure				
Forest tree leaf		1.20	0.60	0.40
Green weeds		0.80	0.30	0.20
Pongamia leaf	<i>Pongamia glabra</i>	3.31	0.44	2.39

4. Crop rotation: Crop rotation can contribute to improved soil physical properties, pest management, nutrient availability, nutrient use efficiency and crop yield. Legumes in the rotation can result in a nitrogen credit for subsequent crops due to biological nitrogen fixation and less nitrogen immobilization compared to a non legume as the preceding crop. However, the nitrogen credit for forage legumes is greater than for grain or vegetable legumes because most of the nitrogen is removed during harvest of the grain. In contrast, forage legumes contribute nitrogen from vegetative biomass, which can be comparable to cover crop nitrogen contribution. Cereal and other non-legume crops in rotation can be useful for building soil organic matter, an important reservoir of nutrients in organic farming.

Nutrient management in organic farms requires long - term planning and a diverse combination of cultural practices and inputs.



CASE STUDIES ON NUTRIENT MANAGEMENT IN MILLETS CONDUCTED AT DIFFERENT AICRP CENTRES

a. Comparison of Organic Vs Inorganic methods of cultivation in small millets:

1. Title of the project : Effect of Organic manures in Long Term Nutrient Management System on Yield, Quality and Soil Health in Finger Millet

Location : Vizianagaram, Andhra Pradesh

Crop : Finger millet

Year of study : 2014-15 to 2019-20

Keeping in view of sustained crop yields as well as soil health, a long term experiment is being conducted at Agricultural Research Station, Vizianagaram for the past six years from 2014 during *Kharif* season to study the effect of use of organic source of nutrients in comparison with conventional method on soil health, Yield and quality of finger millet crop. The organic inputs were supplied in the form of green manure (sunhemp), farmyard manure, neem cake and biofertilizers (*Azospirillum* and Phosphorus Solubilising Bacteria). The results of the mean pooled over a period of six years have been given in the tables below. The performance of the growth parameters, yield attributes, changes in the soil physico chemical properties over the period of time.

Details of the experiment

a) Organic plot:

Green manuring (Sunhemp@50 kg ha⁻¹)

FYM @ 5 t ha⁻¹

Neem cake @ 1t ha⁻¹

Azospirillum@5kg ha⁻¹

PSB@5kg ha⁻¹

b) Inorganic plot

100% RDF N,P,K fertilizers

(Basal:30: 40:30 NPK Kg/ha Top:30 N kg/ha)

Variety : VR 900

Plot size: 0.30 acre

Spacing : 30 cmX10 cm

Irrigation: Rain fed

Results:

Growth and Yield Attributes:

The pooled mean data (Table 1) of all the six years recorded significantly higher grain and straw yields in the conventional plot compared to organic plot. Even after a period of six years the conventional (Inorganic) method of cultivation of finger millet showed 14.40 % higher grain yields when compared to organic method of cultivation. During the initial years the difference in the yields was around 24% and gradually it is in the decreasing trend which may take few more years to be at par with the conventional method of cultivation. While observing the pooled mean data, among the two treatments, significantly highest grain and straw yields were recorded in conventional plot (2749.6 kg ha⁻¹ and 7257.4 kg ha⁻¹) compared to the organic plot (2391.1 q ha⁻¹ and 6711.3 q ha⁻¹). Among the plant growth characters, plant height and leaf length were found significantly high in conventional plot (117.3cm and 35.1cm) compared to the organic plot (110.4cm and 30.9cm). Among the yield contributing characters the number

of productive tillers/plant, ear head length and numbers of fingers/ear were also found highest in conventional plot (3.2, 8.7 cm and 7.8) compared to the organic plot (2.9, 8.7cm and 7.5), but no significant difference was found between the two treatments. Plant growth is dependent on the rate of accumulation of dry matter. The dry matter accumulation may reflect on the economic yield in view of the fact that vegetative parts of the plant serve as a source where as grains are the sink. The increase in the plant growth and yield, owing to the application of inorganic fertilizers may be attributed to the fact that this nutrient being important constituents of nucleotides, proteins, chlorophyll and enzymes involved in various metabolic processes which have direct impact on vegetative and reproductive phase of plants.

Soil Physico-Chemical Properties:

After completion of the sixth year of the experiment it is noted that pH and EC of soil were unchanged over their initial status under both the treatments (Table 2). The Organic Carbon % has shown a increased trend in the organic plot (0.44%) by 0.2% compared to the initial value (0.42%). The Organic Carbon % was increased with application of FYM and continuous *in situ* incorporation of green manures. The buildup of Organic Carbon could be attributed to the manures and subsequently addition of leaf residue and debris of plants. The soil available macronutrients N, P_2O_5 and K_2O showed an increasing trend over their parental status in both the organic and inorganic treatments. The mean pooled data of the soil available Nitrogen and Phosphorus were found significantly high in organic plot (251kg ha^{-1} and 78kg ha^{-1}) compared to the conventional method (224.3 kg ha^{-1} and 71.5 kg ha^{-1}), whereas the soil available Potassium was found high in organic plot (305.2 kg ha^{-1}) compared to the conventional plot (292.3 kg ha^{-1}), but there is no significant difference between the two treatments. The increase in soil available nutrients is due to the enhancement in the efficiency of nutrient use from all the sources through mineralization of soil organic matter, animal excreta, other manures and biofertilizers. The increase in P availability might be due to the release of appreciable quantities of carbon dioxide during decomposition of organic matter which forms the carbonic acid, leading to increased solubility of phosphorous resulting in higher availability. Whereas the higher K status in soil might be due to the organic manures on decomposition released the organic acid which might have mobilized the native or non-exchangeable forms of K and charge the soil solution with K ions, so that it may be readily available. The soil available Zn was found significantly high in the treatment of organic plot (1.35 ppm) compared to the conventional plot (1.19 ppm) and no significant difference was found in soil available Fe, Mn and Cu among both the treatments.

Plant Nutrient Content and Uptake:

The plant N, P and K contents (Table 3) of finger millet are significantly high in organic method of cultivation when compared to the conventional method. But the plant P and K uptake (Table 4) were found significantly highest in conventional plot (14.6 kg ha^{-1} and 33.8 kg ha^{-1}) when compared to the organic plot (11.7 kg ha^{-1} and 28.3 kg ha^{-1}) whereas the plant N uptake even though found higher in the inorganic plot but did not show any significant difference between the two treatments. This could be primarily due to increased availability of the nutrients in the crop root zone which resulted in increased absorption of the elements by the plants as well as higher dry matter production. Whereas the uptake of micronutrients were not significantly influenced by both the treatments.

Grain Bio chemical analysis:

After six of the experiment the bio-chemical components (Table 5) viz., Total anti-oxidant activity, Total Phenol Content, Total sugar, Protein content and Tannin content were analyzed in finger millet grain. It was found that total anti oxidant activity and total phenol content were found highest in inorganic grain whereas Total sugar, Protein content and Tannin content were found highest in organic grain.

Table 1: Effect of organic and inorganic manures on growth, yield of finger millet

Particulars	Organic treatment							Inorganic treatment						
	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	Mean	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	Mean
Plant height(cm)	93.4	86.5	116.2	116.6	124.1	125.4	110.4	96.3	108.9	126.5	116.3	127.5	128.7	117.3
Prod. tillers/ plant	1.6	2.07	3.23	3.33	2.65	4.20	2.85	1.9	2.50	3.77	3.67	2.67	4.70	3.20
Leaf length(cm)	29.5	24.90	32.67	35.80	35.47	27.10	30.91	30.6	35.00	38.90	32.73	35.93	37.57	35.13
Leaf width (cm)	1.1	0.97	1.65	1.43	0.95	1.03	1.19	1.1	1.09	1.37	1.30	0.93	0.98	1.13
Ear head length(cm)	7.8	8.60	7.73	8.20	9.44	10.20	8.66	6.5	9.18	8.90	7.83	9.32	10.47	8.71
No. of Fingers/ear	6.9	6.80	7.00	8.33	7.20	8.60	7.48	7.3	7.93	7.83	7.67	7.00	8.83	7.76
Straw yield (q ha ⁻¹)	7173	7191	7676	4960	7270	5997	6711	7810	8147	8726	4893	7624	6344	7258
Grain yield (q ha ⁻¹)	2230	2648	2429	2116	2344	2584	2392	2520	3284	2926	2043	2768	2956	2749

Table 2: Effect of organic and inorganic manures on soil health at harvest

Particulars	Initial Values	Organic Plot							Inorganic Plot						
		2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	Mean	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	Mean
OC (%)	0.42	0.43	0.44	0.44	0.46	0.46	0.46	0.45	0.42	0.42	0.42	0.42	0.43	0.43	0.43
pH	7.02	6.85	6.57	7.23	7.29	7.10	7.07	7.02	6.31	6.27	6.69	6.84	6.84	7.15	6.68
EC	0.25	0.2	0.20	0.35	0.15	0.26	0.24	0.23	0.21	0.17	0.32	0.16	0.24	0.29	0.23
Ava. N (kg ha ⁻¹)	227	224	249	230	263	275	266	251	213	217	213	227	233	240	224
Ava. P ₂ O ₅ (kg ha ⁻¹)	59	98	77	64	89	74	68	78	86	73	55	87	63	64	72
Ava. K ₂ O(kg ha ⁻¹)	265	312	316	332	295	285	291	305	282	295	294	307	297	279	292
Ava. Zn (ppm)	1.75	1.2	1.37	1.19	1.30	1.75	1.30	1.35	1.1	1.09	1.07	1.27	1.57	1.14	1.19
Ava. Fe (ppm)	10.1	9.5	8.44	8.89	8.93	8.47	9.83	9.84	9.8	8.31	8.78	7.45	8.23	9.03	8.96
Ava. Mn (ppm)	5.95	4.2	5.61	7.14	5.24	6.57	6.06	5.82	4.3	5.67	5.30	4.85	5.20	5.09	5.23
Ava. Cu (ppm)	1.35	0.7	2.21	2.41	1.82	1.89	1.27	1.56	0.8	1.80	2.24	1.65	1.76	1.07	1.52

Table 3: Effect of organic and inorganic manures on Plant nutrient content of finger millet

Particulars	Organic treatment							Inorganic treatment						
	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	Mean	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	Mean
N (%)	1.27	1.28	1.30	1.32	1.35	1.39	1.32	1.18	1.16	1.20	1.23	1.26	1.30	1.23
P (%)	0.17	0.22	0.25	0.26	0.27	0.39	0.26	0.25	0.30	0.27	0.28	0.31	0.34	0.29
K (%)	0.43	0.48	0.54	0.65	0.69	0.76	0.59	0.49	0.54	0.58	0.69	0.75	0.86	0.65
Zn (ppm)	12.8	15.3	20.6	21.2	22.6	25.9	19.7	15.8	16.8	18.5	19.5	20.6	22.3	18.9
Fe (ppm)	20.5	24.0	25.1	26.4	26.6	34.1	26.1	22.3	23.5	23.7	26.2	24.9	28.5	24.9

Table 4: Effect of organic and inorganic manures on Plant nutrient uptake of finger millet

Particulars	Organic treatment							Inorganic treatment						
	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	Mean	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	Mean
N Uptake (kg ha ⁻¹)	59.6	63.1	65.8	59.8	64.1	56.3	61.6	60.7	66.4	70.0	55.1	65.5	57.3	62.5
P Uptake (kg ha ⁻¹)	8.0	10.9	12.5	12.5	12.9	14.1	11.7	13.1	17.1	15.6	12.5	16.2	12.9	14.6
K Uptake (kg ha ⁻¹)	20.2	23.9	27.5	30.9	33.2	35.3	28.3	25.6	30.8	33.8	30.9	39.1	42.4	33.8
Zn Uptake (gm ha ⁻¹)	60.0	75.1	104.3	87.1	98.8	108.6	92.2	81.9	95.6	107.8	87.1	107.2	100.4	96.7
Fe Uptake (gm ha ⁻¹)	96.3	117.6	126.4	117.1	125.7	127.7	119.3	115.2	134.4	138.6	117.1	129.4	109.8	124.1

Table 5: Effect of organic manures and fertilizers on bio-chemical analysis in finger millet

Particulars	Organic treatment	Inorganic treatment
Total anti-oxidant activity by DPPH radicle scavenging activity (mg AAE/100g)	65.81	73.34
Total Phenol Content (mg/100g)	83.93	93.41
Total sugar (%)	68.29	67.22
Protein content (%)	4.82	3.22
Tannin content (mg/100g)	230.0	166.67

2. Title of the project : Organic farming research in small millets – A comparative study

Location : Mandya
Crop : Finger millet
Year of study : 2017-18

Treatments	Grain yield	Straw yield	B:C ratio
T1: Only organics (FYM 5t/ha + VM 1 t/ha + neem cake 500 kg/ha)	3212	6082	0.71
T2: inorganics (RDF NPK only)	3836	6508	2.07
T3: INM (RDF + NPK)	4041	6716	1.71

3. Title of the project : Organic farming research in small millets – A comparative study

Location : Kolhapur
Crop : Finger millet
Year of study : 2017-18

Treatments	Grain yield	Straw yield	B:C ratio
T1: Only organics (FYM 5t/ha + VM 1 t/ha + neem cake 500 kg/ha)	1660	2160	0.77
T2: inorganics (RDF NPK only)	1740	2240	1.29
T3: INM (RDF + NPK)	1987	2487	1.27

4. Title of the project : Organic farming research in small millets – A comparative study

Location : Nandyal
Crop : Foxtail millet
Year of study : 2017-18

Treatments	Grain yield	Straw yield	B:C ratio
T1: Only organics (FYM 5t/ha + VM 1 t/ha + neem cake 500 kg/ha)	1630	2345	1.48
T2: inorganics (RDF NPK only)	1940	2571	2.33
T3: INM (RDF + NPK)	1980	2734	2.15

In common, the highest grain yield was observed at treatment which has an integration of organic and inorganic *i.e.*, INM (RDF+NPK) followed by the application of inorganic (RDF NPK only). Whereas, the lowest grain yield was obtained at only organics (FYM 5.0 t/ha + Vermicompost 1 t/ha + Neem cake 500 kg/ha). All crops *i.e.*, finger millet (rainfed and irrigated), foxtail millet, proso millet and barnyard millet resulted in same trend at all centers. The higher levels of available nutrients N, P₂O₅ and K₂O and higher nutrient uptakes were observed at INM treatment and lower values with only organics. The soil parameters were not affected by treatment effects. B:C ratio was found high at only inorganic treatment for Finger millet (rainfed and irrigated) and Foxtail millet and for proso and barnyard millet, INM was found better.

5. Title of the project : Organic farming research in small millets – A comparative study

Location : Athiyandal
Crop : Barnyard millet
Year of study : 2017-18

Treatments	Grain yield	Straw yield	B:C ratio
T1: Only organics (FYM 5t/ha + VM 1 t/ha + neem cake 500 kg/ha)	1328	2129	0.76
T2: inorganics (RDF NPK only)	1718	2720	2.66
T3: INM (RDF + NPK)	2530	3782	3.13

6. Title of the project : Organic farming research in small millets – A comparative study (proso millet)

Location : Athiyandal
Crop : Proso millet
Year of study : 2017-18

Treatments	Grain yield	Straw yield	B:C ratio
T1: Only organics (FYM 5t/ha + VM 1 t/ha + neem cake 500 kg/ha)	1128	1648	0.64
T2: inorganics (RDF NPK only)	1378	2187	2.14
T3: INM (RDF + NPK)	1790	2678	2.21

Nutrient Management in Organic finger millet cultivation



Insitu cultivation of sunhemp



Incorporation of sunhemp



Organic crop resistant to lodging at the time of cyclone



Inorganic crop lodged due to cyclone

b. Development of organic production technologies in small millets

1. Title of the project : Production technology for organic finger millet

Location : Bengaluru, Coimbatore and Ranichauri
Crop : Finger millet
Year of study : 2006-2007

Treatments	Grain	Straw
T1	2190	3454
T2	1958	2743
T3	1669	2421
T4	2026	2626
T5	1593	2348
T6	1521	2553
T7	1340	2085
T8	1497	2438
T9	1778	2815
SEm±	119	227
CD at 5%	355	667

The results indicated that application of 7.5 t FYM/ha along with RDF NPK recorded significantly higher grain yield compared to other treatments but it was on par with application of poultry manure to supply 100% N and balance of phosphorus through natural minerals and application of recommended FYM and addition of FYM to supply 100% N and balance of phosphorus through natural minerals.

Mean of three years grain yield data indicated that application of poultry manure to supply 100% N gave comparable yields to that of SAU recommendations and only inorganic RDF.

T1: Rec. FYM @7.5 t/ha and inorganic fertilizers to supply NPK 50:40:25 kg/ha

T2: Rec. FYM @7.5 t/ha + additional of FYM to supply 100% N balance of P and K through natural minerals

T3: Vermicompost to supply 100% N and balance of P and K through minerals

T4: Poultry manure to supply 100% N and balance of P and K through minerals

T5: FYM to supply 100% N balance of P and K through natural minerals

T6: 25 kg N through FYM + concentrates to supply balance P and K through minerals

T7: 25 kg N through vermicompost + concentrates to supply balance P and K through minerals

T8: 25 kg N through poultry manure + concentrates to supply balance P and K through minerals

T9: Rec. NPK through chemical fertilizers

2. Title of the project : Organic production package for finger millet

Location : Bengaluru, Coimbatore, Kanke & Ranichauri
 Crop : Finger millet
 Year of study : 2009-2010

Treatments	Grain yield (kg/ha)				Straw yield (kg/ha)			
	2007	2008	2009	Mean	2007	2008	2009	Mean
T1: 100% N through FYM + balance P&K through natural minerals + biofertilizers	2463	2940	2893	2765	3029	2616	4688	3444
T2: 100% N through VC + balance P&K through natural minerals + biofertilizers	2711	2879	305	3065	3469	2821	5566	3952
T3: 100% N through PM + balance P&K through natural minerals + biofertilizers	3945	4012	3806	3921	7619	5572	6185	6459
T4: 50% N through FYM + 50%N PM + balance P&K through natural minerals + biofertilizers + 25 kg N/ha through neem cake @ 7-8 weeks after sowing	2369	3412	2836	3172	4669	3750	3552	3990
T5: 50% N through FYM + 50%N VC + balance P&K through natural minerals + biofertilizers + 25 kg N/ha through neem cake @ 7-8 weeks after sowing	3021	2991	3114	3042	3828	3188	4915	3977
T6: 50% N through PM+50%N VC + balance P&K through natural minerals+biofertilizers +25kg N/ha through neem cake @ 7-8 weeks after sowing	2888	3537	3254	3226	3786	4021	5736	4514
T7: rec. NPK (50:40:25 kg/ha)	2888	2521	3087	2832	4067	2938	5893	4299
T8:rec. FYM (7.5 t/ha) + RDF (50:40:25 kg NPK/ha)	3772	2981	2930	3228	5718	3535	5890	5048
SEm±	223	184	208	-	575	238	545	-
CD at 5%	679	558	631	-	1748	721	1653	-

Mean over years data also indicated that application of poultry manure enriched with biofertilizers to supply 100 per cent recommended dose of nitrogen and balance of phosphorus and potash through natural minerals gave highest grain yield as compared to other treatments.

3. Title of the project : Organic production package for finger millet

Location : Bengaluru, Coimbatore & Kanke

Crop : Finger millet

Year of study : 2011-2012

Treatments	Grain yield (kg/ha)		Straw yield (kg/ha)	
	2010	2011	2010	2011
T1: 100% N through FYM + balance P&K through natural minerals + biofertilizers	3394	1512	3803	4178
T2: 100% N through VC + balance P&K through natural minerals + biofertilizers	3529	1581	5185	4206
T3: 100% N through PM + balance P&K through natural minerals + biofertilizers	3471	1228	3823	3312
T4: 50% N through FYM + 50%N PM + balance P&K through natural minerals + biofertilizers + 25 kg N/ha through neem cake @ 7-8 weeks after sowing	3257	1525	3565	4677
T5: 50% N through FYM + 50%N VC + balance P&K through natural minerals + biofertilizers + 25 kg N/ha through neem cake @ 7-8 weeks after sowing	3603	1725	3549	3573
T6: 50% N through PM + 50%N VC+balance P&K through natural minerals + biofertilizers + 25 kg N/ha through neem cake @ 7-8 weeks after sowing	3549	1428	3816	3266
T7: rec. NPK (50:40:25 kg/ha)	3511	1579	5185	3594
T8:rec. FYM (7.5 t/ha)+RDF (50:40:25 kg NPK/ha)	2949	1552	3926	4047
SEm±	219	95	335	544
CD at 5%	NS	NS	1016	NS

Strip cropping of finger millet 2/3rd and pigeon pea 1/3rd rotating pigeon pea with finger millet and vice versa such that are rotation is complete in 3 years, nutrient supply through FYM and plant protection through bio pesticides gave higher finger millet grain equivalent yield.

4. Title of the project : Production package for organic finger millet under irrigated condition at Mandya

Location : Mandya
 Crop : Finger millet
 Year of study : 2014-2015

Treatments	Yield (kg/ha)						B: C ratio		
	Grain			Starw					
	2012	2013	2014	2012	2013	2014			
T1: 100kg N through FYM	4558	4585	4473	7430	7412	7156	2.74	2.8	2.38
T2: 150kg N through FYM	4659	4638	3840	7594	7625	6143	2.44	2.49	2.19
T3: 100kg N through poultry manure	4386	4538	4001	7149	7171	6401	3.34	3.42	2.06
T4: 150kg N through poultry manure	4750	4990	3764	7742	7792	6022	3.38	3.45	2.08
T5: 100kg N through press mud	4307	4410	4072	7021	7029	6516	2.20	2.33	2.63
T6: 150kg N through press mud	4919	5015	3665	8018	8039	5865	2.10	2.18	2.24
T7: 1/3 rd rec. N from each of FYM, poultry manure and press mud	4502	4513	4119	7339	7336	6591	2.68	2.72	2.46
T8: 100kg N through composted green biomass	3280	3284	3710	5347	5317	5936	2.14	2.2	2.03
T9: 150kg N through composted green biomass	3295	3288	3437	5371	5374	5499	1.90	1.99	1.37
T10: UAS package (7.5 t FYM/ha + 100: 50:50 kg NPK/ha)	5194	5242	3505	8467	8433	5569	2.65	2.7	1.61
SEm±	316	113	190	-	134	300	-	-	-
CD at 5%	940	335	565	-	398	891			

Mean over years data of net monetary returns and B:C ratio indicated that application of FYM to supply 100 kg N through FYM gave higher NMR. But B:C ratio was higher in plots where application of poultry manure was done to supply 100 kg N.

5. Title of the project : Production package for organic finger millet under rainfed conditions at Kolhapur

Location : Kolhapur
 Crop : Finger millet
 Year of study : 2015-2016

Treatments	Grain				Starw				B:C ratio			
	2013	2014	2015	Mean	2013	2014	2015	Mean	2013	2014	2015	Mean
T1: FYM @ 5 t/ha	1030	957	1194	1060	1292	1193	1553	1346	0.74	0.76	0.95	0.82
T2: vermicompost @ 3 t/ha	1350	1337	1361	1349	1813	1833	1770	1805	0.82	0.93	0.94	0.90
T3: poultry manure @ 2.5 /ha	1300	1177	1250	1242	1617	1713	1625	1652	0.99	1.09	1.04	1.04
T4: : FYM @ 2.5 t/ha	980	1015	1212	1069	1230	1685	1573	1496	0.79	0.90	1.06	0.92
T5: vermicompost @ 1.5 t/ha	1140	1126	1185	1150	1433	1407	1540	1460	0.83	0.91	0.96	0.90
T6: 100% N through FYM	1300	1342	1409	1350	1625	1675	1832	1711	0.71	0.84	0.89	0.81
T7: 100% N through vermicompost	1567	1515	1561	1548	1963	1960	2029	1984	0.81	0.91	0.93	0.88
T8: 100% N through poultry manure	1467	1497	1772	1579	1833	1883	2303	2006	1.00	1.14	1.35	1.16
T9: 500% N through FYM + 50% N through vermicompost	1450	1397	1562	1470	1807	1745	2030	1861	0.99	0.85	0.96	0.93
T10: 50% N through FYM+50% N through poultry manure	1553	1443	1551	1516	1983	1853	2016	1951	1.02	1.05	1.13	1.07
T11: RDF (60:30:00 kg/ha)	1967	1836	1925	1909	2460	2200	2502	2387	1.60	1.62	1.64	1.62
T12: control	877	811	833	840	1108	1013	1244	1122	0.82	0.81	0.83	0.82
SEm±	129	143	138	-	162	153	199	-	0.09	-	-	-
CD at 5%	377	420	404	-	475	447	584	-	0.28	-	-	-

Application of recommended dose of fertilizer though gave significantly higher grain yield compared to other organic manures and their doses but it was on par in yield with application of 100 per cent N through Vermicompost/poultry manure or combination of 50 per cent N each from FYM and poultry manure. Mean over three years data also showed the similar trend.

6 Title of the project : Production package for organic finger millet under irrigated condition at Mandya

Location : Mandya
Crop : Finger millet
Year of study : 2015-2016

Treatments	Yield (kg/ha)		B: C ratio
	Grain	Straw	
T1: 100kg N through FYM	4572	6446	2.19
T2: 150kg N through FYM	3799	5357	1.96
T3: 100kg N through poultry manure	3939	5554	1.82
T4: 150kg N through poultry manure	3775	5323	2.06
T5: 100kg N through press mud	4478	6314	2.22
T6: 150kg N through press mud	3716	5239	1.81
T7: 1/3 rd rec. N from each of FYM, poultry manure and press mud	4492	6333	1.91
T8: 100kg N through composted green biomass	3752	5291	1.96
T9: 150kg N through composted green biomass	3551	5006	1.84
T10: UAS package (7.5 t FYM/ha + 100: 50:50 kg NPK/ha)	3607	5086	1.71
SEm±	206	290	-
CD at 5%	611	861	-

Among the different source of organic manures and their combinations tried, application of 100 kg N through FYM gave significantly higher grain yield as compared to other organic manures and their doses tried but it was on par in yield with application of 100 kg N through Press mud and combination of FYM, Poultry manure and press mud to supply 33 percent from each source. Mean over years data also showed similar trend of grain yield.

7. Title of the project : Production package for organic finger millet under rainfed condition

Location : Kolhapur
Crop : Finger millet
Year of study : 2016-2017

Treatments	Grain	Straw	B: C ratio
T1: FYM @ 5 t/ha	1438	1743	0.94
T2: vermicompost @ 3 t/ha	1572	2027	1.03
T3: poultry manure @ 2.5 /ha	1616	1655	1.05
T4: : FYM @ 2.5 t/ha	1291	1527	0.84
T5: vermicompost @ 1.5 t/ha	1365	1362	0.88
T6: 100% N through FYM	1735	2061	1.13
T7: 100% N through vermicompost	1869	2184	1.21
T8: 100% N through poultry manure	1961	1873	1.27
T9: 50% N through FYM + 50% N through vermicompost	1872	2272	1.22
T10: 50% N through FYM + 50% N through poultry manure	1783	1773	1.16
T11: RDF (60:30:00 kg/ha)	2094	2774	1.37
T12: control	1128	1284	0.73
CD at 5%	2.23	3.64	-
CV %	8.09	11.09	-

Significantly higher grain yield and straw yield and B:C ratio was obtained in SAU's recommended package compared to other treatments. However, it was on par with that of application of 150 kg N through Pressmud.

8. Title of the project : Production package for organic finger millet under irrigated condition

Location : Mandya
 Crop : Finger millet
 Year of study : 2016-2017

Treatments	Yield (kg/ha)		B: C ratio
	Grain	Straw	
T1: UAS package (7.5 t FYM/ha + 100: 50:50 kg NPK/ha)	4616	7432	2.81
T2: 100kg N through FYM	4077	6564	2.55
T3: 150kg N through FYM	4165	6706	2.32
T4: 100kg N through poultry manure	4013	6462	2.71
T5: 150kg N through poultry manure	4457	6897	2.76
T6: 100kg N through press mud	3995	6432	2.45
T7: 150kg N through press mud	4640	7307	2.51
T8: 1/3 rd rec. N from each of FYM, poultry manure and press mud	4004	6446	2.58
T9: 100kg N through composted green biomass	3913	5769	2.59
T10: 150kg N through composted green biomass	3947	5960	2.39
SEm±	122.47	240.66	-
CD at 5%	363.88	715.05	-

Significantly higher grain yield and straw yield and B:C ratio was obtained in SAU's recommended package compared to other treatments. However, it was on par with that of application of 150 kg N through Pressmud.

9. Title of the project : Production technology for organic little millet

Location : Dindori
 Crop : Little millet
 Year of study : 2005-06

Treatments	2003		2004		2005		Mean		B: C ratio
	Grain yield (kg/ha)	Straw yield (kg/ha)	Grain yield (kg/ha)	Straw yield (kg/ha)	Grain yield (kg/ha)	Straw yield (kg/ha)	Grain yield (kg/ha)	Straw yield (kg/ha)	
T1	495	648	264	394	512	713	424	585	1.04
T2	544	772	312	454	565	752	474	659	1.24
T3	463	609	468	692	443	636	458	646	0.96
T4	618	858	249	368	638	823	502	683	1.37
T5	638	884	256	379	678	896	524	720	1.46
T6	756	1014	303	452	1174	1034*	530	833	1.60
T7	986	1248	630	940	1174*	1303	881	1164	1.54
T8	586	814	430	638	603	798	540	750	1.20
SEm±	37	49	21	36	32	47	-	-	-
CD at 5%	111	146	62	107	94	140	-	-	-

Organic little millet and pigeon pea inert cropping gave significantly higher little millet grain equivalent yield compared to other treatments but it was on par with treatment receiving little millet with Rec. FYM and inorganic fertilizer NPK. However, mean over year's data indicated that integrated nutrient supply of 7.5 t FYM/ha along with Rec. NPK gave higher grain yield. Little millet-legume two years rotation with recommended organic manures was second best.

T1: LM supplied with only 7.5t FYM/ha

T2: LM supplied with 3.75 t FYM /ha + Rock phosphate + gypsum

T3: LM supplied with 7.5 t FYM/ha + biofertilizer (seed inoculation) *Azospirillum*, *Brassilience* and *aspergillus awamoore* @ 5g/kg of seed

T4: LM + soybean intercropping, 1:1 (incorporation of soybean as green manure and application of 7.5 t FYM/ha)

T5: LM + sunhemp intercropping, 1:1 (intercropping of sunhemp as green manure and application of 7.5 t FYM/ha)

T6: LM + pigeon pea intercropping 2:1 (plant protection measure through bio fertilizers)

T7: LM with recommended FYM + fertilizer

T8: LM – legume two year rotation with recommended organic manures

10. Title of the project : Production package for organic little millet

Location : Dindori, Rewa & Jagadapur

Crop : Little millet

Year of study : 2008-2009

Treatments	Grain yield (kg/ha)				Straw yield (kg/ha)			
	2006	2007	2008	Mean	2006	2007	2008	Mean
T1: LM with 7.5 t FYM/ha	431	538	507	492	1575	1442	1235	1417
T2: LM with 3.75 t FYM/ha + Imali leaves @ 3.75 t/ha + bio fertilizers	331	572	773	559	1157	1371	1349	1292
T3: LM with Imali leaves @2.5 t/ha + sal burnt husk @2.5 t/ha + bio fertilizers	364	709	907	660	997	1440	965	1134
T4: LM with Imali leaves @2.5 t/ha + sal burnt seed @2.5 t/ha + bio fertilizers	479	854	817	717	1256	1749	1053	1353
T5: LM with 2.5 t/ha of sal leaves + sal burnt husk @2.5 t/ha + bio fertilizers	265	543	919	576	1157	1469	1082	1236
T6: LM with sal burnt seed @ 5t/ha + bio fertilizers	281	680	745	569	1422	1471	1340	1411
T7: LM with sal burnt husk @ 5 t/ha + bio fertilizers	245	543	545	444	1013	1348	1454	1272
T8: LM with Imali leaves @ 5 t/ha + bio fertilizers	397	469	446	437	1454	1383	1070	1302
T9: LM with sal leaves @ 5t/ha + bio fertilizers	377	797	556	577	1438	1359	1158	1318
T10: LM with RDF (20:20:10 kg NPK/ha)	-	-	790	790	-	-	1187	1187
SE $\pm$	31	51	28	-	37	87	56	-
CD at 5%	93	154	83	-	111	NS	116	-

Application of sal leaves and sal burnt husk each @ 2.5 t/ha enriched with biofertilizers gave significantly higher grain yield as compared to other treatments tried. However it was on par with application of imali leaves and sal burnt husk each @ 2.5 t/ha enriched with biofertilizers.

11. Title of the project : Production package for organic little millet under rainfed condition

Location : Dindori & Rewa

Crop : Little millet

Year of study : 2011-2012

Treatments	Grain yield (kg/ha)				
	Dindori			Rewa	
	2009	2010	2011	2009	2011
T1: 50% N through VC + balance P&K through natural minerals	646	632	694	428	1141
T2: 100% N through VC + balance P&K through natural minerals	792	848	918	314	1085
T3: 50% N through FYM + balance P&K through natural minerals	599	584	646	583	1224
T4: 100% N through FYM + balance P&K through natural minerals	693	809	878	403	1150
T5: T1+Biofertilizers	730	720	782	478	1306
T6: T2+Biofertilizers	860	993	1050	617	1406
T7: T3+Biofertilizers	660	718	780	403	1415
T8: T4+Biofertilizers	790	970	1026	558	1159
T9:T1 + T3	890	864	926	439	1123
T10: T9+Biofertilizers	940	989	1054	453	1653
T11: 50% N through FYM + sunhemp as green manuring + balance P&K through natural minerals	714	732	769	317	1096
T12: T11+Biofertilizers	780	852	914	472	1159
T13: 100% N through FYM + sunhemp as green manuring + balance P&K through natural minerals	860	988	1050	447	1689
T14: T13+Biofertilizers	930	1096	1172	534	1114
T15: RDF (40;20:10 kg NPK/ha)	1108	1264	1332	833	1808
T16: absolute control	428	504	518	228	1260
SEm±	31	45	48	36	41
CD at 5%	92	134	143	104	118

Application of 100% nitrogen through FYM plus sunhemp as green manure and balance of P and K through natural minerals gave higher grain yield.

12. Title of the project : Production Package for Organic little millet under rain fed conditions in red sandy loam soils of North Coastal Zone

Location : Vizianagaram, Andhra Pradesh
Crop : Little millet
Year of study : 2015-16 to 2019-20

Keeping in view of sustained crop yields as well as soil health, long term experiment is being conducted at Agricultural Research Station, Vizianagaram, Acharya N.G. Ranga Agricultural University, Andhra Pradesh since *Kharif* 2015-16 to 2019-20 for five consecutive years to study the effect of complete organic method of farming in comparison to conventional (inorganic) method on soil health, yield and quality of little millet crop. The organic inputs were supplied in the form of green manures (Sunhemp), farmyard manure, neem cake and biofertilizers (*Azospirillum spp.* and Phosphorus Solubilising Bacteria) whereas the conventional (inorganic) method is by application of recommended dose of inorganic fertilizers (20-20-20 kg NPK/ha). The experiment was laid out with two treatments i.e. organic treated plot and inorganic treated plot (conventional plot). The Recommended Dose of Fertilizer (20:20:20 kg NPK ha⁻¹) was applied to inorganic treated plot in the form of urea in two split doses, Single Super Phosphate and Muriate of Potash as basal dose. The organic plot was incorporated with green manure (sun hemp) and FYM@ 5t ha⁻¹ applied during the last ploughing. Moreover biofertilizers were also applied in the form of *Azospirillum* and *PSB* @ 5 kg ha⁻¹ and top dressed with Neem cake @ 0.5 t ha⁻¹ at the time of maximum tillering.

Results: Among both the treatments after five years (Table1) significantly higher grain yields were recorded in inorganic treated plot when compared to the organic plot. Initially the difference in the grain yield between the two plots was 33% which after five years reduced to 17%. Among the plant growth characters, No. of productive tillers/plant and leaf length were found significantly high in organic plot and Inflorescence length and number of spikes/plant were found significantly high in inorganic plot compared to organic plot .

The physicochemical properties (Table 2) were not significantly influenced by both the treatments. But due to the continuous incorporation of green manures and application of organic manures the organic carbon % has increased by 0.2% in organic treated plot (0.46%) compared to the initial value (0.43%). After 5 years the soil available Nitrogen has shown decrease of 1.25% and 6.67% and soil available Phosphorus has shown an increase of 10.17% and 3.39% over the initial values in organic and inorganic plots respectively. But the soil available potassium has shown a decrease of 5.08% in the inorganic plot over the initial value. Among the micronutrients soil available Zn, Fe, Cu and Mn has shown a decline over the initial in both organic and inorganic plots.

Nitrogen content (Table 3) in grain and straw was found highest in the organic treated plot when compared to the inorganic plot whereas the Phosphorus and Potassium content was found highest in the conventional method even after 5 consecutive years. Whereas the Zn and Fe contents in the grain and straw was found highest in the organic plot. More over the bio-chemical components (Table 4) viz., Total anti-oxidant activity, Total Phenol Content, Total sugar, Protein content and Tannin content were analyzed in little millet grain. It was found that total phenol content was found highest in inorganic grain whereas total anti oxidant activity, total sugar, protein content and tannin content were found highest in organic little millet grain.

Table 1 : Performance of organic trial of little millet (var. OLM 203) over 5 years

Treatments	Grain yield (kg/ha)					Benefit : Cost Ratio				
	2015-16	2016-17	2017-18	2018-19	2019-20	2015-16	2016-17	2017-18	2018-19	2019-20
T1 Organic	501.2	659.3	813.4	698.1	977.7	0.57	0.75	0.93	0.9	1.35
T2 Inorganic	670.4	858.1	835.2	778.4	1147.3	1.28	1.64	1.6	1.73	2.31

Table 2 :Changes in physico-chemical properties of soils in organic trial of little millet.

	Organic					Inorganic					Initial soil Test Val-ues	% increase over initial values	
	2015-16	2016-17	2017-18	2018-19	2019-20	2015-16	2016-17	2017-18	2018-19	2019-20		Or-ganic	Inor-ganic
pH	6.95	7.04	7.2	6.92	7.04	6.9	7.02	7.1	6.88	7.19	6.95	1.37	3.53
EC (dSm ⁻¹)	0.38	0.35	0.15	0.23	0.27	0.39	0.39	0.13	0.22	0.3	0.24	12.50	25.00
O.C. (%)	0.44	0.44	0.45	0.46	0.46	0.42	0.42	0.43	0.43	0.43	0.43	6.98	0.00
Av.N (kg/ha)	201	222	235	242	237	189	205	226	234	224	240	-1.25	-6.67
Av. P ₂ O ₅ (kg/ha)	61	66.7	60.3	63.6	65	57	61	64	61.7	61	59	10.17	3.39
Av. K ₂ O (kg/ha)	316	340	321	326	324	314	314	315	319	299	315	2.86	-5.08
Av. Fe (ppm)	10.25	12.75	10.5	12.13	9.51	9.32	10.13	9.85	11.8	8.29	9.6	-0.94	-13.65
Av.Mn (ppm)	7.32	8.77	5.43	6.43	6.14	7.48	6.7	4.35	2.35	5.71	6.03	1.82	-5.31
Av.Zn (ppm)	1.06	1.18	1.22	0.91	1.16	0.82	0.89	1.15	0.83	0.98	1.22	-4.92	-19.67
Av.Cu (ppm)	3.05	3.36	2.55	2.1	2.24	2.39	2.65	2.42	1.93	2.07	2.35	-4.68	-11.91

Table 3: Effect of organic manures and fertilizers on nutrient content of little millet

S. No.	Parameters	Organic (T1)					Inorganic (T2)				
		2015-16	2016-17	2017-18	2018-19	2019-20	2015-16	2016-17	2017-18	2018-19	2019-20
1	N (%)	0.88	0.89	1.12	1.06	1.25	0.81	0.80	0.96	0.92	0.91
2	P (%)	0.32	0.33	0.34	0.28	0.27	0.35	0.38	0.39	0.32	0.31
3	K (%)	0.64	0.72	0.72	0.74	0.75	0.72	0.78	0.76	0.81	0.80
4	Zn (mg/kg)	19.4	21.5	23.6	28.5	29.5	17.5	19.6	20.3	23.2	25.3
5	Fe (mg/kg)	38.6	39.4	40.6	41.8	42.2	34.5	34.8	35.4	35.0	35.7

Table 4 : Effect of organic manures and inorganic fertilizers on bio-chemical analysis by little millet

Particulars	Organic treatment	Inorganic treatment
Total anti-oxidant activity by DPPH radicle scavenging activity (mg AAE/100g)	88.40	63.05
Total Phenol Content (mg/100g)	52.37	45.59
Total sugar (%)	61.19	64.16
Protein content (%)	6.00	5.90
Tannin content (mg/100g)	393.33	376.67



Application of FYM



Application of Carrier based biofertilizers



Application of Neem cake at tillering stage



Organic little millet crop

13. Title of the project : Organic production package for Little millet under rainfed condition

Location : Dindori, Madhya Pradesh

Crop : Little millet

Year of study : 2007-08 to 2010-11

An experiment on different doses of organic manure in combination with bio-fertilizer and natural mineral nutrients (Rock Phosphate) along with inter cropping of pulses for *in-situ* incorporation to sustaining soil health and productivity of Little millet-pulse cropping system was conducted during 2007-08 to 2010-11. Results revealed that integrated application of 5.0 t FYM/ha with recommended dose of NPK fertilizers gave significantly highest grain yield (1097 Kg/ha), Net income (Rs. 5147.00/ha) and B:C ratio (1.64). Among the various organic production packages, application of enriched FYM with bio-fertilizer to supply 100% recommended nitrogen besides *in-situ* incorporation of sunhemp as a green manure crop and balance of P & K through natural minerals gave higher grain yield (947 kg/ha) as compared to other organic manures. However, it was closely followed by application of enriched vermicompost with bio-fertilizers to supply 100 % recommended N and balance of P & K through natural minerals (912 kg/ha).

14. Title of the project : Production technology for organic foxtail millet

Location : Nandyal

Crop : Foxtail millet

Year of study : 2003-04

Treatments	Grain (kg/ha)	Straw (kg/ha)
T1: 100% RDF	1438	4062
T2: FYM alone (5 t/ha)	792	3063
T3: vermicompost (1t/ha)	1335	3563
T4: biofertilizer inoculation	1238	3529
T5: FYM+ biofertilizer	1495	3925
T6: RDF + FYM	1731	4795
T7: RDF + vermicompost	1532	4362
T8: RDF + biofertilizer	1664	4528
T9: RDF + FYM + biofertilizer	1660	4660
T10: RDF + vermicompost + biofertilizer	1525	4462
T11: foxtail millet followed by pigeon pea	956	3295
T12:foxtail millet + green gram incorporation	995	3506
SEm±	86	401
CD @5%	254	1175

The results revealed that application recommended dose of fertilizer and FYM was beneficial in enhancing the yield significantly over other treatments but it was on par with recommended dose of fertilizer and bio fertilizer application, recommended dose of fertilizer + FYM + biofertilizer application. However, lowest grain yield was recorded in treatment where foxtail millet followed by pigeon pea rotation was followed.

15. Title of the project : Production technology for organic foxtail millet

Location : Nandyal
 Crop : Foxtail millet
 Year of study : 2007-2008

Treatments	Grain yield (kg/ha)				Straw yield (kg/ha)				B:C ratio			
	2005	2006	2007	mean	2005	2006	2007	mean	2005	2006	2007	mean
T1: FYM @ 5t/ha	2214	1148	897	1344	4165	5178	2722	3752	2.18	1.08	0.48	1.09
T2: vermicompost @1 t/ha	1864	1143	920	1258	3332	5297	2892	3602	0.79	0.39	0.26	0.36
T3: bio fertilizer seed inoculation	2057	1118	1079	1360	4093	5356	3089	3898	2.6	1.62	0.66	1.46
T4: FYM 50% + VC 50%	2753	1148	1069	1565	4641	5832	2971	4166	2.06	0.73	0.45	0.90
T5: FYM 50% + T3	1976	1134	996	1357	3451	6011	2812	3902	1.96	1.37	0.57	1.21
T6: FYM + VC 50% + T3	2555	1176	1026	1535	3332	6487	2959	4097	1.45	0.77	0.42	0.77
T7: 100% RDF	3111	1560	1156	1812	4498	7261	3390	4690	2.46	2.26	0.65	1.59
T8: 100% RDF + FYM @ 5 t/ha	3393	1446	1151	1892	4760	7142	3377	4820	2.12	1.40	0.55	1.21
T9: CONTROL	1904	1076	784	1168	3570	5118	2361	3401	2.43	1.64	0.64	1.32
SEm±	185	119	81	-	389	217	251	-	-	-	-	-
CD at 5%	557	359	234	-	1166	653	653	-	-	-	-	-

Mean over years data indicated that the application of recommended FYM along with recommended dose fertilizers gave higher grain yield as compared to other treatments. Whereas, application of recommended dose of fertilizers alone was found to be second best.

16. Title of the project : Production technology for organic foxtail millet under rainfed conditions

Location : Nandyal
 Crop : Foxtail millet
 Year of study : 2009-2010

Treatments	Grain yield		Straw yield		B:C ratio	
	2008	2009	2008	2009	2008	2009
T1: FYM @ 5t/ha	1176	1106	4220	2052	2.60	2.12
T2: vermicompost @1 t/ha	999	965	4020	2103	1.10	0.85
T3: bio fertilizer seed inoculation	1104	1131	3554	2209	3.20	3.08
T4: FYM 50% + VC 50%	1188	1091	4662	2168	2.00	1.48
T5: FYM 50% + T3	1017	1143	3776	2272	2.50	2.56
T6: FYM + VC 50%+ T3	1200	1023	4221	2336	1.90	1.30
T7: 100% RDF	1520	1462	5554	2370	4.40	3.65
T8: 100% RDF + FYM @ 5 t/ha	1495	1563	5710	2995	3.20	2.93
T9: CONTROL	910	828	3554	1969	2.80	2.21
SEm±	160	152	462	252	-	-
CD at 5%	482	455	1386	757	-	-

Mean over years data revealed that application of recommended doses of FYM and fertilizers gave higher grain yield whereas net monetary returns and B:C ratio were higher with application of only inorganic fertilizers. However, among organic sources of nutrients tried, application of FYM and vermicompost to supply 50 per cent recommended dose of nitrogen from each source gave higher grain yield but net monetary returns and B:C ratio were higher with bio fertilizer seed inoculation treatment.

17. Title of the project : Production technology for organic barnyard millet

Location : Ranichauri
Crop : Barnyard millet
Year of study : 2003-04

Treatments	2002		2003		Mean	
	Grain	Straw	Grain	Straw	Grain	Straw
T1	220	4060	1425	5865	823	4963
T2	460	4875	2165	6685	1313	5780
T3	345	4465	1975	6125	1160	5295
T4	285	4455	2060	6350	1173	5403
T5	320	4525	1825	5975	1073	5250
T6	380	4670	1950	6100	1165	5385
T7	550	5060	2150	6425	1350	5743
T8	275	4340	1720	6100	998	5520
SEm±	26	61	-	-	-	-
CD @5%	80	185	180	265	-	-

Mean over years data indicated that application of recommended dose of FYM and fertilizers were essential for realizing higher yield and second best was application of 3.75 t FYM/ha coupled with Mussoorie rock phosphate and gypsum.

T1: BM with only 7.5 t FYM/ha

T2: BM with only 3.75 t FYM/ha + mussoorie rock phosphate + gypsum

T3: BM with only 7.5 t FYM/ha + biofertilizer

T4: BM + soyabean (intercropping) 1:1 (incorporation of soyabean as green manure and 7.5 t FYM/ha)

T5: BM + ricebean (intercropping) 1:1 (incorporation pf ricebean as green manure and 7.5 t FYM/ha)

T6: BM + plant protection measures through biopesticides

T7: BM with RDF + fertilizer

T8: BM- pea two year rotation with rec. organic manures

18. Title of the project : Production technology for organic barnyard millet

Location : Ranichauri
Crop : Barnyard millet
Year of study : 2004-05

Treatments	grain	Straw
Rec. FYM @7.5 t/ha and inorganic fertilizers to supply NPK 50:40:25 kg/ha	1860	4460
Rec. FYM @7.5 t/ha + additional top FYM to supply 50kg N/ha (100% N), balance of P and K through natural minerals	1625	4030
Pine needle compost to supply 50kg N/ha (100% N) AND balance of P and K through minerals	1600	4250
Vermicompost to supply 50 N kg/ha (100% N) AND balance of P and K through minerals	2025	4275
Poultry manure to supply (100% N) AND balance of P and K through minerals	2350	4650
Enriched compost to supply (100% N) and balance of P and K through oilcakes	2300	4665
Enriched pine needle compost to supply 100% of rec. P and K and balance of N through oilcake	2250	4520
25 kg N through FYM + 25 kg N through oilcake and balance P and K through minerals	2480	4850
25 kg N through vermicompost + 25 kg N through oilcake + balance P and K through minerals	2650	4875
25 kg N through poultry manure + 25 kg N through oilcake + balance P and K through minerals	2800	5165
25 kg N through pine needle compost + 25 kg N through oilcake + balance P and K through minerals	2600	4850
Rec. NPK through chemical fertilizers	2800	5050
Control	1225	3450
SEm±	79	154
CD at 5%	230	450

The results indicated that application of 25 kg N through poultry manure and balance of N through oil cake and P and K was supplied through natural minerals gave significantly higher grain yield compared to other treatments but it was on par with plots receiving recommended NPK fertilizer through in organic source and supplying 25 kg N through either vermin compost/ pine needle compost and balance of N through oil cakes along with P and K through natural minerals. Absolute control gave lowest yield.

19. Title of the project : Production technology for organic barnyard millet

Location : Bangalore, Ranichauri, Coimbatore

Crop : Barnyard millet

Year of study : 2004-05

Treat ments	Bangalore				Ranichauri		coimbatore					
	Grain		Straw		Grain	Straw	Grain			Straw		
	2004	2005	2004	2005	2005	2005	2003	2004	2005	2003	2004	2005
T1	3076	2729	3064	5304	2084	4861	2725	2526	2914	5320	4928	5343
T2	2248	2236	3219	3863	1723	4000	2300	2179	2394	4500	3901	4658
T3	2231	1797	3483	2698	2055	4972	2410	1999	2261	4200	3762	4521
T4	2760	2901	3505	3611	2166	5222	2365	1749	2055	4420	3637	4453
T5	2497	2034	3516	4140	2333	5249	2420	1596	1918	4650	3262	4187
T6	2010	1820	3031	3016	1999	4666	2850	2443	2877	6000	4026	4993
T7	2225	1875	2965	3095	1889	4305	3010	2457	2568	6550	4553	4700
T8	2749	2386	4034	4524	2416	5277	2500	2332	2945	4700	4012	5136
T9	2676	3298	3153	5291	2500	5333	850	1249	1033	1750	2846	2473
SEm±	110	166	201	222	124	109	74	57	33	10	81	61
CD at 5%	330	497	602	665	361	323	223	173	101	317	243	197

Third season grain yield data indicated that treatments receiving finger millet-pigeon pea two year rotation with organic nutrition 7.5 t FYM/ha or application of 7.5 t FYM/ha alone or finger millet + pigeon pea inter cropping along with plant protection measures with bio pesticides gave comparable grain yields and were significantly superior over other treatments. However, recommended inorganic nutrition gave lowest yield. Similar trend in yield was observed in mean over years data whereas B:C ratio indicated that finger millet + pigeon pea (8:2) inter cropping was more remunerative because of highest B:C ratio.

T1: Rec. FYM @7.5 t/ha and inorganic fertilizers to supply NPK 50:40:25 kg/ha

T2: Rec. FYM @7.5 t/ha + additional of FYM to supply 100% N balance of P and K through natural minerals

T3: Vermicompost to supply 100% N and balance of P and K through minerals

T4: Poultry manure to supply 100% N and balance of P and K through minerals

T5: FYM to supply 100% N balance of P and K through natural minerals

T6: 25 kg N through FYM + concentrates to supply balance P and K through minerals

T7: 25 kg N through vermicompost + concentrates to supply balance P and K through minerals

T8: 25 kg N through poultry manure + concentrates to supply balance P and K through minerals

T9: Rec. NPK through chemical fertilizers

20. Title of the project : Production package for organic barnyard millet

Location : Ranichauri
Crop : Barnyard millet
Year of study : 2008-2009

Treatments	Grain yield (kg/ha)				Straw yield (Kg/ha)			
	2006	2007	2008	Mean	2006	2007	2008	Mean
T1:vermicompost @ 2 t/ha	1083	1813	1353	1416	3389	4907	1533	4276
T2: vermicompost @ 4 t/ha	1249	1920	1447	1539	3500	5023	5160	4561
T3: FYM @ 8 t/ha	889	1730	1545	1388	3111	4762	4340	4071
T4: Chullu cake @ 4 t/ha	999	1910	1688	1532	3220	5089	4973	4427
T5: Enriched manure @ 8 t/ha	1278	1990	1675	1648	3611	5106	4890	4536
T6:VC + FYM (50% N from each other)	1417	2190	1844	1817	3334	6545	5637	5172
T7: VC + CC (50% N from each other)	1944	2330	2319	2198	4889	9745	7797	7477
T8: VC+ CC (50% N from each other) + Azatobacter	2111	2450	2506	2356	5667	10133	7810	7870
T9:VC+ EM (50% N from each other)	1333	2060	1837	1743	4172	5046	6323	5180
T10:VC+ FYM (50% N from each other) + Azatobacter	1372	2100	1983	1818	4556	5469	6016	5347
T11: recommended NPK through inorganic source	2172	2493	2390	2352	6567	10210	8995	8591
T12: control	660	1149	1097	969	2889	3214	3300	3134
SEm±	92	106	86	-	337	496	452	-
CD at 5%	269	312	252	-	989	1454	1325	-

Application of vermicompost and chullu cake to provide 50% N from each source along with biofertilizer seed treatment gave significantly higher grain yield and B:C ratio as compared to other treatments tried. However, it was on par with application of vermicompost + chullu cake to provide 50% N from each source and recommended dose of NPK alone whereas absolute control gave lowest yield.

21. Title of the project : Production package for organic barnyard millet under rainfed conditions

Location : Ranichauri
 Crop : Barnyard millet
 Year of study : 2009-2010

Treatments	Grain Yield (kg/ha)	Straw Yield (kg/ha)
T1:vermicompost @ 2 t/ha	984	4164
T2: vermicompost @ 4 t/ha	1078	4091
T3: FYM @ 8 t/ha	1176	3971
T4: Chullu cake @ 4 t/ha	1319	4604
T5: Enriched manure @ 8 t/ha	1426	4521
T6:VC + FYM (50% N from each other)	1475	5268
T7: VC + CC (50% N from each other)	1949	5341
T8: VC+ CC (50% N from each other) + Azatobacter	2137	5428
T9:VC+ EM (50% N from each other)	1467	4654
T10:VC+ FYM (50% N from each other) + Azatobacter	1614	5247
T11: recommended NPK through inorganic source	2021	5526
T12: control	72	2931
SEm±	86	452
CD at 5%	252	1325

Application of vermicompost and chullu cake to supply 50% recommended dose of nitrogen from each source and seed treatment with bio fertilizer gave significantly higher grain yield as compared to other treatments but it was on par with application of vermicompost and chullu cake to supply 50% recommended dose of nitrogen from each source and application of recommended dose of inorganic fertilizers.

22. Title of the project : production technology for organic kodo millet

Location : Dindori
 Crop : Kodo millet
 Year of study : 2003-05

Treat-ments	2003		2004		2005		Mean		2005 B:C ratio
	Grain yield (kg/ha)	Straw yield (kg/ha)	Grain yield (kg/ha)	Straw yield (kg/ha)	Grain yield (kg/ha)	Straw yield (kg/ha)	Grain yield (kg/ha)	Straw yield (kg/ha)	
T1	728	978	586	840	745	920	686	913	1.08
T2	982	1274	621	940	728	4980	777	2398	1.13
T3	813	1095	864	1068	978	1346	885	1170	1.39
T4	1018	1281	494	723	674	823	729	942	1.02
T5	1056	1186	512	798	634	936	734	973	0.97
T6	962	1269	594	815	646	878	734	987	1.49
T7	1582	1958	1186	1643	1238	1668	1335	1756	-1.45
T8	804	1036	846	1043	923	1266	858	1115	1.35
SEm±	64	93	38	45	69	96	-	-	-
CD at 5%	191	278	103	103	196	287	-	-	-

Application of recommended FYM along with recommended inorganic nutrition gave higher kodo grain yield whereas inter cropping of kodo + Pigeon pea in 2:1 row proportion gave higher B:C ratio 1.49 besides giving higher kodo millet grain equivalent yield. However, kodo millet + sunhemp inter cropping in 1:1 row proportion and incorporation of sunhemp as green manure and application of 7.5 t FYM/ha gave lower grain yield.

T1: KM supplied with only 7.5t FYM/ha

T2: KM supplied with 3.75 t FYM /ha + Roc phosphate + gypsum

T3: KM supplied with 7.5 t FYM/ha + bio fertilizer (seed inoculation) Azoospirillum, brassilience and aspergillus awamooore @ 5g/kg of seed

T4: KM + soybean intercropping, 1:1(incorporation of soybean as green manure and application of 7.5 t FYM/ha)

T5: KM + sunhemp intercropping, 1:1 (intercropping of sunhemp as green manure and application of 7.5 t FYM/ha)

T6: KM + pigeon pea intercropping 2:1 (plant protection measure through biofertilizers)

T7: KM with recommended FYM + fertilizer

T8: KM – legume two year rotation with recommended organic manures

23. Title of the project : Production technology for organic kodo millet

Location : Dindori, Jagadapur and Rewa

Crop : Kodo millet

Year of study : 2006-2007

Treatments	Grain	Straw	B:C ratio
T1	2300	3226	0.74
T2	1767	2478	0.47
T3	1967	2758	0.41
T4	2533	3553	0.71
T5	2267	3179	0.64
T6	1966	2758	0.42
T7	2533	3553	0.47
T8	2733	3834	0.6
T9	1767	2478	1.31
SEm±	94	250	-
CD at 5%	282	750	-

Application of poultry manure to supply 25 kg N and balance of N, P and K were applied through concentrated cakes and natural minerals gave significantly higher grain yield compared to other treatments but it was on par with application of poultry manure to supply 100% N through and balance of P & K through natural minerals and application of vermin compost to supply 25 kg N and balance of N through concentrated cakes and P & K through natural minerals.

T1: Rec. FYM @7.5 t/ha and inorganic fertilizers to supply NPK 50:40:25 kg/ha
T2: Rec. FYM @7.5 t/ha + additional of FYM to supply 100% N balance of P and K through natural minerals
T3: Vermicompost to supply 100% N and balance of P and K through minerals
T4: Poultry manure to supply 100% N and balance of P and K through minerals
T5: FYM to supply 100% N balance of P and K through natural minerals
T6: 25 kg N through FYM + concentrates to supply balance P and K through minerals
T7: 25 kg N through vermicompost + concentrates to supply balance P and K through minerals
T8: 25 kg N through poultry manure + concentrates to supply balance P and K through minerals
T9: Rec. NPK through chemical fertilizers

24. Title of the project : Production package for organic kodo millet

Location : Dindori & Jagadapur
Crop : Kodo millet
Year of study : 2007-2008

Treatments	Grain	Straw	B:C ratio
T1: 50% N through VC + balance P&K through natural minerals	728	1082	0.97
T2: 100% N through VC + balance P&K through natural minerals	1372	1927	1.37
T3: 50% N through FYM + balance P&K through natural minerals	703	1104	1.00
T4: 100% N through FYM + balance P&K through natural minerals	1133	1640	1.26
T5: T1+Biofertilizers	836	1463	1.09
T6: T2+Biofertilizers	1458	2122	1.43
T7: T3+Biofertilizers	804	1234	1.12
T8: T4+Biofertilizers	1266	1837	1.38
T9:T1 + T3	894	1267	0.94
T10: T9+Biofertilizers	989	1405	1.02
T11: 50% N through FYM + sunhemp as green manuring + balance P&K through natural minerals	929	1369	1.16
T12: T11+Biofertilizers	1056	1521	1.29
T13: 100% N through FYM + sunhemp as green manuring + balance P&K through natural minerals	1281	1781	1.16
T14: T13+Biofertilizers	1352	1960	1.21
T15: RDF (40;20:00 kg NPK/ha)	1588	2242	1.99
T16: absolute control	558	792	1.11
SEm±	33	-	-
CD at 5%	98	-	-

Application of enriched vermicompost to supply 100% N and balance of P and K through natural minerals gave significantly higher grain yield, NMR and B:C ratio over other treatments. However, it was found to be on par with application of vermicompost to supply 100% N and balance of P and K through natural minerals. Whereas application of RDF gave significantly higher grain yield, NMR and B:C ratio.

25. Title of the project : Production package for organic kodomillet

Location : Dindori, Rewa and Jagadapur
Crop : Kodomillet
Year of study : 2008-2009

Treatments	Yield (kg/ha)	
	Grain	Straw
T1: KM with 7.5 t FYM/ha	1716	3763
T2: KM with 3.75 t FYM/ha + Imali leaves @ 3.75 t/ha + bio fertilizers	1637	3355
T3: KM with Imali leaves @ 2.5 t/ha + sal burnt husk @2.5 t/ha + bio fertilizers	1663	2452
T4: KM with Imali leaves @ 2.5 t/ha + sal burnt seed @2.5 t/ha + bio fertilizers	1431	2625
T5: KM with 2.5 t/ha of sal leaves + sal burnt husk @2.5 t/ha + bio fertilizers	1604	3400
T6: KM with sal burnt seed @ 5t/ha + bio fertilizers	1357	2642
T7: KM with sal burnt husk @ 5 t/ha + bio fertilizers	1508	2913
T8: KM with Imali leaves @ 5 t/ha + bio fertilizers	1736	3446
T9: KM with sal leaves @ 5t/ha + bio fertilizers	2239	5235
T10: KM with RDF (20:20:10 kg NPK/ha)	2362	5358
SEm±	74	132
CD at 5%	220	392

Application of Sal leaves enriched with biofertilizers at 5 tons/ha gave significantly higher grain yield as compared to other sources of organic manures used. However, application of RDF gave on par yield to that of plots receiving Sal leaves at 5 tons/ha enriched with biofertilizers.

26. Title of the project : Organic production package for kodo millet under rainfed condition

Location : Dindori, Madhya Pradesh
Crop : Kodo millet
Year of study : 2007-08 to 2010-11

An experiment on different doses of organic manure in combination with bio-fertilizer and natural mineral nutrients (Rock Phosphate) along with inter cropping of pulses for *in-situ* incorporation to sustaining soil health and productivity of kodo millet-pulse cropping system was conducted during 2007-08 to 2010-11. Results revealed that, application of enriched FYM with bio-fertilizer to supply 100% recommended nitrogen besides *in-situ* incorporation of sunhemp as a green manure and balance of P & K through natural minerals gave higher grain yield (1404 kg/ha) as compared to other organic manures tried, but it was on par with application of enriched vermicompost with bio-fertilizers to supply 100 % recommended N and balance of P & K through natural minerals (1381 kg/ha).

27. Title of the project : Organic production package for kodo millet under rainfed conditions

Location : Dindori & Rewa

Crop : Kodo millet

Year of study : 2009-2010

Treatments	Grain yield (kg/ha) 2009	
	Dindori	Rewa
T1: 50% N through VC + balance P&K through natural minerals	884	1111
T2: 100% N through VC + balance P&K through natural minerals	1204	1194
T3: 50% N through FYM + balance P&K through natural minerals	780	1292
T4: 100% N through FYM + balance P&K through natural minerals	1105	1236
T5: T1+Biofertilizers	948	1181
T6: T2+Biofertilizers	1316	1722
T7: T3+Biofertilizers	844	1306
T8: T4+Biofertilizers	1210	1264
T9:T1 + T3	1080	1333
T10: T9+Biofertilizers	1202	1292
T11: 50% N through FYM + sunhemp as green manuring + balance P&K through natural minerals	1010	958
T12: T11+Biofertilizers	1140	1097
T13: 100% N through FYM + sunhemp as green manuring + balance P&K through natural minerals	1270	1536
T14: T13+Biofertilizers	1380	1389
T15: RDF (40;20:00 kg NPK/ha)	1580	2125
T16: absolute control	690	1056
SEm±	42	122
CD at 5%	125	352

Among organic source of nutrients tried, application of enriched vermicompost to supply 100 per cent recommended dose of nitrogen and balance of P and K through natural minerals gave higher grain yield. However, application of recommended dose of nutrients through inorganic fertilizers gave highest grain yield.

28. Title of the project : Production technology for organic kodo millet under rainfed conditions

Location : Dindori & Rewa

Crop : Kodo millet

Year of study : 2011-2012

Treatments	Grain yield (kg/ha)		NMR (Rs./ha)		B:C ratio	
	2010	2011	2010	2011	2010	2011
T1: 50% N through VC + balance P&K through natural minerals	920	980	1572	2172	2.15	1.22
T2: 100% N through VC + balance P&K through natural minerals	1289	1320	3916	4226	1.37	1.47
T3: 50% N through FYM + balance P&K through natural minerals	880	870	1208	1108	1.14	1.18
T4: 100% N through FYM + balance P&K through natural minerals	1190	1240	2716	3216	1.22	1.35
T5: T1+Biofertilizers	1010	1060	2606	2910	1.28	1.32
T6: T2+Biofertilizers	1435	1470	5519	5669	1.51	1.53
T7: T3+Biofertilizers	972	975	2168	2098	1.16	1.26
T8: T4+Biofertilizers	1309	1390	4034	4668	1.35	1.44
T9:T1 + T3	1216	1238	3116	4736	1.29	1.26
T10: T9+Biofertilizers	1338	1382	4428	5968	1.37	1.33
T11: 50% N through FYM + sunhemp as green manuring + balance P&K through natural minerals	1040	1105	2764	3014	1.30	1.31
T12: T11+Biofertilizers	1146	1250	3896	4236	1.42	1.41
T13: 100% N through FYM + sunhemp as green manuring + balance P&K through natural minerals	1408	1490	5038	5258	1.23	1.39
T14: T13+Biofertilizers	1585	1680	7013	6963	1.64	1.47
T15: RDF (40;20:00 kg NPK/ha)	1888	1970	12808	10828	1.80	1.83
T16: absolute control	632	740	292	1372	1.02	1.16
SEm±	36	48	-	-	-	-
CD at 5%	107	143	-	-	-	-

Mean over locations data across years indicated that among the organic manures and their combinations, application of FYM to supply 100 per cent nitrogen, sunhemp as green manure and balance of phosphorus and potassium through natural minerals gave higher grain yield.

c. Effect of organic manures in small millets

1. Title of the project : Effect of organic manures with different levels of nitrogen on yield of finger millet under rainfed conditions

Location : Bengaluru
Crop : Finger millet
Year of study : 1987-88

Treatment	Grain yield (kg/ha)				Straw yield (kg/ha)			
	FYM (kg/ha)				FYM (kg/ha)			
	0	5000	750 enriched	Mean	0	5000	750 enriched	Mean
0	2225	2454	2199	2293	2521	3009	2752	2761
20	2894	3447	3215	3185	3447	3652	3833	3644
40	3100	3760	3413	3424	3601	4218	3833	3901
Mean	2740	3220	2942		3190	3626	3489	
	FYM	N	FYM X N		FYM	N	FYM X N	
SEd±	137	137	237		125	125	127	
CD @5%	290	290	-		266	266	-	

The results revealed that FYM @ 5 tons/ha was significantly superior in yield compared to no FYM but was on par with enriched FYM @ 750 kg/ha. Among the nitrogen levels, there was linear response upto 40 kg N/ha. The interaction between FYM and nitrogen was non-significant for yield.

2. Title of the project : Investigations on enrichment and method of application of compost

Location : Bengaluru
Crop : Finger millet
Year of study : 2000-2001

Treatments	Yield (kg/ha)	
	Grain	Straw
T1: 50% NPK alone	2063	2463
T2: 100% rec. NPK alone	2695	3549
T3: Broadcasting 2.5 t un-enriched FYM/ha + 50% rec. NPK	1768	2381
T4: Band placement 2.5 t un-enriched FYM/ha + 50% rec. NPK	1892	3122
T5: Broadcasting 2.5 t un-enriched FYM/ha + 100% rec. NPK	2512	3495
T6: Band placement 2.5 t un-enriched FYM/ha + 100% rec. NPK	2715	3678
T7: Broadcasting 2.5 t enriched FYM/ha + 50% rec. NPK	2307	3281
T8: Band placement 2.5 t enriched FYM/ha + 50% rec. NPK	2641	3578
T9: Broadcasting 2.5 t enriched FYM/ha + 100% rec. NPK	2436	3155
T10: Band placement 2.5 t enriched FYM/ha + 100% rec. NPK	3036	3594
T11: Broadcasting 7.5 t un-enriched FYM/ha + 50% rec. NPK	1682	2686
T12: Band placement 7.5 t un-enriched FYM/ha + 50% rec. NPK	2153	2983
T13: Broadcasting 7.5 t un-enriched FYM/ha + 100% rec. NPK	2336	3039

T14: Band placement 7.5 t un-enriched FYM/ha + 100% rec. NPK	2961	4087
T15: Broadcasting 7.5 t enriched FYM/ha + 50% rec. NPK	2106	3472
T16: Band placement 7.5 t enriched FYM/ha + 50% rec. NPK	3073	3941
T17: Broadcasting 7.5 t enriched FYM/ha + 100% rec. NPK	2473	3717
T18: Band placement 7.5 t enriched FYM/ha + 100% rec. NPK	3380	4742
SEm±	164	335
CD @5%	506	930

Enrichment with 100% recommended fertilizer P and K to 7.5 t FYM/ha and band placement has enhanced yield of finger millet which was on par with enrichment of 100% P and K to 2.5 t FYM/ha and band placement. Band placement of FYM has enhanced the yield of finger millet compared to broadcasting. Recommended NPK 100% dose has given higher yield compared to 50% recommended NPK. Straw yield also followed similar trend as that of grain yield.

3. Title of the project : Effect of *Sesbania* mulching on productivity of finger millet

Location : Almora
Crop : Finger millet
Year of study : 2016-2017

Treatments		Grain yield	Straw yield
M1	FYM (5 t/ha)	2309	7257
	FYM (50%) + RDF (50%)	2469	6650
	RDF (100%)	2325	6342
	Control	813	2204
M2	FYM (5 t/ha)	2084	5186
	FYM (50%) +RDF (50%)	2103	5802
	RDF (100%)	2483	7293
	Control	503	1532
Location mean		1885.86	5283
CD at 5%		170.23	590.08
Mulching (M)			
M1		1979	5613
M2		1793	4953
CD at 5%		138.55	873.53
Fertility level (F)			
FYM (5 t/ha)		2196	6221
FYM (50%) + RDF (50%)		2286	6226
RDF (100%)		2404	6817
Control		658	1868
CD at 5%		120.37	417.25

M1: line sowing of finger millet and broadcasting of sesbaniasesbania (to be cut at 25-30 DAS and used as mulch)

M2: finger millet without mulch

Significantly higher grain yield and straw yield was obtained in Sesbania mulching. Whereas 100 per cent RDF gave significantly higher grain yield compared to absolute control. Interaction between mulching and fertilizer levels was significant on grain yield, mulching Sesbania with 50% RDF and 50% FYM gave higher yield.

4. Title of the project : Effect of organic foliar spray on growth and yield of ragi (*Eleusine coracana*) in Tiruvannamalai District

Location : Athiyandal, Tamil Nadu

Crop : Finger millet

Soil application of Farm yard manure with sunhemp insitu incorporation at 45 Days after planting with 3 % panchakavya spray at 45 Days after planting/ sowing recorded higher grain yield of 2420 kg/ha and straw yield of 3549 kg/ha , net returns (Rs.55372/ha) and B:C ratio (2.84).

5. Title of the project : Effect of organic manures in small millets

Location : Ranichauri, Uttarakhand

Crop : Finger millet

- ◆ Application of 7.5 t/ha FYM gave significantly higher yield as compare to no use of FYM in finger millet based cropping system.
- ◆ Three to four hand weeding required to control weeds throughout the crop season and to get maximum return in millet production.
- ◆ For finger millet production under organic mode, application of FYM followed by vermicompost gave significantly higher yield. Incorporation of green manure sanai and dhaincha gave higher grain yield compare to other green manuring crop.
- ◆ Inter cropping of finger millet and soybean, 1:1 row and incorporation of soybean as green manure along with 7.5 t/ha of pine needle compost were essential for realizing yield at par with the application of 3.7 t FYM/ha coupled with mussoorie rock phosphate and gypsum.
- ◆ Finger millet +soybean-oat and finger millet+ soybean- barley cropping sequence was the best choice with the application of farm yard manure @ 7.5 t/ha.
- ◆ Application of farm yard manures @ 2.5 t/ha+ vermicompost @ 1.0 t/ha + chullu cake (apricot cake 1.0 t/ha) recorded maximum yield in finger millet-barley cropping system.
- ◆ Application of vermicompost + chullu cake (apricot) each 50 % N from each sources + seed inoculation with *Azotobacter* gave higher production in finger millet.

6. Title of the project : Effect of organic manures in finger millet

Location : Waghai, Gujarat

Crop : Finger millet

Year of study : 2018

The farmers of South Gujarat heavy rain fall zone I (AES-I) growing finger millet variety GN-4 during kharif season are recommended to fertilize the crop with 50 % Nitrogen through FYM + 25 % Nitrogen through biocompost + 25 % Nitrogen through castor cake + *Azotobacter* @ 2 l/ha + PSB @ 2 l/ha for getting higher yield and net income.

7. Title of the project : Effect of organic manures with different levels of fertilizers on yield of little millet

Location : Similiguda
Crop : Little millet
Year of study : 1986-87

Fertilizers N:P:K (kg/ha)	Grain yield (kg/ha)			
	Organic manures (FYM) (q/ha)			
	0	50	100	Mean
0:20:0	237	350	381	323
20:20:0	329	535	432	462
40:20:0	453	576	597	542
60:20:0	535	586	679	600
Mean	389	512	522	
	OM	Fert.	OM x Ferti	
SEd±	22	28	44	
CD @5%	67	78	NS	

The data revealed that, there was linear response in yield as the nitrogen dose was increased from 0-60 kg/ha. 60 kg N/ha and 40 kg N/ha were on par in yield but both were significantly superior compared to 20kg N/ha and no nitrogen. FYM @ 100 q/ha produced significantly higher yield compared to control, but was on par with FYM @ 50 q/ha.

8. Title of the project : Effect of organic manures with different levels of nitrogen on yield of little millet under rainfed conditions

Location : Bengaluru, Similiguda
Crop : Little millet
Year of study : 1987-88

Treatments	Yield (kg/ha)					
	Bangalore		Similiguda		Mean over locations	
	Grain	Straw	Grain	Straw	Grain	Straw
FYM Levels (kg/ha)						
0	1287	2212	1385	2932	1336	2572
5000	1492	2438	1675	3834	1584	3136
750 enriched	1322	2246	1632	3830	1477	3038
SEd±	26	106	36	87		
CD @5%	55	NS	102	269		
Nitrogen (kg/ha)						
0	1331	2069	141	2716	1381	2393
20	1368	2435	1542	3728	1455	3082
40	1402	2392	1719	4152	1561	3272
SEd±	26	106	36	87		
CD @5%	52	225	102	269		
FYM X Nitrogen						
SEd±	45	184	58	160		
CD @5%	94	NS	NS	NS		

Farm yard manure @ 5 tons/ha gave significantly higher grain yield compared to enriched FYM @ 750 kg/ha and no FYM. Nitrogen @ 40 kg/ha gave significantly higher grain yield compared to no nitrogen but was on par with 20 kg N/ha. The interaction between FYM and nitrogen was significant for yield.

9. Title of the project : Effect of organic manures with different levels of nitrogen on yield of foxtail millet

Location : Nandyal
 Crop : Foxtail millet
 Year of study : 1986-87

Treatments	yield (kg/ha)	
	Grain	Straw
Nitrogen (kg/ha)		
0	1865	2380
20	2249	3215
40	2286	3443
60	2227	3472
SEd±	46	54
CD @5%	134	158
FM levels (q/ha)		
0	2093	3040
50	2175	3151
100	2205	3184
SEd±	39	47
CD @5%	NS	NS
CV %	6.3	5.20

Grain yields between 20 and 40 and 60 kg N/ha were on par and all three levels were significantly superior compared to no nitrogen. Among the FYM levels, 100 q/ha ranked first followed by 50 kg/ha. Interaction between nitrogen and FYM was not significant on grain yield.

10. Title of the project : Effect of organic manures with different levels of nitrogen on yield of foxtail millet under rainfed conditions

Location : Bengaluru & Nandyal
Crop : Foxtail millet
Year of study : 1987-88

Treatments	Bangalore								Nandyal							
	Grain yield (kg/ha)				Straw yield (kg/ha)				Grain yield (kg/ha)				Straw yield (kg/ha)			
	FYM (kg/ha)				FYM (kg/ha)				FYM (kg/ha)				FYM (kg/ha)			
	0	5000	750 enriched	Mean	0	5000	750 enriched	Mean	0	5000	750 enriched	Mean	0	5000	750 enriched	Mean
0	3498	3910	3553	3654	2909	3235	3009	3051	2136	2468	2528	2377	3600	5230	5606	4812
20	4090	4296	3858	4081	3362	3488	3183	334	2400	2739	2754	2631	5411	5683	5811	5635
40	4025	4385	4128	4179	3246	3619	3331	3399	2837	2566	3057	2830	5683	5532	5736	5650
Mean	3871	4197	3846		3172	3447	3174		2458	2591	2780		4898	5482	5718	
	FYM	N	FYM X N		FYM	N	FYM X N		FYM	N	FYM X N	FYM	N	FYM X N		
SEd±	71	71	123		62	62	107									
CD @5%	151	151	NS		131	131	NS		210	210	NS		544	544	32	

Among the FYM levels, 5 tons/ha gave significantly higher grain yield compared to enriched FYM @ 750 kg/ha and no FYM. Nitrogen dose of 40 kg N/ha gave significantly higher grain yield compared to no nitrogen but was on par with 20 kg N/ha. Interaction between FYM and nitrogen levels was non-significant for yield.

11. Title of the project : Effect of organic manures on yield of foxtail millet

Location : Hanumanamatti, Karnataka
Crop : Foxtail millet
Year of study : 2000-2001

Treatments	Grain yield (kg/ha)				Mean
	1998	1999	2000	2001	
Control	540	807	782	752	720
FYM 2.5 t /ha	960	1282	1088	851	1045
FYM 5 t/ha	1130	1438	1310	971	1212
20 kg N/ha	1290	1640	1585	1111	1406
40 kg N/ha	1350	1766	1685	1416	1554
FYM 2.5 t /ha +20 kg N/ha	1610	1998	1893	1887	1847
FYM 2.5 t /ha +40 kg N/ha	1680	1817	1742	1911	1787
FYM 5 t /ha +20 kg N/ha	1730	1632	1627	1990	1748
FYM 5 t /ha +40 kg N/ha	1760	1734	1745	2180	1855
SEm±	60	140	60	52	
CD @5%	175	350	168	156	

The results indicated that application of 5 t FYM/ha along with 40 Kg N/ha gave significantly higher grain yield followed by plots receiving 5 t FYM/ha with 20 Kg N/ha.

12. Title of the project : Effect of organic manures on yield of foxtail millet

Location : Hanumanamatti, Karnataka
Crop : Foxtail millet
Year of study : 2002-2003

Treatments	Grain yield (kg/ha)					Mean
	1998	1999	2000	2001	2002	
Control	540	807	782	752	330	642
FYM 2.5 t /ha	960	1282	1088	851	545	945
FYM 5 t/ha	1130	1438	1310	971	625	1095
20 kg N/ha	1290	1640	1585	1111	646	1254
40 kg N/ha	1350	1766	1685	1416	708	1385
FYM 2.5 t /ha +20 kg N/ha	1610	1998	1893	1887	757	1629
FYM 2.5 t /ha +40 kg N/ha	1680	1817	1742	1911	806	1591
FYM 5 t /ha +20 kg N/ha	1730	1632	1627	1990	836	1563
FYM 5 t /ha +40 kg N/ha	1760	1734	1745	2180	856	1655
SEm±	60	140	60	52	8	
CD @5%	175	350	168	156	28	

Response was higher for inorganic fertilizer than organic manures. It was clear that, when these two form of nutrients were combined yield increased substantially.

13. Title of the project : Optimizing appropriate nutrient management techniques for tenai (*Setaria italica*) under rainfed ecosystem

Location : Athiyandal, Tamil Nadu
Crop : Foxtail millet

Application of enriched FYM +RDF + *Panchagavya* 1 Spray at 20 DAS registered higher grain yield, straw yield , net return and B:C ratio followed by enriched FYM +RDF and enriched FYM + *Panchagavya* 2 Spray at 20 and 40 DAS in foxtail millet.

14. Title of the project : Effect of organic manures with different levels of nitrogen on yield of proso millet

Location : Dholi
Crop : Proso millet
Year of study : 1986-87

Fertilizers N:P:K (kg/ha)	Grain yield (kg/ha)			
	Organic manures (FYM) (q/ha)			
	0	50	100	Mean
0:20:0	1203	1319	1426	1316
20:20:0	2111	2134	2277	2174
40:20:0	2199	2370	2509	2359
60:20:0	2037	2486	2583	2369
Mean	1888	2077	2199	
	OM	Fert.	OM x Ferti	
SEd±	95	109	188	
CD @5%	NS	489	653	
CV %		15.8		

Among the nitrogen levels 60kg N/ha produced significantly higher grain yield compared to control but was on par with 40 kg N/ha and 20 kg N/ha. Among the organic manure tried, there was no significant difference in yield. However, FYM @ 100 Q/ha ranked first in grain yield. The interaction between organic manures and nitrogen was significant on yield

15. Title of the project : Effect of organic manures with different levels of nitrogen on yield of proso millet under rainfed condition

Location : Dholi
Crop : Proso millet
Year of study : 1987-88

Nitrogen levels (kg/ha)	Grain yield (kg/ha)				Straw yield (kg/ha)			
	Organic manures (FYM) (q/ha)				Organic manures (FYM) (q/ha)			
	0	20	40	Mean	0	20	40	Mean
0	740	1759	2083	1537	1415	2971	3773	2720
20	68	1296	1851	1265	1462	2547	3584	2531
40	694	1527	1898	137	1462	2877	3348	2562
60	740	1388	1805	1311	1415	2547	3160	2374
Mean	706	1493	1909	-	1439	2736	3466	
	FYM	N	FYM x N		FYM	N	FYM x N	
SEd±	46	39	64		108	94	187	
CD @5%	136	118	NS		316	274	548	

FYM @ 10 tons/ha gave significantly higher grain yield compared to no FYM but was on par with 5 tons FYM/ha. 60 kg N/ha gave significantly higher grain yield compared to other levels of nitrogen but was on par with 40 kg N/ha. The interaction between FYM and nitrogen was non-significant for yield.

16. Title of the project : Effect of organic manures with different levels of nitrogen on yield of proso millet under rainfed condition

Location : Bengaluru
Crop : Proso millet
Year of study : 1988-89

FYM levels (tons/ha)	Grain yield (kg/ha)					Straw yield (kg/ha)				
	Nitrogen (kg/ha)					Nitrogen (kg/ha)				
	0	20	40	60	Mean	0	20	40	60	Mean
0	833	1782	1991	2222	1707	2802	3243	3197	3522	3191
5	1065	1968	2315	2	1939	3171	3243	3287	3333	3257
10	1204	1991	2361	2500	2014	2919	3058	3297	3660	3234
Mean	1034	1914	2222	2376		2964	3181	3260	3505	
	FYM	N	FYM x N			FYM	N	FYM x N		
SEd±	81	93	161			226	260	451		
CD @5%	168	194	NS			NS	NS	NS		

FYM @ 10 tons/ha was significantly superior in yield compared to no FYM but was on par with FYM @ 5 tons/ha. Among the nitrogen levels, 60 kg/ha was significantly superior in yield compared to other levels of nitrogen but was on par with 40 kg N/ha. The interaction between FYM and nitrogen levels was non-significant for yield.

17. Title of the project : Effect of organic manures with different levels of nitrogen on yield of kodo millet under rainfed condition

Location : Dindori
Crop : Kodo millet
Year of study : 1987-88

Varieties	Grain yield (kg/ha)			
	Nitrogen levels (kg/ha)			
	0	20	40	Mean
PSC 1	339	727	525	531
PSC-3	517	703	673	631
PSC-4	342	716	832	630
Local Check	493	870	1055	806
Mean	423	754	776	
	V	N	NXV	
SEd±	95	82	164	
CD @5%		170	339	

Among the FYM levels, 5 tons/ha produced significantly higher grain yield compared to no FYM but was on par with 750 kg enriched FYM. There was linear and significant response for nitrogen upto 40 kg N/ha. The interaction between FYM levels and nitrogen was non-significant for yield.

d. Role of Biofertilizers in small millets

1. Title of the project : Effect of biofertilizers on yield of ragi under rainfed conditions

Location : GKV, Bengaluru

Crop : Finger millet

Year of study : 1986-87

Biofertilizers	Grain yield (kg/ha)				Mean	Straw yield (kg/ha)				Mean
	Organic manure levels (FYM)					Organic manure levels (FYM)				
	0 tons/ha		5 tons/ha			0 tons/ha		5 tons/ha		
	Nitrogen (kg/ha)					Nitrogen (kg/ha)				
	25	50	25	50		25	50	25	50	
B0:Control	1426	1428	1517	1698	1517	1993	2211	2250	2415	2217
B1:Azotobactor	1628	1584	1602	1827	1660	2383	2420	2676	2525	2476
B2:Azospirillum	1587	1671	1717	1925	1725	2447	2627	2574	2593	2560
B3:Phosphoous solubilizing fungus	1654	1616	1813	2026	1777	2591	2330	2574	2734	2551
B4:B1+B2	1703	1731	1964	2016	1854	2557	2544	2724	2636	2610
B5:B2+B3	1754	1778	2008	2029	1892	2599	2651	2725	2873	2712
Mean (N)	1625	1635	1770	1920		2425	2464	2583	2629	
Mean (OM)	1650		1845			2445		2606		

Grain yield (kg/ha)								Straw yield (kg/ha)						
	FYM	N	B	FYM x N	FYM x B	N x B	FYM x N x B	FYM	N	B	FYM x N	FYM x B	N x B	FYM x N x B
SEd±	20	5	29	8	42	42	57	46	171	86	62	122	122	173
CD @	86	14	59	22	85	NS	NS	NS	NS	173	NS	NS	NS	NS
CV (%)	4.16							8.38						

The data revealed that the seed inoculation of bio fertilizers increased grain yield significantly compared to control. Among the bio fertilizers, Phosphate Solubilizing fungus and *Azospirillum* were on par in yield and both were significantly superior to *Azotobacter*. Combination of phosphate solubilizing fungus with *Azotobacter* and *Azospirillum* were on par with each other but were significantly superior to seed inoculation with single Biofertilizer. The organic manure @ 5 tons/ha produced significantly higher grain yield compared to control and had a positive interaction with bio fertilizers.

2. Title of the project : Use of biofertilizers in finger millet

Location : Bengaluru and Simliguda

Crop : Finger millet

Year of study : 1987-88

Treatments	Yield (kg/ha)					
	Grain		Straw		Mean over locations	
	Bangalore	Similiguda	Bangalore	Similiguda	Grain	Straw
B0:Control	5968	1611	10285	2899	3790	6592
B1:Azotobacter	6425	1899	11148	3290	4162	7219
B2:Azospirillum	6786	2215	11954	3668	4501	7811
B3:Phosphorous solubilizing fungus	6921	2313	12000	3933	4617	7967
B4:B1+B2	7292	2418	12654	4142	4855	8397
B5:B2+B3	7684	2596	13419	4409	5140	8914
SEd±	29	44	301	36		
CD @5%	59	123	607	103		
Organic manures (kg/ha)3492						
0	6704	2008	11630	3492	4356	7561
5000	6988	2342	122249	3955	4665	8102
SEd±	20	51	123	36		
CD @5%	86	175	528	123		
Nitrogen (kg/ha)						
25	6672	1928	11850	3156	4300	7503
50	7021	2422	11969	4291	4722	8130
SEd±	5	51	59	36		
CD @5%	14	175	83	123		

Combination of *Azospirillum* and *Aspergillus* seed inoculation produced higher grain yield which was followed by *Azotobacter* + *Aspergillus* seed inoculation. Among single biofertilizers, *Aspergillus* seed inoculation was found superior in yield followed by *Azospirillum* seed inoculation. FYM @ 5 tons/ha and nitrogen @ 50 kg/ha produced higher grain yields compared to their respective lower doses.

3. Title of the project : Effect of biofertilizers on yield of finger millet under rainfed condition

Location : Bengaluru and Simliguda

Crop : Finger millet

Year of study : 1988-89

Treatments	Yield (kg/ha)					
	Grain		Straw		Mean over locations	
	Bangalore	Similiguda	Bangalore	Similiguda	Grain	Straw
B0:Control	3482	2318	5561	4562	2900	5062
B1:Azatobactor	3839	2487	6150	4927	3153	5539
B2:Azospirillum	3925	2566	6264	5281	3246	5773
B3:Phosphoous solubilizing fungus	4507	2705	6972	5613	3606	6293
B4:B1+B2	4641	2876	7089	5915	3759	6552
B5:B2+B3	4915	3051	7594	6272	3983	6933
CD @5%	161	33	332	90		
Organic manures (kg/ha)						
0	4005	2546	6428	5252	3276	5840
5000	4432	2788	6812	5605	3610	6254
CD @5%	302	26	344	121		
Nitrogen (kg/ha)						
25	4069	2526	6412	5209	3298	5811
50	4369	2808	6829	5648	3589	3699
CD @5%	42	26	154	121		

The data indicated that seed inoculation with biofertilizers increased the yields significantly over control and among them *Aspergillus awamori* ranked first in grain yield, when they were tried separately. However, seed inoculation with combination of two biofertilizers was more effective and between the two combinations, *Azospirillum brasilense* + *Aspergillus awamori* produced highest yield.

4. Title of the project : Effect of biofertilizers on yield of ragi under rainfed condition

Location : Bengaluru, Berhampur, Dindori, Kanke
 Crop : Finger millet
 Year of study : 1989-90

Treatments	yield (kg/ha)	
	Mean over locations	
	Grain	Straw
T1: 100% recommended NPK P as superphosphate	4339	4121
T2: 100% recommended NPK, 50% P as superphosphate and 50% P as rock phosphate	4605	4340
T3: 100% recommended NPK P as superphosphate + <i>Agrobacterium radiobacter</i> seed inoculation	4867	4434
T4: 100% recommended NPK P as superphosphate + <i>Aspergillus awamui</i> seed inoculation	4633	4406
T5: T2 + <i>Agrobacterium radiobacter</i> seed inoculation	4854	4521
T6: T2+ <i>Aspergillus awamui</i> seed inoculation	4725	4454
T7: T2 + <i>Agrobacterium radiobacter</i> seed inoculation + <i>Aspergillus awamui</i> seed inoculation	4782	408

Mean over locations indicated that 50% phosphorus supplied as single superphosphate and remaining 50% phosphorus as rock phosphate produced highest grain yield and seed treatment of *Agrobacterium radiobacter* was more effective compared to *Aspergillus awamori*. However, the combination of *Agrobacterium radiobacter* and *Aspergillus awamori* seed inoculation did not increase the grain yield over the single biofertilizer.

5. Title of the project : Effect of biofertilizers on yield of finger millet under rainfed condition

Location : (Bengaluru, Berhampur, Dindori, Kanke)
 Crop : Finger millet
 Year of study : 1990-91

Treatments	Grain yield (Mean over locations)
T1: 100% recommended NPK P as superphosphate	2177
T2: 100% recommended NPK, 50% P as superphosphate and 50% P as rock phosphate	2407
T3: 100% recommended NPK P as superphosphate + <i>Agrobacterium radiobacter</i> seed inoculation	2436
T4: 100% recommended NPK P as superphosphate + <i>Aspergillus awamui</i> seed inoculation	2314
T5: T2 + <i>Agrobacterium radiobacter</i> seed inoculation	2430
T6: T2+ <i>Aspergillus awamui</i> seed inoculation	2369
T7: T2 + <i>Agrobacterium radiobacter</i> seed inoculation+ <i>Aspergillus awamui</i> seed inoculation	2575

Seed coating with *Agrobacterium radiobacter* and *Aspergillus awamori* produced highest yield and all treatments receiving biofertilizers have yielded more compared to control.

6. Title of the project : Studies on use of biofertilizers on grain yield of finger millet under rainfed conditions

Location : Bengaluru, Berhampur, Kanke
Crop : Finger millet
Year of study : 1991-92

Treatments	Mean over locations
	Grain yield (kg/ha)
T1: 100% recommended NPK P as superphosphate	2830
T2: 100% recommended NPK, 50% P as superphosphate and 50% P as rock phosphate	3186
T3: 100% recommended NPK P as superphosphate + <i>Agrobacterium radiobacter</i> seed inoculation	3243
T4: 100% recommended NPK P as superphosphate + <i>Aspergilhis awamui</i> seed inoculation	3011
T5: T2 + <i>Agrobacterium radiobacter</i> seed inoculation	3188
T6: T2+ <i>Aspergilhis awamui</i> seed inoculation	3093
T7: T2 + <i>Agrobacterium radiobacter</i> seed inoculation+ <i>Aspergilhis awamui</i> seed inoculation	3291

It was observed that application of P in the form of both super phosphate and rock phosphate found to give higher yield at all the centres than application of P in the form of superphosphate alone. Higher response at Kanke and Bangalore was observed to seed inoculation with *Agrobacterium radiobacter* was found

7. Title of the project : Response of finger millet to different levels of fertilization and bio fertilizers under irrigated conditions

Location : Mandya
Crop : Finger millet
Year of study : 2003-2004

Treatments	yield (kg/ha) 2003	
	Grain	Straw
T1: 100% Rec fertilization	5162	7743
T2: 125% Rec fertilization	5291	7936
T3: 150% Rec fertilization	5634	8451
T4: 175% Rec fertilization	5653	8479
T5: 200% Rec fertilization	5759	8638
T6: 100% Rec fertilization + Azospirillum (Dipping seedlings)	5625	8437
T7: 100% Rec fertilization + Azatobacter (Dipping seedlings)	5384	807

T7: 100% Rec fertilization + Azatobacter (Dipping seedlings)	5384	807
T8: 50% Rec N + full P&K fertilization + Azospirillum (Dipping seedlings)	4833	7249
T9: 75% Rec N + full P&K fertilization + Azospirillum (Dipping seedlings)	5023	7534
T10: 50% Rec N + full P&K fertilization + Azatobacter (Dipping seedlings)	4893	7339
T11: 75% Rec N + full P&K fertilization + Azatobacter (Dipping seedlings)	5074	7611
T12: Control	3508	5263
SEm±	88	133
CD @5%	259	389

Dipping seedlings with Azospirillum and 100 percent recommended fertilization was significantly superior in yield over 100 per cent recommended NPK resulting in 463 kg higher yield besides a low cost input.

8. Title of the project : Response of finger millet to different levels of fertilization and bio fertilizers under irrigated conditions

Location : Mandya
Crop : Finger millet
Year of study : 2004-2005, 2005-2006

Treatments	Yield (kg/ha) 2004		Yield (kg/ha) 2005	
	Grain	Straw	Grain	Straw
T1: 25% Rec fertilization	2467	4200	2151	3442
T2: 50% Rec fertilization	3215	6480	2408	3854
T3: 75% Rec fertilization	3445	5360	2788	4461
T4: 100% Rec fertilization	3660	6220	3256	5211
T5: 100% Rec fertilization + Azospirillum (Dipping seedlings)	4393	7470	3754	5974
T6: 100% Rec fertilization + Azatobacter (Dipping seedlings)	3944	6700	3432	5491
T7: 50% Rec fertilization + Azospirillum (Dipping seedlings)	3282	5580	2762	4419
T8: 50% Rec fertilization + Azatobacter (Dipping seedlings)	3534	6180	2435	3897
T9: 75% Rec fertilization + Azospirillum (Dipping seedlings)	3545	6020	3075	4920
T10: 75% Rec fertilization + Azatobacter (Dipping seedlings)	3655	6210	2825	4453
SEm±	2310	389	142	186
CD @5%	687	1160	423	552

The results revealed that dipping seedlings with *Azospirillum brasilense* along with recommended NPK nutrition gave significantly higher grain yield as compared to 100 percent recommended NPK alone. Between the bio fertilizers tested, treatment with *Azospirillum brasilense* gave higher yield as compared to Azotobacter. Increase in yield due to increase in NPK fertilizer dose from 25 percent Rec. NPK up to 100 percent Rec. doses was noticed. Similar trend as in grain yield was also observed in straw yield.

Second year results indicated that application of 100 % Rec. NPK along with dipping seedlings in Azospirillum bio fertilizers was beneficial in enhancing productivity which was in confirmation with the first year results and second best yielder was dipping seedlings with Azotobacter bio fertilizer along with 100% Rec. NPK. Between the bio fertilizers, Azospirillum gave higher yields at all the fertility gradients as compared to Azotobacter. Application of 75% Rec. NPK along with dipping seedlings with Azospirillum bio fertilizer gave on par yields to that of recommended fertilization without bio fertilizers besides having higher benefit cost ratio.

9. Title of the project : Efficacy of Biofertilizer in sustaining the productivity of finger millet + pigeon pea (8:2) inter cropping systems at Bangalore

Location : GKVK, Bengaluru
Crop : Finger millet
Year of study : 2014-2015

Treatments	FMGEY (kg/ha)
T1: sole crop of FM @ 100% RDF	4398
T2: sole crop of FM @ 50% RDF	4042
T3: T2+ AMF	4233
T4:T2 + PGPR	4056
T5: T2 + AMF+ PGPR	4533
T6: Sole crop of pigeon pea @ 100% RDF	4942
T7: Sole crop of pigeon pea @ 50% RDF	4371
T8: T7 + AMF	4499
T9: T7+ PGPR	4397
T10: T7+ AMF + PGPR	5010
T11: FM + pigeon pea (8:2) inter cropping + 100% RDF	4983
T12: FM + pigeon pea (8:2) inter cropping + 50% RDF	4537
T13: T12 + AMF	4802
T14: T12 + PGPR	4748
T15: T12 + AMF + PGPR	5198
T16: FM + pigeon pea (8:2) inter cropping + 50% RDF + AMF	4777
T17: FM + pigeon pea (8:2) + 50% RDF + AMF + PGPR	4996
T18: FM + pigeon pea (8:2) + no fertilizer + no Biofertilizer	3260
T19: FM + pigeon pea (8:2) + no fertilizer + AMF	3489
T20: T19 + PGPR	3708
SEm±	137
CD at 5%	388

Finger millet grain equivalent yield data indicated that, application of 50 percent RDF along with seed treatment of Rhizobia/PGPR + soil application of AMF to poly bag pigeon pea seedlings and intercropping with finger millet (2:8) gave significantly higher yield compared to other treatments.

10. Title of the project : Studies on seed palleting with biofertilizer and micronutrients on growth and yield of finger millet

Location : Kanke, Kolhapur

Crop : Finger millet

Year of study : 2015-2016

Treat ments	Kanke								Kolhapur							
	Grain				Straw				Grain				Straw			
	2013	2014	2015	Mean	2013	2014	2015	Mean	2013	2014	2015	Mean	2013	2014	2015	Mean
T1	1223	1240	1245	1236	2134	2160	2184	2159	1210	1130	1162	1167	1510	1413	1452	1458
T2	1712	1717	1722	1717	2379	2402	2429	2403	1450	1363	1334	1382	1808	1708	1668	1728
T3	2435	2445	1972	2284	2623	2646	2938	2736	1650	1578	1687	1638	2050	1989	2108	2049
T4	1779	1794	1804	1792	3169	3199	2922	3097	1600	1527	1509	1545	1990	1912	1886	1929
T5	1857	1873	1913	1881	2891	2912	2679	2827	1700	1623	1676	1666	2117	2030	2095	2081
T6	1868	1874	1904	1882	2490	2513	2719	2574	1650	1556	1658	1621	2058	1948	2073	2026
T7	1912	1928	1970	1937	2890	2897	2932	2906	1710	1633	1639	1661	2130	2062	2048	2080
T8	1934	1943	2111	1996	2890	2901	2948	2913	1940	1792	2055	1929	2500	2243	2606	2450
T9	2068	2081	2479	2209	2890	2908	3231	3010	2170	2019	2300	2163	2700	2282	2915	2632
T10	2501	2513	1537	2517	3491	3497	3533	3507	2060	1934	2199	2064	2573	2432	2790	2598
SEm±	128	126	126	-	189	198	190	-	137	85	201	-	170	113	258	-
CD at 5%	381	375	373	-	562	590	565	-	406	252	598	-	505	336	765	

T1: control

T2: FYM

T3: FYM + NPK

T4: T2 + *Azospirillumbrasilence* + *Bacillus magathelium* + *Pseudomonas fluroscence*

T5: T4 + ZnSO₄ @ 3g/kg seed

T6: T4 + boron @ 3g/kg seed

T7: T4 + ZnSO₄ + boron

T8: T7 + 50% RDF

T9: T7 + 75 %

T10: T7 + 100% RDF

Mean over locations over the years data indicated that 100 per cent RDF along with micro nutrients and seed treatment with biofertilizers gave higher grain yield followed by 75 per cent RDF along with micro nutrients and seed treatment with biofertilizers.

11. Title of the project : Efficacy of plant growth promoting Rhizobacteria (PGPR) on growth and yield of finger millet under rainfed conditions

Location : Almora
Crop : Finger millet
Year of study : 2015-2016

Treatments	Grain			Straw		
	2014	2015	Mean	2014	2015	Mean
T1: control	3292	2442	2867	5483	3467	4475
T2: <i>Pseudomonas sp.</i> Strain PGERs 17	3967	2722	3345	6767	3975	5371
T3: <i>Pseudomonas sp.</i> Strain PPERs 23	3958	2788	3373	7250	4425	5838
T4: <i>Pseudomonas putida</i> . Strain PBRs 5	3158	2199	2679	6908	3350	5129
T5: <i>Pseudomonas putida</i> Strain PGR4	3233	2593	2913	6475	4275	5375
T6: <i>Pseudomonas lurida</i> Strain NPRs 15	3183	2085	2634	6058	3708	4883
T7: <i>Pseudomonas sp.</i> Strain NPRs 3	3500	2453	2977	6000	3833	4917
T8: <i>Pseudomonas sp.</i> Strain NARs 9	3767	2620	3194	7408	3908	5658
T9: <i>Pseudomonas fluorescence</i> Strain PPRs 4 4634		2642	2220	2431	6100	3167
SEm±	110	65	-	168	216	-
CD at 5%	329	266	-	505	647	-

Grain yield indicated that application of *Pseudomonas* strain PPERs 23 gave significantly higher grain yield as compared to other *Pseudomonas* strains but it was on par with strains PGEs 17, NARs 9 and PGP 4. Mean over two seasons data indicated that strain PPERs 23 gave higher grain yield followed by PPER 17



Carrier based biofertilizer



Liquid Biofertilizer

12. Title of the project : Studies on seed palleting with biofertilizer and micro nutrients on growth and yield of finger millet

Location : Kanke, Bengaluru

Crop : Finger millet

Year of study : 2016-2017

Treatments	Kanke			Bengaluru		
	Grain	Straw	B:C ratio	Grain	Straw	B:C ratio
T1: control	2064	3348	1.23	1383	2057	1.15
T2: FYM	2737	4830	1.52	1393	2265	1.15
T3: FYM + NPK	3237	5029	1.45	1487	2346	1.14
T4:T2 + <i>Azospirillum brasillense</i> + <i>Bacillus magathelium</i> + <i>Pseudomonas fluorescens</i>	2858	4367	1.54	1860	2222	1.49
T5: T4 + ZnSO ₄ @ 3g/kg seed	3176	4566	1.76	1464	2107	1.17
T6: T4 + boron @ 3g/kg seed	3255	4698	1.84	1440	2331	1.15
T7: T4 + ZnSO ₄ + boron	3335	4632	1.85	1581	2218	1.24
T8: T7 + 50% RDF	3176	5029	1.54	1590	2206	1.27
T9: T7 + 75 % RDF	3335	5558	1.58	1746	2265	1.41
T10: T7 + 100% RDF	4147	5490	2.01	1792	2361	1.29
SEm±	3132	4755	1.63	1574	2238	1.25
CD at 5%	529	872	0.42	235	222	0.26

Mean over location data indicated that seed treatment with bio fertilizers and micro nutrients with 100 per cent RDF application gave higher grain yield and B:C ratio compared to absolute control but it was on par with other treatments. Significantly higher grain yield and B:C ratio was obtained in Bengaluru by application of farm yard manure with micro nutrients and bio fertilizer seed treatment, whereas grain yield of 100.92% and B:C ratio of 2.01 was recorded in seed treatment with bio fertilizers and micronutrients with 100 per cent RDF application.

13. Title of the project : Effect of plant growth promoting Rhizobacteria on growth, nutrient uptake and yield of finger millet

Location : Almora
Crop : Finger millet
Year of study : 2016-2017

Treatments	Grain (kg/ha)	Straw (kg/ha)
T1: control	2348	5808
T2: <i>Pseudomonas sp.</i> Strain PGERs 17	2716	6417
T3: <i>Pseudomonas sp.</i> Strain PPERs 23	2554	6292
T4: <i>Pseudomonas putida</i> . Strain PBRs 5	2532	5792
T5: <i>Pseudomonas putida</i> Strain PGR4	2593	5917
T6: <i>Pseudomonas lurida</i> Strain NPRp 15	2383	5875
T7: <i>Pseudomonas sp.</i> Strain NPRs 3	2614	6125
T8: <i>Pseudomonas sp.</i> Strain NARs 9	2627	6250
T9: <i>Pseudomonas fluorescence</i> Strain PPRs 4	2153	4683
CD at 5%	214.68	434.66
CV%	4.96	4.25

Significantly increased grain yield of 15.67 per cent and straw yield was obtained by application of *Pseudomonas* strain PGERs 17 compared to absolute control.

14. Title of the project : Studies on seed palleting with biofertilizer and micronutrients

Location : Kanke
Crop : Finger millet
Year of study : 2017-2018

Treatments	Kanke	
	Grain	Straw
T1: control	1019	3347
T2: FYM (7.5 t/ha)	1463	4828
T3: FYM + NPK (50:40:25)	2124	5026
T4: T2 + <i>Azospirillum brasiliense</i> + <i>Bacillus magathelium</i> + <i>Pseudomonas fluorescens</i>	2310	4365
T5: T4 + ZnSO ₄ @ 3g/kg seed	2124	4563
T6: T4 + boron @ 3g/kg seed	2257	4696
T7: T4 + ZnSO ₄ + boron	2011	4630
T8: T7 + 50% RDF	2050	5026
T9: T7 + 75 % RDF	2265	5556
T10: T7 + 100% RDF	2328	5622
Local mean	1995	4766
CD at 5%	312	935

An experiment was conducted at Kanke to know the effect of seed coating of biofertilizer viz., *Azospirillum brasiliense*, *Bacillus megathelium* and *Pseudomonas fluorescens* along with micronutrients Zinc sulphate and boron @ 3 g/kg seed were evaluated at different fertility levels 50, 75 and 100 per cent RDF. Results indicated that the application of 100 per cent RDF, micronutrients and seed treatment with biofertilizers gave significantly higher grain yield which was found to be 128 and 59 per cent higher than absolute control and application of recommended dose of FYM only (7.5 t/ha), respectively. The straw yield also showed the similar pattern.

15. Title of the project: Effect of inoculation of P solubilizing microorganisms on little millet under rainfed conditions

Location : Dindori
Crop : Little millet
Year of study : 1990-91

Treatments	Grain yield (kg/ha)
T1: 100% recommended NPK P as superphosphate	628
T2: 100% recommended NPK, 50% P as superphosphate and 50% P as rock phosphate	658
T3: 100% recommended NPK P as superphosphate + <i>Agrobacterium radiobacter</i> seed inoculation	727
T4: 100% recommended NPK P as superphosphate + <i>Aspergillus awamui</i> seed inoculation	748
T5: T2 + <i>Agrobacterium radiobacter</i> seed inoculation	950
T6: T2+ <i>Aspergillus awamui</i> seed inoculation	754
T7: T2 + <i>Agrobacterium radiobacter</i> seed inoculation+ <i>Aspergillus awamui</i> seed inoculation	761
SEd±	163
CD @5%	NS

There was no significant difference in yield among the treatments. However, application of recommended NPK where P was supplied through 50% as SSP and remaining 50% as rock phosphate along with seed inoculation of *Agrobacterium radiobacter* produced highest yield.

16. Title of the project : Effect of inoculation of P solubilizing microorganisms on yield of Foxtail millet

Location : Nandyal, Andhra Pradesh

Crop : Foxtail millet

Year of study : 1989-90

Treatments	Yield (kg/ha)	
	Grain	Straw
T1: 100% recommended NPK P as superphosphate	2222	3444
T2: 100% recommended NPK, 50% P as superphosphate and 50% P as rock phosphate	2444	3111
T3: 100% recommended NPK P as superphosphate + <i>Agrobacterium radiobacter</i> seed inoculation	2111	3555
T4: 100% recommended NPK P as superphosphate + <i>Aspergilhis awamui</i> seed inoculation	2555	3222
T5: T2 + <i>Agrobacterium radiobacter</i> seed inoculation	2444	3222
T6: T2+ <i>Aspergilhisawamui</i> seed inoculation	2333	3333
T7: T2 + <i>Agrobacterium radiobacter</i> seed inoculation+ <i>Aspergilhis awamui</i> seed inoculation	2444	3555
SEd±	190	641
CD @5%	NS	NS

The data revealed that there was no significant differences in yield among the treatments. However, 100% recommended NPK where P was supplied as superphosphate along with seed inoculation of *Aspergillus awamori* produced more grain yield.

17. Title of the project : Effect of inoculation of P solubilizing microorganisms on yield of Setaria

Location : Nandyal, Andhra Pradesh

Crop : Foxtail millet

Year of study : 1990-91

Treatments	Grain yield
T1: 100% recommended NPK P as superphosphate	1500
T2: 100% recommended NPK, 50% P as superphosphate and 50% P as rock phosphate	1550
T3: 100% recommended NPK P as superphosphate + <i>Agrobacterium radiobacter</i> seed inoculation	1480
T4: 100% recommended NPK P as superphosphate + <i>Aspergilhis awamui</i> seed inoculation	1700
T5: T2 + <i>Agrobacterium radiobacter</i> seed inoculation	1570
T6: T2+ <i>Aspergilhis awamui</i> seed inoculation	1680
T7: T2 + <i>Agrobacterium radiobacter</i> seed inoculation+ <i>Aspergilhis awamui</i> seed inoculation	1600
SEd±	207
CD @5%	NS

Though there was non-significant difference in yield among the treatments, seed treatment of *Aspergillus awamori* and application of 100% phosphorus in the form of 50%SSP and 50% as rock phosphate gave highest yield.

18. Title of the project : Effect of biofertilizers on yield of barnyard millet under rainfed conditions

Location : Ranichauri
Crop : Barnyard millet
Year of study : 1991-92

Treatments	1991
	Grain yield (kg/ha)
100% recommended NPK P as SSP	1080
100% recommended NPK P as MRP	1150
100% recommended NPK 50%P as SSP +50%P as MRP	1185
100% recommended NPK P as SSP + <i>Agrobacteriumradiobacter</i> seed inoculation	1260
100% recommended NPK P as SSP + <i>Aspergilhisawamui</i> seed inoculation	1220
100% recommended NPK P as MRP + <i>Agrobacteriumradiobacter</i> seed inoculation	1295
100% recommended NPK P as MRP + <i>Aspergilhisawamui</i> seed inoculation	1335
100% recommended NPK P as SSP + MRP + <i>Agrobacteriumradiobacter</i> seed inoculation	1528
100% recommended NPK P as SSP + MRP + <i>Aspergilhisawamui</i> seed inoculation	1569
100% recommended NPK P as SSP + MRP + <i>Aspergilhisawamui</i> seed inoculation+ <i>Aspergilhisawamui</i> seed inoculation	1420
SEd±	-
CD @5%	95

Application of P in the form of superphosphate and rock phosphate in 50:50 proportion and inoculating seed either with *Agrobacterium radiobacter* or with *Aspergillus awamori* gave higher yield than without inoculation but with only P in the form of superphosphate or rock phosphate or in combination of both.

19. Title of the project : Effect of *Azospirillum brasilense* on yield of prosomillet under rainfed conditions

Location : GKV, Bengaluru
 Crop : Proso millet
 Year of study : 1986-87

Treatments	Yield (kg/ha)	
	Grain	Straw
T1: Control	1157	3333
T2: Sees inoculation with <i>Azospirillum</i>	1296	3356
T3: Soil application of <i>Azospirillum</i> in furrows	1319	3356
T4: application of <i>Azospirillum</i> with FYM in furrows	1356	3379
T5: 10 kg N/h + T2	1898	3518
T6: 10 kg N/ha+T3	1875	3657
T7: 10 kg N/ha+T4	1921	3703
T8: 10 kg N/ha+alone	2129	4583
SEd±	77	420
CD @5%	233	NS
CV %	8.3	19.9

The results indicated that 20kg N/ha alone gave significantly higher grain yield compared to other treatments, but was on par with application of *Azospirillum* mixed with FYM to the furrows plus 10 kg N/ha.

20. Title of the project : Studies on the effect of P-solubilizing microorganisms on kodo millet under rainfed conditions

Location : Dindori
 Crop : kodo millet
 Year of study : 1989-90

Treatments	Grain Yield (kg/ha)
T1: 100% recommended NPK P as superphosphate	1742
T2: 100% recommended NPK, 50% P as superphosphate and 50% P as rock phosphate	2235
T3: 100% recommended NPK P as superphosphate + <i>Agrobacterium radiobacter</i> seed inoculation	2651
T4: 100% recommended NPK P as superphosphate + <i>Aspergillus awamui</i> seed inoculation	2576
T5: T2 + <i>Agrobacterium radiobacter</i> seed inoculation	2932
T6: T2+ <i>Aspergillus awamui</i> seed inoculation	2954
T7: T2 + <i>Agrobacterium radiobacter</i> seed inoculation+ <i>Aspergillus awamui</i> seed inoculation	3190
SEd±	131
CD @5%	340

21. Title of the project : Effect of P-solubilizing microorganisms on yield of kodo millet under rainfed conditions

Location : Dindori
Crop : kodo millet
Year of study : 1990-91

Treatments	Grain Yield (kg/ha)
T1: 100% recommended NPK P as superphosphate	1909
T2: 100% recommended NPK, 50% P as superphosphate and 50% P as rock phosphate	2045
T3: 100% recommended NPK P as superphosphate + <i>Agrobacterium radiobacter</i> seed inoculation	2152
T4: 100% recommended NPK P as superphosphate + <i>Aspergillus awamori</i> seed inoculation	2545
T5: T2 + <i>Agrobacterium radiobacter</i> seed inoculation	2697
T6: T2+ <i>Aspergillus awamori</i> seed inoculation	2818
T7: T2 + <i>Agrobacterium radiobacter</i> seed inoculation+ <i>Aspergillus awamori</i> seed inoculation	3425
SEd±	189
CD @5%	414

Application of 100% NPK where P was supplied as 50% SSP and 50% as rock phosphate along with seed treatment of *Agrobacterium radiobacter* and *Aspergillus awamori* produced highest yield compared to other treatments. However, biofertilizer seed treatment enhanced yield over respective no fertilizer treatments

22. Title of the project : Effect of P-solubilizing microorganisms on yield of kodo millet under rainfed conditions

Location : Dindori
Crop : Kodo millet
Year of study : 1991-92

Treatments	Grain Yield (kg/ha)			
	1991	1990	1989	Mean
T1: 100% recommended NPK P as superphosphate	1584	1909	1742	1745
T2: 100% recommended NPK, 50% P as superphosphate and 50% P as rock phosphate	1411	2045	2235	1897
T3: 100% recommended NPK P as superphosphate + <i>Agrobacterium radiobacter</i> seed inoculation	1703	2152	2651	2169
T4: 100% recommended NPK P as superphosphate + <i>Aspergillus awamori</i> seed inoculation	1971	2545	2576	2364
T5: T2 + <i>Agrobacterium radiobacter</i> seed inoculation	1893	2697	2932	2507
T6: T2+ <i>Aspergillus awamori</i> seed inoculation	2069	2818	2954	2614
T7: T2 + <i>Agrobacterium radiobacter</i> seed inoculation+ <i>Aspergillus awamori</i> seed inoculation	2288	3425	3190	2968
SEd±	79	134	131	-
CD @5%	255	414	340	-

Inoculation of seeds with *Agrobacterium radiobacter* and *Aspergillus awamori* with P in the form of superphosphate and rock phosphate (50:50 proportion) gave significantly higher yield than without inoculation but with P either in super phosphate form or in combination with rock phosphate.

e. Integrated Nutrient Management studies in small millets

1. Title of the project : Integrated Nutrient Management in finger millet

Location : Kolhapur, Maharashtra
Crop : Finger millet
Year of study : 2014-15 to 2017-18

The experiment on production package of organic sources in finger millet was conducted for four years from 2014-15 to 2017-18 under AICRP on Small millets with combination of organic and inorganic nutrients. The highest grain and straw yield, gross and net monetary returns as well as B:C Ratio was recorded in recommended dose of fertilizer (60:30:00 kg NPK ha⁻¹). However, it was statistically on par with application of 50 % nitrogen through FYM and 50% nitrogen through Vermicompost

2. Title of the project : Integrated Nutrient Management in Finger millet

Location : Waghai, Gujarat
Crop : Finger millet
Year of study : 2018

Farmers of South Gujarat heavy rainfall zone I (AES I & III) and South Gujarat Zone II (AES I) growing finger millet are advised to adopt integrated nutrient management system for getting higher yield and net profit. Component of Integrated Nutrient Management are: 30 kg N, 20 kg P₂O₅ and organic Bio compost 2 t/ha. , Apply Azotobacter 2 kg/ha. + PSB 2 kg/ha. as soil application

3. Title of the project : Integrated Nutrient Management in finger millet

Location : Waghai, Gujarat
Crop : Finger millet
Year of study : 2019

The farmers of south Gujarat heavy rain fall zone I (AES-I) growing finger millet during kharif season are recommended to apply FYM @5 t/ha or incorporate forest tree leaves @5 t/ha in absence of FYM one month before transplanting and fertilizer the crop with 75% RDF (30: 15: 00 NPK kg/ha) for getting higher grain yield and net income.

4. Title of the project : Integrated Nutrient Management in Barnyard millet

Location : Ranichauri, Uttarakhand
Crop : Barnyard millet

Maximum grain and straw yield of barnyard millet under integrated application of fertilizer {RDF (NPK)+ FYM @ 7.5 ton/ha} while lowest in organic fertilization (FYM 7.5 ton/ha + vermicompost 1 ton/ha + neem cake 500 kg/ha)

5. Title of the project : Nutrient management under organic conditions in Barnyard millet (var. PRJ 1)

Location : Ranichauri, Uttarakhand
Crop : Barnyard millet

From the means of three consecutive years data, it was observed that barnyard millet variety PRJ-1 gave significantly higher grain yield with the application of vermicompost and wild apricot cake to provide 50% nitrogen from each source along with *Azotobacter* seed treatment. The data revealed that Vermicompost + wild apricot cake (50% of N from each source) + *Azotobacter* produced significantly higher yields both in grain and straw. The recommended dose of NPK and Vermicompost + wild apricot cake (50% of N from each source) along with *Azotobacter* seed treatment produced grain yield were at par. The benefit cost ratio of different treatments indicated that the recommended dose of NPK showed highest B: C ratio followed by Vermicompost + Chullu cake (50% of N from each source) + *Azotobacter*.

6. Title of the project : Integrated Nutrient Management in Proso millet

Location : Ranichauri, Uttarakhand
Crop : Proso millet

Maximum grain and straw yield of proso millet under integrated application of fertilizer [RDF(NPK)+ FYM @ 5 ton/ha] while lowest in organic fertilization (FYM 5 ton/ha + vermicompost 1 ton/ha + neem cake 500 kg/ha)

7. Title of the project : Integrated Nutrient Management in Kodo millet

Location : Dindori, Madhya Pradesh
Crop : Kodo millet
Year of study : 2003-2004

In a study carried out during *kharif* 2003 & 2004 on Integrated Nutrient Management, intercropping and crop rotations in combination revealed that application of 7.5 t FYM/ha along with recommended dose of fertilizer (NPK) gave significantly higher grain yield (1384 kg/ha) followed by application of 7.5 t FYM /ha + seed treatment with bio fertilizer (839kg/ha) and Kodo millet-Legumes two year rotation with recommended dose of organic manures (825 kg/ha).

8. Title of the project : Integrated Nutrient Management in Kodo millet

Location : Dindori, Madhya Pradesh
Crop : Kodo millet
Year of study : 2004-05 to 2009-10

Nutrient management in kodo millet based cropping systems was conducted during 2004-05 to 2009-10 cropping seasons on a fixed site where mono cropping of kodo millet , kodo millet-soybean and kodo millet-niger two years rotation were evaluated to different levels of organics (with and without FYM @ 5 t/ha) and inorganics (No fertilizer, 50% RDF and 100% RDF) to develop an appropriate nutrient management package for enhancing productivity of kodo millet based cropping systems. The results revealed that application of 5.0 t FYM/ha gave higher grain yield (1068 kg/ha) as compared to no FYM (871 kg/ha). Among the fertilizer levels tested, application of 100% recommended dose of fertilizer gave significantly higher grain yield (1209 kg/ha) as compared to 50% recommended dose of fertilizer (1027 kg/ha) and no fertilizer (744 kg/ha). Among the cropping systems, KM- Soybean 6 Years rotations gave higher KM grain yield equivalent (KMGYE- 1149 Kg/ha) over kodo millet – niger two years rotations (1041 kg/ha).

f. Intercropping studies in small millets:

1. Title of the project : Intercropping studies in barnyard millet

Location : Ranichauri, Uttarakhand

Crop : Barnyard millet

FYM @ 7.5t/ha application gave higher grain yield 2534 kg/ha & 8635 kg/ ha straw yield as compared to No FYM (1935 & 7150 kg/ha). Among the cropping sequence Barnyard millet +Rice bean- Radish gave maximum barnyard millet equivalent yield (6074 kg/ha) but maximum benefit cost ratio was obtained under Barnyard millet+ Rice bean -Lentil. Therefore on the basis of benefit cost ratio Barnyard millet + Rice bean - Lentil is the most profitable cropping sequence under rainfed mid hill condition of Uttarakhand



Finger millet: Red gram (8:2)



Finger millet: Maize (8:2)



Finger millet: Green gram (8:2)



Finger millet: Bhendi (8:2)

g. In situ incorporation of green manures

1. Title of the project : Improving the productivity of finger millet by *insitu* incorporation of green manures

Location : Vizianagaram, Andhra Pradesh

Crop : Finger millet

Year of study : 2015

Continuous cultivation of crops in every season makes the soil nutrient exhaustive which leads to drastic deterioration of the soil fertility status. Hence green manures initially grown *insitu* and later incorporated before flowering stage into the soil will help to rejuvenate the soil health. Further incorporation of these legume crops into the soil allows the nutrients held within the green manures to be released and made available to the succeeding crops. Hence study was conducted at Agricultural Research Station, Vizianagaram to evaluate the effect of *insitu* incorporation of green manures on yield attributes. The results of the study showed that significantly higher grain yield (mean 30.5q/ha) was observed with 100% RDF N,P,K + diancha incorporation and this is followed by 100% RDF N,P,K + sunhemp incorporation (mean 28.2q/ha) and 100% RDF N,P,K + green gram incorporation (mean 26.6q/ha). Yield reduction was observed at 50 % RDF N,P,K fertilizers application in spite of incorporating green manures (sunhemp /diancha/green gram) (mean grain yield of 13.6, 16.1 and 11.3q/ha, respectively) when compared to 100% RDF N,P,K fertilizers application alone without incorporating any green manure (20.8q/ha). However, diancha green manure incorporation at 75 % RDF N,P,K fertilizers recorded significantly higher grain yield (mean 26.6q/ha) over to that of 100 % RDF N,P,K fertilizers alone (mean 20.8q/ha). Post harvest soil analysis showed increase in soil organic carbon content and availability of N,P,K nutrients at 100% RDF N,P,K fertilizers + green manures (sunhemp/diancha/green gram) incorporation over to that of 100 % RDF N,P,K fertilizers alone.



Insitu cultivation of different green manures for incorporation

CHAPTER IV: RESIDUE MANAGEMENT IN ORGANIC MILLET CULTIVATION

Increased concern for the environment and increased emphasis on sustaining soil productivity has resulted a major interest in the maintenance and improvement of soil organic matter in recent years. Proper management of crop residues is thus essential in developing countries, especially in tropical climates to maintain soil quality. According to an estimate, more than 1000 Mt of cereal residues are being produced in the developing world (FAO, 1995). If crop residues could be better managed, this would have the direct effect on improving crop yields by increasing soil nutrient availability, decreasing erosion, improving soil structure and water holding capacity.

Asia is a major producer of crop residues. 52.6% of the world residue production is in Asia. In India, approximately 500 to 550 million tons of crop residue are generated every year and in that coarse cereals contribute 71.0 million tons. Among cereal crops, rice, wheat, Maize and millets together contributed 70% of crop residue followed by fibre crops (13%). Of the total crop residue, majority part is used as fodder and fuel for domestic and industrial purposes. However, there is still a surplus of 140 million ton out which 92 million tons is burned every year.

Of the total straw generated from coarse cereals some quantity is being utilized as an animal fodder and rest of the straw will be removed out of the field and subjected to burning or leaving as a waste.

a. Adverse impacts of crop residue burning on environment:

The burning of crop residues generates numerous environmental problems. The main adverse effect of crop residue burning include the emission of green house gases (GHGs) that contributes to the global warming, increased levels of particulate matter (PM) and smog that cause health hazards.

Heat generated from burning of residues increases sub soil temperatures to nearly 33.8 – 42.2°C at 10mm depth, and long term effects can even reach up to 15cm of the top soil. Frequent burning reduces nitrogen and carbon potential of the soil and kills the microflora and fauna beneficial to the soil and further remove large portion of the organic matter. With the crop residue burning the carbon – nitrogen equilibrium of the soil is completely lost.

According to NPMCR (National Policy for Management of Crop Residue), burning of 1ton of straw accounts for the loss of entire amount of organic carbon, 5.5 kg of nitrogen, 2.3 kg of phosphorus, 25 kg potassium and 1.2kg sulphur.

Nutritional Composition of Crop Residue:

On an average crop residue of different crops contain approximately 80% of nitrogen (N), 25% Phosphorus (P), 50% Sulphur (S) and 20% Potassium (K). If the residue is retained in the soil itself, it can enrich the soil with C, N, P, K and S as well.

Sustainable Management Practice for Crop Residue:

There are several other options for managing surplus crop residues, apart from using as an animal fodder and fuel in domestic and industrial purposes.

1. Crop residues can be used as mulch material in the field to prevent the weed growth and also to control evaporation losses of moisture from soil.
2. Residues can be composted and used as a main source of nutrients in organic farming.
3. Residues can be used for biogas production.
4. Residues can be used as bedding material for animals.

5. Residues can also be used as raw material for mushroom culture.
6. Residues can be incorporated into the soil immediately after harvesting of ear heads. However, it will take long time for decomposition as it is rich in cellulose and lignins. Hence machines like straw shredders can be used to chop the crop residue into small pieces which in turn helps in faster decomposition. To hasten the decomposition process, cow dung or cow urine or microbial decomposers can be used. They will reduce the decomposition time and hence provide opportunity to go for second crop immediately.

Effect of residue application on Soil Properties:

1. Soil Organic matter :

For a given climatic region and soil type, the rate of addition of carbon inputs is an important factor determining the amount of organic matter that can be maintained in the soil.

2. Soil available nutrients :

Incorporation of crop residues enhances the nutrient contents of N, P, K, S and other micro nutrients like Zn, Cu, Fe and Mn contents.

3. Aggregation :

Straw incorporation helps in the formation and stability of aggregates through increase in microbial cells, excreted microbial products and decomposition products released during the death of the micro organisms. Cereal crop residues provide persistent aggregate stabilizing agents compared to green manures or grain legumes.

4. Porosity:

Residue application in the field increases soil porosity. In a long term field study in China, rice straw incorporation increased the porosity and decreased the bulk density of paddy soils (Xu and Yao, 1988).

5. Hydraulic Conductivity and Infiltration Rate:

Crop residues affect hydraulic conductivity and infiltration by modifying soil structure, proportion of macropores and aggregate stability.

6. Bulk Density:

Incorporation of crop residues reduces bulk density, penetration resistance and compaction in soils.

7. Microbial Population:

Residue incorporation in the soil always leads to increased bacterial and fungal activities. Protein decomposing microbes increase during the early stages followed by increase in populations of cellulose decomposing micro organisms and Sulphate reducing micro organism after lag phase.

8. Enzyme activities:

Enzyme activities indicate the activities of micro organisms in soils. Crop residue incorporation significantly increase dehydrogenase, urease, phosphatase enzymes, etc. in the soil.

In some cases, straw incorporation may lead to reduced crop yields because of the release of phytotoxic compounds during decomposition or immobilization of soil and fertilizer nitrogen, particularly when the crop is planted immediately after straw incorporation. The effect of residue incorporation on the succeeding crop depends on the amount of residues and time and method of incorporation. Overall, the long term effects of crop residue incorporation are generally expected to be beneficial to the soil and to the succeeding crops.

CASE STUDIES ON RESIDUE MANAGEMENT IN SMALL MILLETS CONDUCTED AT DIFFERENT AICRP CENTRES

1. Title of the project : Residue Utilization at Agricultural Research Station, Vizianagaram, Andhra Pradesh

Location : Vizianagaram, Andhra Pradesh

Crop : Finger millet

Year of study : 2019-20

Agricultural Research Station, Vizianagaram is having 20 acres of cultivated area. Out of which 4-5 acres during *kharif* and 2-3 acres during rabi season is being allocated for finger millet seed production. Harvesting of earheads followed by cutting of straw and throwing it away out of the fields is a common practice. This operation requires nearly 25-30 number of human laborers per acre of land per day which costs around Rs.7500-9000/-. Besides 6-8 t/ha of crop residue is being lost every season which gradually leads to soil degradation, erosion of surface soil, reduced soil nutritional status (NPK), low organic matter content (<0.4%) and poor water and nutrient holding capacities. In order to match the lost nutrients and to maximize the productivity of this degraded land excessive fertilizers are being applied resulting in increased cost of cultivation with little fertilizer factor productivity. Later in 2018, millet straw shredder was purchased and is being utilized for shredding of finger millet straw into small pieces. It covers one acre within 45 minutes to 1 hour. Shredded material after shredding can be left as such on the soil surface which can act as mulch material during summer to protect the soil or it can be incorporated into the soil by using rotavator. With the utilization of shredder all the crop residue generated in the field can be completely incorporated into soil which helps in enriching the soil in the long run. Further it saves lot of time and money incurred in cutting of straw and transportation out of the field. Hence, straw shredder is extensively being utilized at Agricultural Research Station, Vizianagaram.

2. Title of the project : Crop Residue Management for Nutrient management and Improving Soil Productivity in Finger millet of North Coastal Zone

Location : Vizianagaram, Andhra Pradesh

Crop : Finger millet

Year of study : 2019-20

Crop residues of agricultural crops are important resources, not only as sources of nutrients for succeeding crops but also for succeeding crops but also for improved soil quality and crop productivity. Hence, an experiment was conducted at Agricultural Research Station, Vizianagaram during 2019-20 to study the impact of residue management practices on nutrient requirement of finger millet. The results revealed that the increased grain and straw yields were observed with the treatment 100% RDF N,P,K which was found on par with 75% RDF + Incorporation of ragi straw shredded material + Urea @ 5 kg/ha + SSP @ 5 kg/ha fertilizers. The treatment with Incorporation of ragi straw shredded material + Urea @ 5 kg/ha + SSP @ 5 kg/ha fertilizers + 75% RDF has shown the highest grain (2207 kg/ha) and straw yields (7482 kg/ha) which was on par with 100% RDF (2155.6 kg/ha and 7288.9 kg/ha respectively). The B:C ratio (1.88:1) was also found highest in the treatment with Incorporation of ragi straw shredded material + Urea @ 5 kg/ha + SSP @ 5 kg/ha fertilizers + 75% RDF. Post harvest soil analysis showed that soil available N and P_2O_5 was found significantly high in the treatment in which microbial consortium (decompo A + decompo B) was applied for decomposition of incorporated ragi straw shredded material whereas the soil available potassium has shown no significant difference between the treatments.

Residue Management in Finger millet



Cutting of small millet straw residue into pieces with shredder



Line transplanting of finger millet in soil incorporated with crop residue



Use of decomposing material (Decompo A+ B) for fast decomposing of finger millet shredded material

CHAPTER V: INSECT PEST MANAGEMENT IN ORGANIC MILLET CULTIVATION

By default millets are resilient to pest and disease attacks. However, due to changes in climatic conditions over a period of time, insect pests and crop plants have been subjected to many changes resulting in increased appearance of pest and disease attacks on millet crops also.

In organic farming, variety of strategies have to be integrated to maintain insect pests at acceptable levels and minimize the chance that insects will adapt to any one management strategy. Here, the goal is not to eliminate all the pests, but rather to reduce pest populations to tolerable levels. The strategies used against insect pests include:

1. Use of resistant or tolerant varieties
2. Cultural methods
3. Physical methods
4. Mechanical methods
5. Biological methods and
6. Chemical methods.

Integrated pest management (IPM) requires an understanding of the ecology of the cropping system including that of the pests and their environment. Field scouting is to be conducted at regular intervals throughout the crop season or during critical stages of crop growth to estimate the severity of pest infestations. If the insect-pest population exceeds the economic threshold level, the following management strategies have to be adopted.

1. Resistant / Tolerant Varieties:

It is an important management option for the control of insect pests. A resistant variety may be less preferred by the insect pest or adversely affect the development and survival of insect pests, or sometimes the plant may tolerate the damage without any economic loss in yield or quality.

S.No	Name of the crop	Resistant / Tolerant variety	Resistant to Pest/Disease
1.	Finger millet	GPU 28, GPU 45, GPU 48, PRM 1, KOPM 235, VL 348, Co 15, VL 379, DHFM 78-3, Vegavathi, Suwarnamukhi	Blast
		VL 379, Vegavathi	Sheath Blight
		BM9-1	Brown spot
		Chilika, KMR 340, Chattisgarh Ragi-2	Stem Borer
		Depoli-2	Aphid
2.	Foxtail millet	SiA 3085	Blast and downey mildew
		RAU (RajendraKauni 1-2)	Leaf blast, rust, smut, brown spot, downy mildew and leaf blight

3.	Little millet	OLM 203, OLM 217	Grain smut
		OLM 217	Rust
		OLM 203	Blast
		OLM 36	Brown spot, Sheath blight
		OLM 208, OLM 217, JK 36, GV 2, JK 4, DHLM-14-1, PRC 10,11, Dindori 2-1	Shoot fly
4.	Proso millet	PRC 1	Leaf blight resistant
		GPUP 8, TNAU 164	Brown spot, Rust
		BR 6, Co 1, GPUP 21, TNAU 151, TNAU 164	Shoot fly
5.	Kodo millet	JK 155, JK 48, JK 13, JK 106	Head smut
		JK 155, JK 13, JK 106, JK 65, JK 98, JK 137	Shoot fly
6.	Barnyard millet	VL Madira 172, Kanchan	Grain smut

2. Cultural Methods:

- Deep summer ploughing will expose the pupal stages, larval stages and other dormant stages of insect pests to hot sun.

Eg : Deep ploughing is particularly helpful in reducing the populations of grasshoppers and hairy caterpillars in finger millet.

- Trimming of the field bunds to eliminate the weeds as many weeds act as alternate or collateral host for the plant pathogens and insects and thus help in perpetuation and spread of pests and diseases.

Eg 1: Rust, Smut, Ergot and Downy mildew pathogens of various millets infect weed species like *Cyanodon dactylon*, *Sorghum halopense*, *Oxalis corniculata*, *Digitaria marginata*, *Pennisetum sp.* and *Eragrostis tenifolia* and help them overwinter (Reed *et al.*, 2000; Marley, 1995).

Eg 2: Sorghum shoot fly and gall midge infests weeds like *Brachiaria distacha*, *Panicum repens*, *Setaria intermedia*, *Cyperus sp.* and *Sorghum halopense* and survive there in until new crop will be ready (Bilbro, 2008).

Eg 3. Elimination of weeds reduces the shoot fly infestation in kodo millet. One hand weeding at 20-25DAS checks the attack of the shoot fly and resulted in economically higher yield (Singh *et al.*, 1993).

Eg 4. Proper and timely weeding of the crop reduces the damage by *Mythimna separata* (Sharma and Davies, 1982) and *Spodoptera sp.* (Ndoye *et al.* 1986).

- Removal and burning of crop residues and weeds if they are infected with pests or diseases in the previous season.

- d. Green manuring
- e. Seed/Seedling treatment before sowing or transplanting of millets with biocontrol agents like *Trichoderma viride*, *Pseudomonas fluorescens*, *Bacillus sp.*, etc.
- f. Timely sowing is necessary to make the millet plants less susceptible to pest and disease attack.
- g. Adoption of optimum plant to plant / row to row spacing.
- h. Adopting proper crop rotation.
- i. Growing of inter or mixed cropping.
Eg. Little millet+Soybean/Radish reduce the shoot fly occurrence
Kodumillet+Pulses/Oil seeds-reduces the infestation, development and movement of shoot fly
- j. Use of trap crops around millet crops.
- k. Use of appropriate quality of organic manures.
- l. Proper water management.
- m. Regular monitoring of pests by using different traps.
Eg. Shoot fly population in little millet can be monitored by fish meal trap

Effect of green manuring on pests and diseases:

It is evident that green manure crops not only improve the soil health and fertility but also helps to manage the agricultural pests.

Effect of green manuring on fungal diseases:

- a. The effect of green manuring with *Sesbania aculeata* L. significantly reduced the mycelia growth and sclerotia production of three potential soil borne plant pathogens viz., *Rhizoctonia solani*, *Sclerotium rolfsii* and *Sclerotinia sclerotiorum*.
- b. Sunhemp (*Crotalaria juncea* L) green manure during its decomposition reduces the soil borne pathogens. Yadava (2001) found significant reduction of sclerotia and mycelia of *Sclerotium rolfsii* and *Rhizoctonia solani* pathogens. He recorded *T. harzianum*, *Gliricidium virens*, *A. niger*, *P. citrinum* and *Curvularia lunata*, the most effective fungal decomposers of sunhemp and are effective against *Sclerotium rolfsii* and *Rhizoctonia solani*. Similar effect of green manuring of *Crotalaria juncea* on soil borne pathogens *Sclerotium rolfsii* and *Rhizoctonia solani* was observed.

Effect of green manuring on nematodes:

- a. *Crotalaria* is a poor host to many plant parasitic nematodes including *Meloidogyne sp.*, *Rotylenchulus reniformis*, *Radopholus similis*, *Belanolaimus longicaudatus* and *Heterodera glycines* (Wang, 2002).
- b. Incorporation of green leaf manures viz., *Potria* (*Thespesia populnea*), *Calotropis* (*Calotropis gigantia*), *Neem* (*Azadiracta indica*), *Gliricidia* (*Gliricidia maculata*) and *Glycosmis pentaphylla* helps to reduce most serious “root knot nematodes” (Pakeerathan et al. 2009).
- c. *Crotalaria sp.* can reduce the detrimental nematodes viz., *Meloidogyne javanica* and *M. incognita* and increasing the level of beneficial mycorrhizal fungi *Glomus sp.* (Germani & Plenchette, 2004).

3 Mechanical Methods:

- a. Collection and destruction of eggs, egg masses, larvae and pupae of insect pests.
- b. Mass trapping of pest adults through light traps or pheromone traps.
- c. Yellow pan water traps have been found useful to trap thrips and aphids.
- d. Removal or destruction of leaf portion / plant parts infested with insect pests.

4. Biological Methods:

i) For pest management:

- a. Biological control is defined as the reduction of pest populations by natural enemies. Natural predators of the insect pests such as lady bird beetles, grasshoppers, frogs, snakes, spiders, etc., will be helpful in destroying the insect-pests.
 - o Lady bird beetles help in destroying aphids that feed on plant sap.
 - o Spiders prey on moths, leaf hoppers, aphids, bugs, etc.
- b. Release of *Trichogramma* sp. @ 40,000 -50,000 eggs / ha is highly effective as it parasitize the eggs and adults of many pests, especially eggs of moths and butterflies.
- c. *Apantiles* sps, *Chelonus blackburni* @ 15000 – 20000 eggs / ha *Chrysoperla* sp.@5000 eggs/ha at 15 days after sowing and again at 30 days after sowing can effectively control many insect pests.
- d. Pathogens of the insect pests : These are the disease causing bacteria and virus which can be used to kill the insects. Application of formulations viz., *Metarrhizium anisopliae*, *Beauveria bassiana*, *Noumeria rileyi*, *Verticillium* sp. can manage their specific host pests.
 - o *Metarrhizium anisopliae* is an entomopathogenic fungus known to infect insects from more than seven orders.
 - o *Beauveria bassiana* is considered as non selective pesticides because it parasitizes a very high range of arthropod hosts like termites, thrips, white flies, aphids and different beetles.
 - o *Noumeria rileyi* is also entomopathogenic fungi effective against *Spodoptera litura*, *Spodoptera frugiperda* and other insects belonging to Noctuidae family.
 - o *Verticillium* sps is an effective nematode parasite which control *Meloidogyne* sps .

ii) For Disease Management :

- a. ***Trichoderma viride*:** It is a biological pesticide effective against fungal attacks like wilts, rusts, root rot and foot rot disease etc. in millets. It can be used in different ways like seed treatment, seedling treatment, nursery treatment and main field treatment.
- b. ***Pseudomonas fluorescens*:** It is a bacterium that lives near the roots of plants and produce secondary metabolites like antibiotics that suppress soil borne pathogens.
- c. Viral bio pesticides of Nuclear Polyhedrosis Viruses (NPV) and Granulosis Virus (GV) are also capable of controlling pests and diseases.

5. Use of botanicals:

Certain products derived from plants can effectively control insect pests.

- Formulations containing Azadirachtin obtained from neem tree have been found effective against aphids, mites and caterpillars.
- Spray of Lemon grass oil, Citronella oils, crude extracts of *Tagetes sp* have good control over some pests.
- Extracts of some common weeds viz., *Polygonum runcinatum*, *Artemisia vulgaris*, *Eupatorium glandulosum*, *Urtica sp.*, *Lantana sp.*, etc., are effective against some sucking pests.
- Crush 1kg Ipomea leaves, hot chilli 500 gm, garlic 500 gm, neem leaves 5 kg in 10 liters cow urine. Boil the suspension till it becomes half. Filter the extracts. 2-3 liters extract diluted to 100 liters and is used in one acre against leaf rollers and stem borers.

Effect of neem on pests and diseases:

Different parts of the neem tree can effect more than 200 insect sp., some nematodes, fungi, bacteria and viruses. Neem contains several active chemicals like Azadirachtin, Salanin, Meliantriol and Nimbin. All these chemicals act on insects by blocking their endocrine system, which resulting in behavioural and physiological aberrations making the insects so confused in brain and body that they cannot reproduce and there by their populations plummet. However, it is remarkable benign to spiders, butterflies and insects such as bees, lady bugs and wasps.

Mode of action of neem products:

- Disrupting or inhibiting the development of eggs, larvae or pupae.
- Blocking the molting of larvae and nymphs.
- Disrupting mating and sexual communication.
- Repelling larvae and adults.
- Deterring females from laying eggs.
- Sterilizing adults.
- Poisoning larvae and adults.
- Deterring feeding.
- Blocking the ability to swallow.
- Inhibiting the formation of chitin.

Effect of neem cake:

Neem cake is an organic byproduct of seed oil production. It is used as a natural fertilizer to increase the plant growth as well as root development as it is a potential source of nutrients like N: 2-3%, P_2O_5 : 1%, K_2O : 1.4%, S: 1.072-1.36 %. It has antifungal properties and also protects the plants from nematodes and white ants.

Use of Animal Products:

- Cow urine - Cow urine when diluted with water in the ratio of 1:20 is used as foliar spray to control pathogens and insects.
- Neem – Cow urine extract: Crush 5 kg Neem leaves and add 5liters cow urine and 2kg cow dung. Ferment for 24hours and filter the extract and dilute to 100 liters. This is effective against sucking pests and mealy bugs.

CASE STUDIES ON ECO-FRIENDLY MANAGEMENT OF PESTS AND DISEASES CONDUCTED AT DIFFERENT AICRP CENTERS

1. Title of the project : Blast management of finger Millet using Bioagents

Location : Vizianagaram, Andhra Pradesh
Crop : Finger millet
Year of study : 2014-15

Although, finger millet is tolerant to several abiotic and biotic stresses, blast disease caused by *Magnaporthe grisea* is one of the most destructive disease which leads to significant yield loss under favourable conditions. Among the different bioagents tested for control of blast disease for organic cultivation, it was observed that seed treatment with *Pseudomonas fluorescence* @ 10 gm/kg seed followed by two times (stages) need based spray of *Pseudomonas fluorescence* @ 10gm/lit water has shown an effective control. This recommendation has been recognized at National level in the year 2011 and wide spread among the finger millet growing farmers especially organic growers all over the Nation. The same critical intervention has been applied in the long term organic experiment in the research station and also TSPs and FLDs conducted in the farmer fields of North Coastal Zone of Andhra Pradesh.

2. Title of the project : Eco friendly management of Banded/Sheath Blight in six small millets

Location : Vizianagaram, Andhra Pradesh
Crop : Finger millet, Little millet, Foxtail millet, Barnyard millet, Proso millet and Kodo millet
Year of study : 2015-2017

Banded sheath blight caused by *Rhizoctonia solani* is another emerging malady in small millets commonly referred to as Nutri-cereals. Among the different treatments tested in all the six small millets for three consecutive years from 2015-2017, soil application of value added *Pseudomonas fluorescence* + *Trichoderma asperellum* + *Bacillus subtilis* @ 1kg each talc formulation mixed in 25kg FYM or Vermicompost incubated for 15 days is recommended for management of banded/ sheath blight in all six millets (Ref: 26-01-2018-CS-FFC dt 25.05.2018 of the Assistant Director General (FFC), ICAR, New Delhi).

3. Title of the project : Bio control agents for blast management in finger millet

Location : Ranichauri, Uttarakhand
Crop : Finger millet

In organically grown finger millet, two sprays of *P. fluorescens* @ 0.6 % (first spray at 50% flowering followed by second spray 10 days later) is recommended for the control of neck and finger blast diseases. Seed treatment with *P. fluorescens* @ 6g/kg seed and two sprays of *P. fluorescens* @ 0.6% (first at 50% flowering and second after 10 days) is recommended for all three blast diseases in finger millet. It is advised to mix sticker in solution. Seed treatment with *Trichoderma harzianum* @ 6g/kg seed and two sprays of *P. fluorescens* @ 0.6% (first at 50% flowering and second after 10 days) is recommended for all three blast diseases in finger millet. It is advised to mix sticker in solution.

4. Title of the project : Varietal mixture for blast management in finger millet

Location : Ranichauri, Uttarakhand
Crop : Finger millet

The local ragi cultivars though susceptible to blast are widely grown because of farmer's preference. VL 149, which was resistant to blast earlier, is losing resistance due to development of newer races. Therefore, keeping the need to extend the life span of VL 149 as well as to reduce blast damage in the local, a preferred variety an experiment was planned and conducted for three consecutive years. Of all treatments, cultivar mixture of VL 149 & local cultivar (1:1) of resistant and susceptible cultivars of finger millet was found best which led to reduction of blast incidence in the mixture than that in both the varieties individually. At the same time, culture mixture yielded better than the individual variety. No spray application of any chemical is required for the management of disease if this practice is followed.

5. Title of the project : Semi organic experiment for blast management in finger millet

Location : Ranichauri, Uttarakhand
Crop : Finger millet
Year of study : 2017-2019

On the basis of mean of three years' data (2017-2019), seed treatment with carbendazim@ 2g/kg seed + 2 foliar sprays with *Pseudomonas fluorescens* @ 1 g/lit water, first at panicle initiation stage and second at grain filling stage has been found most promising for suppressing all the three blasts in finger millet i.e. leaf, neck and finger followed by the seed treatment with *Pseudomonas fluorescens* @ 10 g /kg seed + 2 foliar sprays with *Pseudomonas fluorescens* @ 10 g /lit water, first at panicle initiation stage and second at grain filling stage and seed treatment with *Bacillus subtilis* @ 10 g/kg seed + 2 foliar sprays with *Bacillus subtilis* @ 10 g/lit water, first at panicle initiation stage and second at grain filling stage. Seed treatment with carbendazim @ 2g/kg seed + two foliar sprays with *Pseudomonas fluorescens* @ 10 g/liter water, first at panicle initiation stage and second at grain filling stage also gave 45.88 per cent higher grain yield and 38.64 per cent higher fodder yield over control followed by the seed treatment with *Pseudomonas fluorescens* @ 10 g /kg seed + 2 foliar sprays with *Pseudomonas fluorescens* @ 10 g / lit water, first at panicle initiation stage and second at grain filling stage with 38.70 per cent higher grain yield and 35.13 percent higher fodder yield over control.

6. Title of the project : Eco-friendly management of banded sheath blight caused by *Rhizoctonia solani* in barnyard millet

Location : Ranichauri, Uttarakhand
Crop : Barnyard millet

The soil application of value added FYM(FYM pre-colonized by *Pseudomonas fluorescens* + *Trichoderma viridi* and *Bacillus subtilis* @ 1 kg each talc formulation mixed in 25 kg FYM or Vermicompost incubated for 15 days prior to application) is recommended for managing banded sheath blight and enhancing grain and fodder yield in barnyard millet. Seed bio-priming with *Trichoderma asperellum* Th-14 and application of value added FYM (pre colonized FYM by Th-14) is recommended for mid hill farmers to enhance the barnyard millet's yield and yield contributing traits at the same time managing sheath blight and brown leaf spot diseases under rainfed conditions of Uttarakhand.

7. Title of the project : Bio agents for management of grain smut & leaf spot in barnyard millet

Location : Ranichauri, Uttarakhand

Crop : Barnyard millet

Use of *Pseudomonas fluorescens* as seed treatment + soil application with value added FYM/ Vermicompost + single spray @ 6 g/litre at the time of 50 % flowering is recommended for grain smut and brown leaf spot diseases in barnyard millet.

8. Title of the project : Maintenance of repository of microbial inoculants

Location : Ranichauri, Uttarakhand

Crop : Barnyard millet

Ranichauri AICRP(SM) centre has maintained a bio-control repository consisting of 42 *Pseudomonas* isolates and thirty *Trichoderma* isolates for the bio-management of small millets diseases keeping the organic scenario of millets cultivation in state Uttarakhand viz., ❖ *Trichoderma asperellum* (ITCC- 7903):- effective against banded sheath blight disease of barnyard millet.❖ *Pseudomonas fluorescens* (UUHF Psf- 4): - effective against foot rot and blast diseases in finger millet and against grain smut and brown spot diseases of barnyard millet.❖ *Pseudomonas fluorescens* (UUHF Psf-1):- effective against *Pyricularia grisea* causing blast disease in finger millet. The *Pseudomonas* strain PRS 9 identified as Phosphorous solubilizer in acid soils

9. Title of the project : Eco friendly management of Shoot fly in Kodo millet & Little millet

Location : Dindori, Madhya Pradesh

Crop : Kodo millet & Little millet

Year of study : 2008-2010, 2015-2017

Early sowing, with the onset of monsoon results in low infestation of shoot fly in Kodo millet as compared to late sowing. Inter cropping of pigeon pea with kodo millet in the ratio of 2:1 substantially reduces the shootfly infestation as compared to sole crop of Kodo millet. The experiment (2008 to 2010) on organic pest management revealed that soil application of Karanj cake and spraying of Karanj oil @ 4 % or NSKE @ 5% were significantly effective to control kodo millet shootfly. The experimental data (2015-2017) revealed that spraying of 1500 ppm Azadirachtin at 15 DAS reduced the shoot fly incidence (6.74%) and it was on par with spraying of 5% NSKE at 15 DAS (7.21%). The control plot recorded higher dead hearts i.e. 17.06%.

CHAPTER VI: WEED MANAGEMENT IN ORGANIC FARMING

A weed is any plant, native or non-native that interferes with the crop production by doing more harm than good and has a habit of encroaching where it is not wanted. Weeds not only compete with crops for light, moisture, space, nutrients, etc. but also harbour the pests and diseases which are harmful to the crop plants. Hence proper weed management is essential to boost the yields of millet crops.

As millets are grown predominantly during hot and humid rainy season, weeds grow profusely and deprive these crops of vital nutrient and moisture and thus reduce the yield considerably. Most of the millet lands suffer from poor soil fertility due to their marginal nature; the removal of nutrient by the weed further deteriorates the situation. Millets are basically drought tolerant crops. However, there are certain weeds in millets that can thrive in lower amounts of water than millets, eg: The water requirement for production of unit dry matter for barnyard grass is less than that of many types of millets (Shipley and Wiese, 1969).

Wider row spacing and slow initial growth in millets encourage weed growth during the early crop growth period. Therefore early control of weeds within the critical period of crop weed competition is considered necessary to optimize the yield. For example yield losses caused by weeds in some millets, Sorghum – 15 -83% , Pearl millet – 16-94%, Finger millet – 55-61%.

Dactyloctenium aegyptium, *Eleusine indica*, *Sorghum halepense*, *Cyperus rotundus*, *Cynodon dactylon*, *Euphorbia hirta*, *Solanum nigrum*, *Striga litura*, *Trianthema portulacastrum*, *Echinichloa colona*, *Commelina benghalensis*, *Ageratum conyzoides*, etc. are some of the dominant weeds in millet fields.

Since traditional times, hand weeding, inter cultivation with small implements like kurpis, hoes, etc. are very common among farmers. Though these methods are effective, these should be integrated with other weed management practices to increase the effectiveness and to reduce the cost.

Important management practices for reducing the weed problem in organic farming:

1. Deep summer ploughing exposes the weed seeds to hot sun and make them loose viability.
2. Organic manures should be thoroughly decomposed to ensure the destruction of viable weed seed.
3. All the agricultural implements should be properly cleaned before and after using in the field.
4. Use of weed – suppressive cover crops in rotation to suppress the problematic weeds. Eg :Sunhemp -millets.
5. Intercropping – growing different plant types together enhances weed control by increasing shade and increasing crop competition. Eg: Finger millet + Redgram(8:2),
6. Do not allow weeds to form seed heads and perennial rooting structures in the cropping system. One single season of allowing weeds to set seeds may create years of weed management problems.

7. Communal approach of weed management gives good result in eliminating or minimizing the potential spread of noxious and problematic weeds.
8. Line sowing of millet crops facilitates inter cultural operations with wheel hoes, row weeders, cono weeders, rota weeders, etc. These implements also facilitate incorporation of weeds which in turn helps to enhance soil fertility.
9. Mulching is an effective way for weed control. Crop straws can be used as mulch material. Mulching not only restrict weed growth but also conserve soil moisture.
10. Irrigation – drip irrigation is more efficient in controlling weeds as it minimizes weed germination due to targeted application of water to root zone of crop plants.
11. Plant breeding approach: Development of varieties with increased competitive ability and enhanced weed suppressive qualities.
12. Green manures in weed management: Green manures are primarily grown for their contributions to soil fertility; however, they can also play an important role in managing weeds in organically cultivated millet crops.
 - i. Green manure crops like smother crops, out-compete the weeds and deprive them of water, nutrients and sunlight.
 - ii. Green manure crops diversify the cropping system and hence they narrow the opportunities for weeds to become adapted to that niche cropping cycle.
 - iii. Green manure exhibits allelopathic effect i.e., they release certain chemicals that inhibit germination, growth and development of weeds. Several green manure crops such as Sunhemp, Lucern, Alfalfa, Cowpea and Velvet bean found to have allelopathic effect on weeds.
 - iv. Agronomic practices involved in termination of green manure crop like tillage, mowing and incorporation inhibits the germination of weeds.
13. Bio herbicides / Allelopathy: Weed infestation can be managed by using allelopathic crop as cover crop, intercrop or green manures, mulch, smother crops or grown in rotational sequences.
 - i. An allelochemical or bioherbicide “Sorgoleone” released by sorghum living root hairs is responsible for suppression of broad leaved and grassy weeds in the field.
 - ii. A water extract of mature sorghum plants obtained after 24hr soaking in water, called “Sorgaab”, is used as a natural herbicide.
 - iii. Some of the natural products exploited as commercial herbicides are Triketone, Cinmethylin, Bialaphos, Glufosinate and Dicamba.

CHAPTER VII: NEW TRENDS IN SMALL MILLET CULTIVATION

a. Guli/ Netti method of cultivation of finger millet:

Small millets are low yielders when compared to other cereal crops. Small grain size and less tiller number may be the reason for low yields of millet crops. Increase in seed size may not be advantageous to increase the yield but increase in productive tillers and number of earheads per plant will significantly increase the yields. System of Rice Intensification has tremendous effect on tiller number due to optimum spacing availed by each individual plant. Influenced by the SRI principles, farmers of Haveri district of Karnataka has developed Guli/Netti method of finger millet cultivation in order to increase the productive tillers per plant during 1980's. It resulted in yield increase up to 5-10 q/ha through significant increase in tiller number per plant.

This system involves two very interesting bullock-drawn implements that farmers had adopted for following purposes.

- i.) **Koradu / Wooden log/wooden plank:** is a hallowed out wooden log, about 6-7 feet long and 15-18 inch width attached by ropes to the yoke of a team of bullock. It is pulled across the field several times between 15 and 45 DAT, to bend over the young plants without breaking them off, causing the mildly traumatized plants to put out more tillers and more roots.
- ii.) **YadeKunte / Intercultivation implement:** It is a blade mounted on the end of a long handle pulled along the wide space (45cm) between rows and between plants, in order to eliminate weed competition and also to aerate the soil.

Methodology:

1. In Guli method, 15-20 day old seedlings are used for transplanting as they put forth more number of tillers than over aged seedlings.
2. Plant spacing will depend upon soil fertility and soil type. In well fertile soils, 45cm x 45cm or 60cm x 60cm is optimum spacing. Wide spacing between the plants allows the roots to spread properly.
3. Guli method includes putting a handful of compost or manure into each pit/guli along with the seedlings to boost soil fertility.
4. Two saplings will be planted at each inter section point. As only two saplings are placed in one pit/guli, they absorb maximum nutrients.
5. Between 15 and 45 DAT, the Koradu is dragged 3-6 times across the field in different directions.
6. The inter cultivation equipment Yedukunte will be used for weeding. It breaks, lifts and aerate the surface soil as it cuts through the roots of weeds, burying them in the soils.
7. Normally farmers in Haveri district of Karnataka used to get only 1.25 to 3.5 t/ha, however, in guli method, yield has increased to 4.5-5 t/ha.

Differences between Conventional and Guli method of finger millet cultivation:

S.No.	Conventional method	Guli ragi method
1.	Seed requirement is 5-6 kg/acre	Seed requirement is 1 kg/acre
2.	Yield : 800-1000 kg/ac	Yield : 1800-2000 kg/ac
3.	Spacing is 22.5 x 10cm or 30 cm x 10cm	Spacing is 45 x 45 cm or 60 x 60cm
4.	Pest and disease attacks are more due to high plant density.	Pest and disease attacks are less due to wide spacing.
5.	FYM requirement is 5 Cartloads /acre	FYM requirement is 10 Cartloads of /acre
6.	Chemical fertilizers will be used along with FYM.	Only organic manures will be used.
7.	Tillers/hill varies between 4-6	Tillers/hill varies between 25-30
8.	No use of Koradu or Yadukunte. Inter cultivation will be done manually or chemically.	Koradu will be used between 15-45 DAT. For inter cultivation, Yedukunte will be used.

Advantages of Guli method:

1. Seed requirement is less (1 kg/acre) when compared to conventional method (5-7 kg/ac) of sowing.
2. This method found to be yielding well in areas with poor rainfall as the presence of Farm Yard manure in pits increases water holding capacity and nutrient holding capacity.
3. In this system, individual plant produces more tillers and more number of ear heads. Tiller number varies from 40-50 /hill in high fertile soils/black soils and 14-16/hill in low fertile or light textured soils.
4. Bending of the plants with Koradu, traumatizes their stems at ground level, which might accelerate the plant's meristematic tissues to produce new tillers and roots.
5. Inter cultivation with Yedukunte not only helps in weed control but also aerates the root zone soil and helps in production of more adventitious roots. More aeration in the root zone also helps in increased availability of nutrients and reduced incidence of soil borne diseases.
6. The larger root system helps in increased absorption of nutrients by plants which in turn helps in better filling of grains.
7. Due to application of huge quantities of farm yard manure, soil health will be improved in the long run.
8. Lodging problem is less due to maximum stem girth in guli method.
9. Pest and disease problems are less due to wide spacing.

Disadvantages of Guli method:

1. Guli method is cumbersome process, requires more input of labour for marking lines, making pits at intersection points and filling of pits with FYM, pulling of Koradu and Yedukunte.
2. It requires huge quantities of well decomposed FYM, which is very costly and its transportation and application in pits are also cumbersome.

3. Adequate care must be taken while handling the young seedlings in guli method.
4. Weed problem is more due to wider spacing.
5. Adequate care must be taken with regard to method of pulling of Koradu and stage of the crop.
6. Maturity of all earheads on the same plant is not uniform. Hence, it requires multiple harvests, which in turn increases cost of cultivation.
7. Total crop duration will be extended when compared to conventional method.
8. In light soils, adoption of 45cm x 45cm or 60cm x 60 cm is not advantageous as the tiller number/hill does not exceed fifteen. Hence it is unable to compensate the yield loss caused by reduced plant density.

CASE STUDIES OF GULI METHOD OF CULTIVATION IN DIFFERENT AICRP CENTERS

1. Title of the project : Validation of Guli method in Andhra Pradesh

Location : Vizianagaram, Andhra Pradesh
 Crop : Finger millet
 Year of study : 2018-2019

In Andhra Pradesh, guli/SMI method of finger millet cultivation is widely popularized among farming community by Non Governmental Organizations, Agricultural Technology management Agency (ATMA) and Department of Agriculture. Demonstrations in farmer's fields on guli method of finger millet cultivation are being taken up in the districts like Srikakulam, Vizianagaram, Visakhapatnam, East Godavari, Ananthapur, Chittoor and Kurnool districts. In some places, this Guli method is clubbed with Zero Budget Natural Farming of Subhas palekar. Adoption of spacing and performance of guli method depend upon soil fertility and soil type. In low fertile or light soils 25 x 25 cm or 30 x 30 cm is giving good results and in highly fertile or heavy soils, 45 cm x 45 cm or 60cm x 60cm is giving good result. Agricultural Research Station, Vizianagaram for the first time initiated validation of guli method in Andhra Pradesh during *rabi* 2018-19 in red sandy loam soils. The same work was also continued in *kharif*, 2019. Experimental results revealed that general recommended spacing of 30 cm x 10 cm gave 15.6% and 18.9 % higher yields over 45 cm x 45 cm guli planting and normal planting respectively during *rabi*, 2018-19. Similarly during *kharif* 2019-20, 30 cm x 10 cm spacing recorded 19.4% and 6.5% higher grain yield when compared to 45cm x 45cm or 30cm x 30cm spacing of guli method. Tillers/plant were significantly high in square planting, whereas the number of productive tillers per m² were significantly high in 30cm x 10cm spacing. Considering the economics of both methods B:C ratio was also significantly high in recommended spacing of 30cm x 10cm. However, pulling of Koradu over the crop has positive impact on yield attributes and hence it helped in increasing the grain yield by 5.4%.

2. Title of the project : Validation of Guli method in Karnataka:

Location : GKV, Bengaluru
Crop : Finger millet
Year of study : 2018-2019

In a series of experiments conducted by Dev Kumar at GKV, Bengaluru on effect of spacings, farm yard manure levels and NPK fertilizer levels on growth and yield of finger millet on Sandy loam soils revealed that;

- Square planting with application of 10-12 t/ha of farm yard manure recorded maximum number of productive tillers and grain yield/ha compared to line method with 10-12 t/ha of farm yard manure.
- In another experiment, 30cm x 10cm spacing has resulted in significant increase in grain yield compared to square planting.

3. Title of the project : Validation of Guli method of planting in Eastern dry zone of Karnataka

Location : UAS, GKV, Bengaluru
Crop : Finger millet
Year of study : 2018-2019

A field experiment was conducted at Zonal Agricultural Research Station, UAS, GKV, Bengaluru by Prakasha *et al.*, 2015 on sandy loam soil during 2015. Treatments consisted of two spacings (45 x 45 cm and 60 x 60 cm) with three levels of NPK (0%, 75% and 100% RDF) compared with farmer's method of guli planting (60 x 60 cm with only farm yard manure) with remaining package of practices being common in both the treatments. Significantly higher grain and straw yield (41.60 q/ha and 60.80 q/ha respectively) were recorded with spacing of 60 x 60 cm with 100% RDF followed by 60 x 60 cm with 75% RDF. Higher grain yield was attributed to more number of productive tillers per hill, more dry matter production and grain yield per hill and higher 1000 grain weight. Higher net return (Rs. 44789/-) and B:C ratio (2.2) were recorded in 60 x 60cm spacing with 75% RDF due to on par grain and straw yields and also saving in cost of fertilizers.

4. Title of the project : Validation of System of Millet Intensification in Uttarakhand:

Location : Uttarakhand
Crop : Finger millet
Year of study : 2018-2019

In a demonstration on application of SRI principles to finger millet conducted by an NGO People's Science Institute (PSI) during 2007 revealed that, transplanting of 15 days old seedlings at 20cm x 20cm spacing increased the yield by 33% over the conventional method of finger millet planting. In the next year (2008) even under unfavorable weather conditions, square planting (20cm x 20cm) method gave 60% higher yield (2.4 t/ha) when compared to conventional method(1.5t/ha). Though it is not a demonstration of guli method but it showed the effect of spacing on finger millet yields.

Guli Method /Netti Method



Preparation of field for Guli cultivation



Filling of pits with FYM/Compost before transplanting



Passing of wooden log over crop



Finger millet crop in Guli method



**Guli /Pit method
(45 cmX 45 cm)**

**Recommended
spacing
(30cmx 10cm)**



**Guli / Pit
method
(45 cmX 45
cm)**

**Square
planting**

**Normal
trans-
planting**

b. Cultivation of finger millet under Rice fallow condition in Andhra Pradesh

Rice- Pulse (Black gram / Green gram) relay cropping system is well flourished system in Andhra Pradesh. Minimum input requirement and high remunerative price of pulses have left this system highly successful for decades. Further it also provides the farmers food and nutritional security.

However due to increased number of extreme weather events in recent times, timely sowing of rice and pulses is not happening resulting in increased pests and disease problems (Yellow Vein Mosaic Virus disease and Powdery mildew), moisture stress and ultimately leading to crop failure in pulses. Lack of pest and disease tolerant varieties in pulses and extreme sensitivity to drought or excess rainfall further aggravate the problem.

Under these circumstances, Agricultural Research Station, Vizianagaram has started introducing finger millet in Rice fallows as a relay crop to compensate the yield losses caused by the failure of pulses. Millets as climatic resilient and nutritive crops can assure farmers economic, food and nutritional security as of pulses. They can withstand extreme weather events; less affected by pest and diseases and can be grown with low inputs similar to pulses. Further, introduction of millets into PDS (Public Distribution System) and announcement of MSP (Minimum Support Price) assures price stabilization of millets in the market. Hence Paddy – millet relay system can replace the rice – pulse system in areas where there is frequent failure of pulse crop.

In India 11.65 M.ha of rainfed rice fallow area is located in different states like Andhra Pradesh, Assam, Bihar, Chattisgarh, Jharkhand, Karnataka, Madhya Pradesh, Odisha and West Bengal (Singh *et al.*, 2012). This area offers a huge potential for growing *rabi* crop. However, most of this area has been left vacant during *rabi* due lack of irrigation facilities. Adoption of short duration climate resilient crops like millets can be successfully grown in these rice fallow areas with the adoption of in-situ and ex-situ soil and moisture conservation practices.



Sowing of finger millet under zero tillage

APPENDIX-I

Area, Production and Productivity of Small Millets in India (1950-1951 to 2017-2018-2nd Advance Estimates)

Year	Area (In ' 000 Hectare)	Production (In ' 000 Tonne)	Productivity (In Kg./Hectare)
1950-1951	4605	1750	380
1951-1952	4764	1915	402
1952-1953	5044	1926	382
1953-1954	5677	2477	436
1954-1955	5630	2495	443
1955-1956	5335	2070	388
1956-1957	4976	1930	388
1957-1958	4870	1733	356
1958-1959	5159	2179	422
1959-1960	5148	2025	393
1960-1961	4955	1909	385
1961-1962	4868	2050	421
1962-1963	4772	1855	389
1963-1964	4621	2022	438
1964-1965	4558	1964	431
1965-1966	4564	1555	341
1966-1967	4584	1488	325
1967-1968	4857	1907	393
1968-1969	4746	1804	380
1969-1970	4733	1732	366
1970-1971	4783	1988	416
1971-1972	4477	1669	373
1972-1973	4265	1552	364
1973-1974	4567	1966	431
1974-1975	4466	1613	361
1975-1976	4672	1924	412
1976-1977	4680	1752	374
1977-1978	4574	2070	453
1978-1979	4397	1894	431
1979-1980	4002	1425	356
1980-1981	3976	1574	396
1981-1982	3787	1638	433
1982-1983	3500	1229	351
1983-1984	3638	1676	461

Year	Area (In ' 000 Hectare)	Production (In ' 000 Tonne)	Productivity (In Kg./Hectare)
1984-1985	3214	1194	372
1985-1986	3155	1217	386
1986-1987	2975	1162	391
1987-1988	2901	1169	403
1988-1989	2743	1164	424
1989-1990	2703	1306	483
1990-1991	2447	1190	486
1991-1992	2088	882	423
1992-1993	1983	869	438
1993-1994	1888	917	486
1994-1995	1792	798	445
1995-1996	1662	779	469
1996-1997	1601	728	455
1997-1998	1529	645	422
1998-1999	1495	671	449
1999-2000	1411	618	438
2000-2001	1424	587	412
2001-2002	1311	577	440
2002-2003	1201	459	383
2003-2004	1191	564	473
2004-2005	1101	478	434
2005-2006	1064	472	443
2006-2007	1010	480	475
2007-2008	1039	551	530
2008-2009	905	445	491
2009-2010	831	382	460
2010-2011	800	442	553
2011-2012	798	452	565
2012-2013	754	436	578
2013-2014	682	430	630
2014-2015	590	386	654
2015-2016	650	391	602
2016-2017	619	442	714
2017-2018	546	439	804

Note : # : 2nd Advance Estimates.

Source : Ministry of Agriculture & Farmers Welfare, Govt. of India. (ON2271) & Past Issues.

APPENDIX-II

Selected State-wise Area, Production and Productivity of Small Millets in India (2017-2018)

States/UTs	Area (In ' 000 Hectare)	Production (In ' 000 Tonne)	Productivity (In Kg./Hectare)
Andhra Pradesh	21.00	16.00	762
Arunachal Pradesh	26.77	27.30	1020
Assam	5.23	3.06	584
Bihar	4.40	3.31	752
Chhattisgarh	89.47	21.13	236
Goa	-	-	-
Dadra and Nagar Haveli	0.00	0.00	-
Gujarat	13.00	15.73	1210
Himachal Pradesh	4.10	3.31	807
Jharkhand		0.00	-
Jammu and Kashmir	6.19	3.08	497
Karnataka	34.00	26.01	765
Kerala	0.02	0.02	719
Madhya Pradesh	148.00	144.00	973
Maharashtra	42.00	20.83	496
Meghalaya	2.92	2.78	951
Nagaland	10.52	11.70	1112
Punjab		0.00	-
Odisha	29.18	14.94	512
Puducherry	0.07	0.19	2783
Rajasthan	10.48	7.34	701
Sikkim	2.47	2.55	1031
Tamil Nadu	25.26	31.28	1238
Telangana		0.00	-
Tripura	0.49	0.39	783
Uttar Pradesh	9.00	5.85	650
Uttarakhand	59.00	76.00	1288
West Bengal	2.70	2.20	815
India	546.27	438.99	804

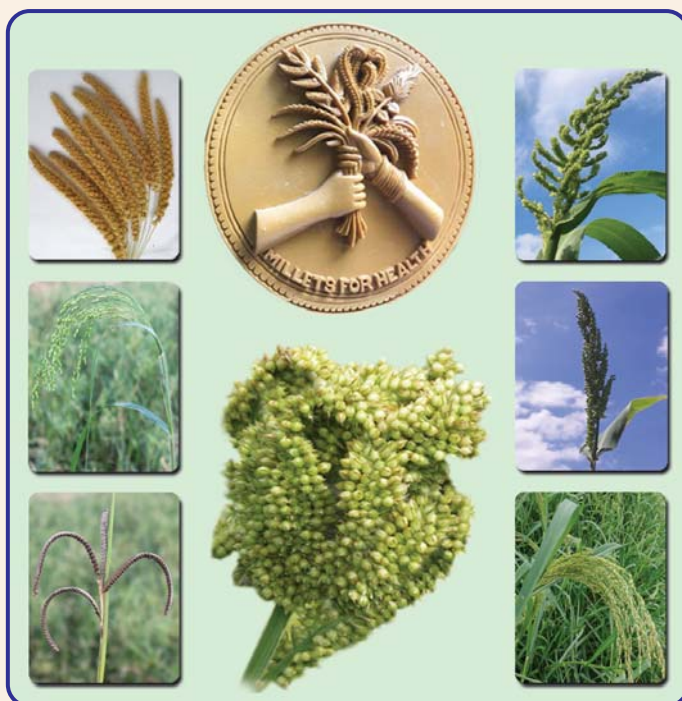
Source: Ministry of Agriculture & Farmers welfare, Govt. of India. (ON1953)

APPENDIX-III

India Position in Production of Millet in the World (2017)

Rank	Top 10 Countries	(In Tonne)
		Production
1	India	11560000
2	Niger	3790028
3	China	1996856
4	Mali	1806559
5	Nigeria	1500000
6	Ethiopia	1077616
7	Sudan	954000
8	Burkina Faso	828234
9	Chad	660175
10	Senegal	568939
	Other Countries	3716612
	World	28459020

Source: Food and Agriculture Organization (FAO). (ON1407)



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◆ Source of Case studies : AICRP Annual reports from 1986 to till date



G.I. = 38.97

FINGERMILLET (*Eleusine coracana*)

English name: Finger millet, Birds foot millet, African millet

Origin: India

Finger millet is commonly known as Ragi. It is an important minor millet grown in India. It is a staple food crop in many hilly regions of the country. In fact, it is the main cereal crop for monsoon season in some hilly areas. It is predominantly grown as a dry land crop in Karnataka, Andhra Pradesh, and Tamil Nadu. Ragi is relished mostly by the rural population of southern India for the nutritious meal it provides. It is grown both for grain and forage. In northern hills, grains are eaten mostly in the form of 'Chapaties' and halwa. In South India, grains are used in many preparations like cakes, puddings, sweets etc.



G.I = 50.8

KORRA (*Setaria italica*)

English Names: Italianmillet, Foxtailmillet, Germanmillet

Origin: China

It is grown in many of the Asian, African and American countries especially where the climatic hazards do not permit the cultivation of other cereals. It is grown as a rainfed crop in India. The grains are cooked like rice and then utilized. In some areas the grains are ground to flour and used in the form of chapaties.



G.I = 58.5

ARIKA (*Paspalum scrobiculatum*)

English name: Kodomillet

Origin: South-East Asia

Kodomillet is a highly drought resistant crop. It is the coarsest of all the food grains. The grain is covered with a horny seed coat which should be removed before cooking. Immature and molded grains are supposed to be poisonous. Kodo grain is easily preserved and proves as a good famine reserve. The grain is recommended as a substitute for rice to patients suffering from diabetes disease. Arika is grown mostly in A.P., Maharastra, Karnataka, Tamilnadu and U.P.



G.I = 52.5

VARIGA (*Panicum miliaceum*)

English names : Prosomillet or Commonmillet

Origin : India

This is an important minor millet grown in India. The crop is able to evade drought by its quick maturity. Being a short duration crop (80-90 days) with relatively low water requirement, it escapes drought and, therefore, offers better prospects for intensive cultivation in dry land areas. Where irrigation facilities are available, this is profitably grown as summer catch crop in high intensity rotations. It is quick growing grain crop, often grown during emergencies, famines or when sowings of other crops is abnormally delayed. Grains are cooked like rice, flour is used for making "Chapaties". It can be used in making 'Kheer'. It is a good poultry feed. Straw is good fodder for cattle.



G.I=52.11

SAMA (*Panicum sumatrense*)

English name: Littlemillet

Sama is known as Poorman's crop. It grows on the soils which are unsuitable for other crops. Like most millets, it is also a good famine food as it can produce some grain under drought conditions when all the other crops fail to produce. It has a potential as fodder crop as it has been reported that some of the little millet types mature in about 45-50 days. The husked grain of this millet is cooked almost in a manner similar to rice or rotis can be prepared with this flour.



G.I=34.2

OODA (*Echinochloa frumentacea*)

English name : Barnyardmillet

Barnyardmillet cultivated for food and fodder by hilly and tribal communities in Asia particularly in India, China and Japan. The crop has wide adaptability and occupies a special place in marginal rainfed areas because of its short life cycle. It has been identified as a suitable choice for climate-resilient agriculture. High nutrient content and antioxidant effects make it to be considered as a functional food crop.

*Glycemic Index (G.I.) = Conversion factor of carbohydrates to glucose with in the human body

